Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```python
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as f
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_wl_2h_D2_energy', 'r_NH3_wl_2h_D2_energy_rel', 'r_temperature_wl_2h_D2_energy', 'r_temperature_wl_2h_D2_energy_rel', 'r_umidity_wl_2h_D2_energy', 'r_umidity_wl_2h_D2_energy_rel']

1.2 Metadata

Nombre de lignes df	7793
Nombre de lignes df clean	7793
Nombre de features	47
% de NaN	0.0
Taille dataset entraînement	5455
Taille dataset test	2337
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_D1_energy_rel, r_NH3_wl_2h_A_energy_rel, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D2_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_NH3_wl_8h_A_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D2_energy, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel, r_umidity_wl_8h_D1_energy, r_umidity_wl_8h_D2_energy, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_energy, r_umidity_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy_rel, r_umidity_wl_8h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

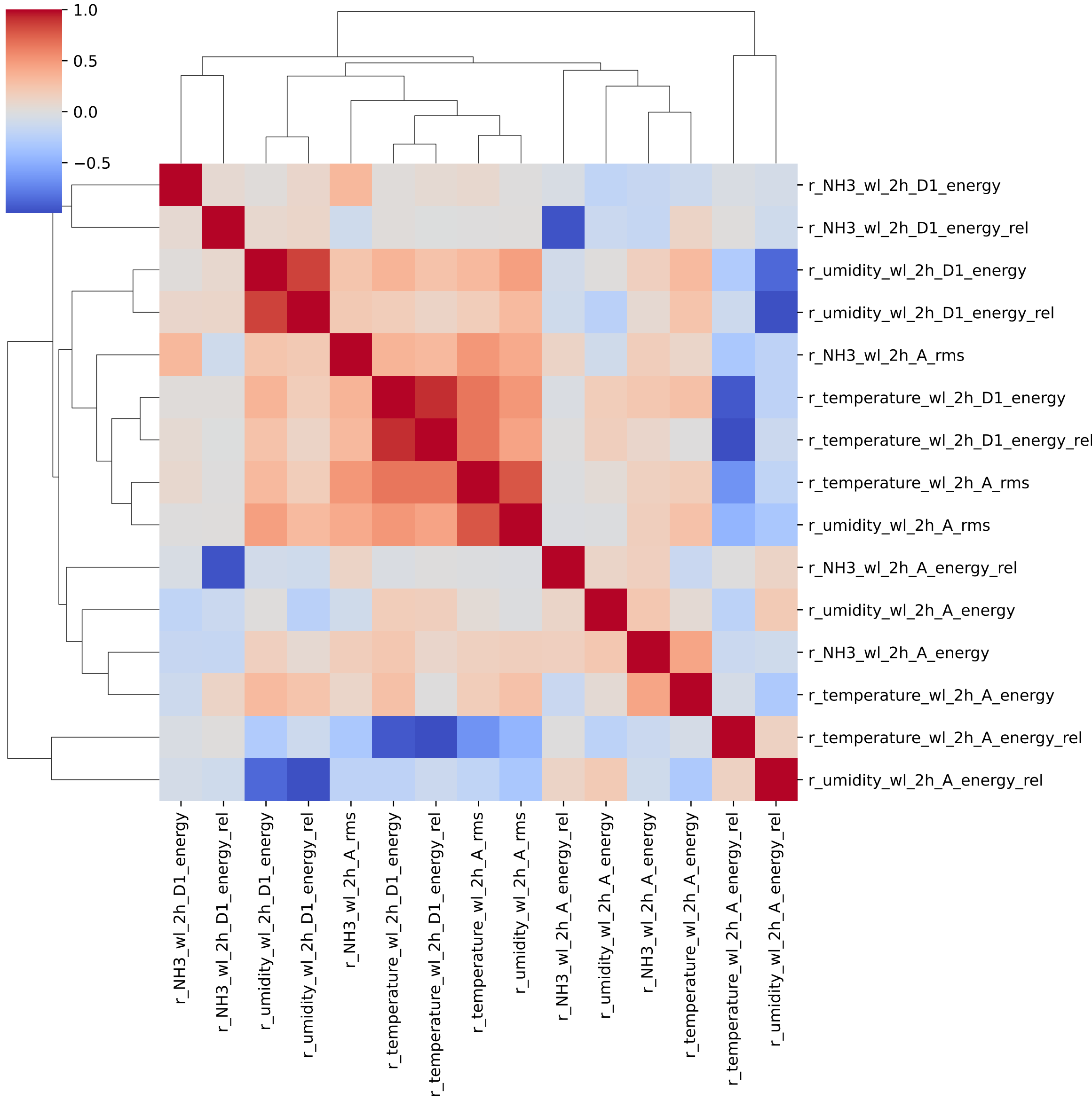
cluster 1: r, *, N, H, 3 ; **cluster 2:** r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ; **cluster 4:** r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_wl_2h_D1_energy ; **cluster 7:** r_NH3_wl_2h_A_energy ; **cluster 8:** r_NH3_wl_2h_A_rms ; **cluster 9:** r_NH3_wl_2h_A_energy_rel, r_NH3_wl_2h_D1_energy_rel ; **cluster 10:** r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_D1_energy, r_temperature_wl_2h_A_energy_rel ; **cluster 11:** r_temperature_wl_2h_A_energy ; **cluster 12:** r_temperature_wl_2h_A_rms ; **cluster 13:** r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_D1_energy_rel, r_umidity_wl_2h_A_energy_rel ; **cluster 14:** r_umidity_wl_2h_A_energy ; **cluster 15:** r_umidity_wl_2h_A_rms ; **cluster 16:** r_NH3_wl_8h_D1_energy ; **cluster 17:** r_NH3_wl_8h_D2_energy ; **cluster 18:** r_NH3_wl_8h_D3_energy ; **cluster 19:** r_NH3_wl_8h_A_energy ; **cluster 20:** r_NH3_wl_8h_A_rms ; **cluster 21:** r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_A_energy_rel, r_NH3_wl_8h_D2_energy_rel ; **cluster 22:** r_NH3_wl_8h_D3_energy_rel ; **cluster 23:** r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D2_energy_rel, r_temperature_wl_8h_D1_energy, r_temperature_wl_8h_D1_energy_rel, r_temperature_wl_8h_D2_energy ; **cluster 24:** r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_D3_energy_rel ; **cluster 25:** r_temperature_wl_8h_A_energy, r_umidity_wl_8h_A_energy ; **cluster 26:** r_temperature_wl_8h_A_rms ; **cluster 27:** r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_A_energy_rel, r_umidity_wl_8h_D1_energy ; **cluster 28:** r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D2_energy ; **cluster 29:**

r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_D3_energy_rel ; **cluster 30:** r_umidity_wl_8h_A_rms ;

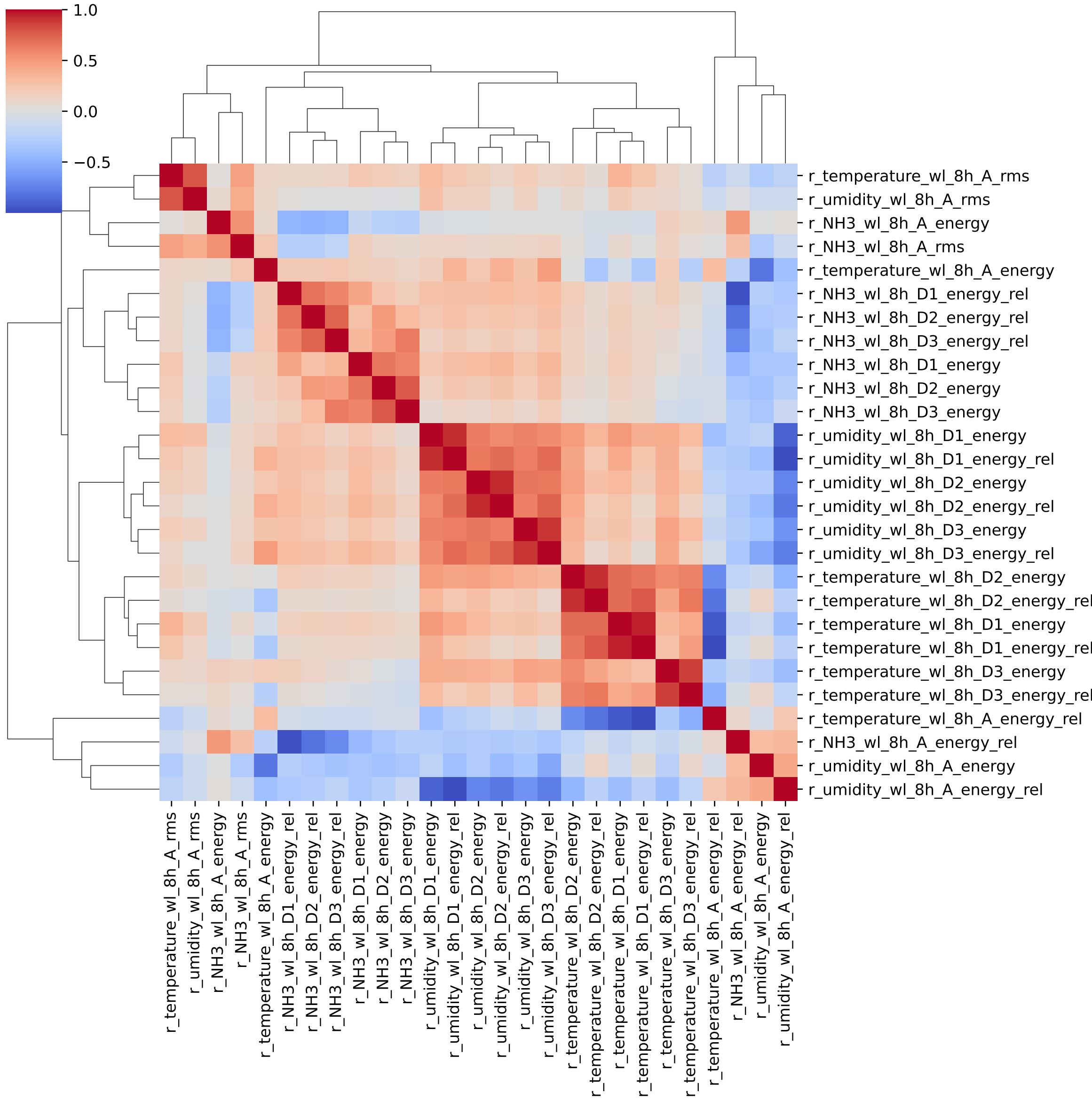
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_rms



1.0
0.5
0.0
−0.5
r_NH3_wl_2h_D1_energy
r_NH3_wl_2h_D1_energy_rel
r_umidity_wl_2h_D1_energy
r_umidity_wl_2h_D1_energy_rel
r_NH3_wl_2h_A_rms
r_temperature_wl_2h_D1_energy
r_temperature_wl_2h_D1_energy_rel
r_temperature_wl_2h_A_rms
r_umidity_wl_2h_A_rms
r_NH3_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy
r_NH3_wl_2h_A_energy
r_temperature_wl_2h_A_energy
r_temperature_wl_2h_A_energy_rel
r_umidity_wl_2h_A_energy_rel



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_2h_D1_energy, r_NH3_wl_2h_A_energy, r_NH3_wl_2h_A_rms, r_NH3_wl_2h_A_energy_rel, r_temperature_wl_2h_D1_energy_rel, r_temperature_wl_2h_A_energy, r_temperature_wl_2h_A_rms, r_umidity_wl_2h_D1_energy, r_umidity_wl_2h_A_energy, r_umidity_wl_2h_A_rms, r_NH3_wl_8h_D1_energy, r_NH3_wl_8h_D2_energy, r_NH3_wl_8h_D3_energy, r_NH3_wl_8h_A_energy, r_NH3_wl_8h_A_rms, r_NH3_wl_8h_D1_energy_rel, r_NH3_wl_8h_D3_energy_rel, r_temperature_wl_8h_A_energy_rel, r_temperature_wl_8h_D3_energy, r_temperature_wl_8h_A_energy, r_temperature_wl_8h_A_rms, r_umidity_wl_8h_D1_energy_rel, r_umidity_wl_8h_D2_energy_rel, r_umidity_wl_8h_D3_energy, r_umidity_wl_8h_A_rms

Len(Features_X): 30

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

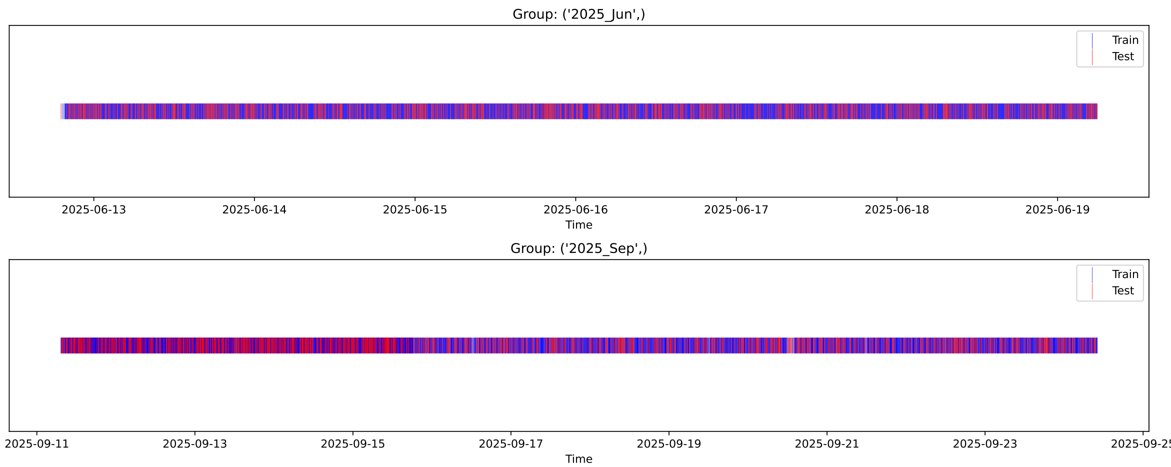
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5455 samples Test: 2338 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 2/5

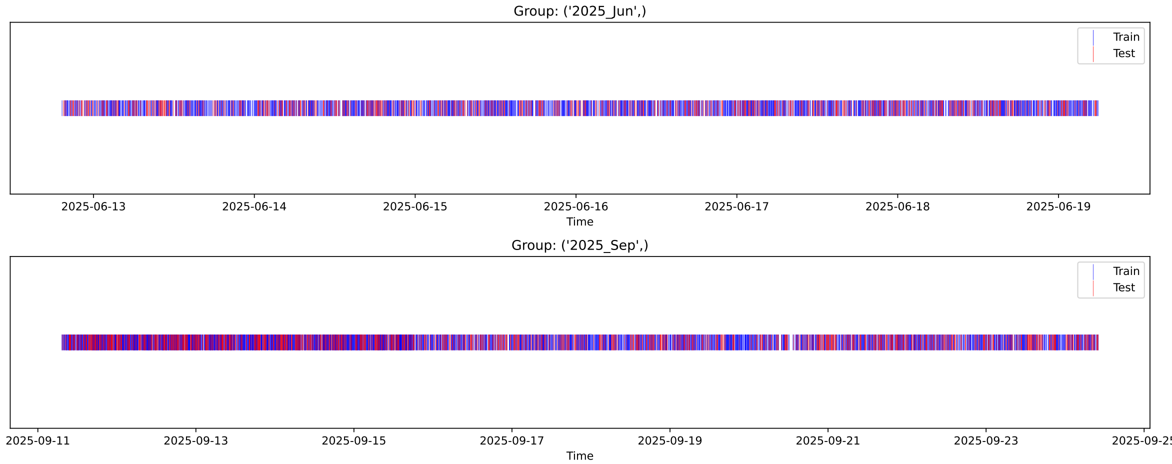
Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 3/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

Fold 4/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3818 samples Test: 1637 samples Excluded: 0 samples Window exclusion: 0h

3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

Fold 0 — y_test: mean=2.33, std=0.78, y_train: mean=2.35, std=0.81

Fold 1 — y_test: mean=2.34, std=0.79, y_train: mean=2.34, std=0.80

Fold 2 — y_test: mean=2.32, std=0.77, y_train: mean=2.35, std=0.81

Fold 3 — y_test: mean=2.33, std=0.78, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.76, y_train: mean=2.34, std=0.82

train scores CV: [0.56723788 0.55640366 0.56235584 0.55076586 0.56210594]

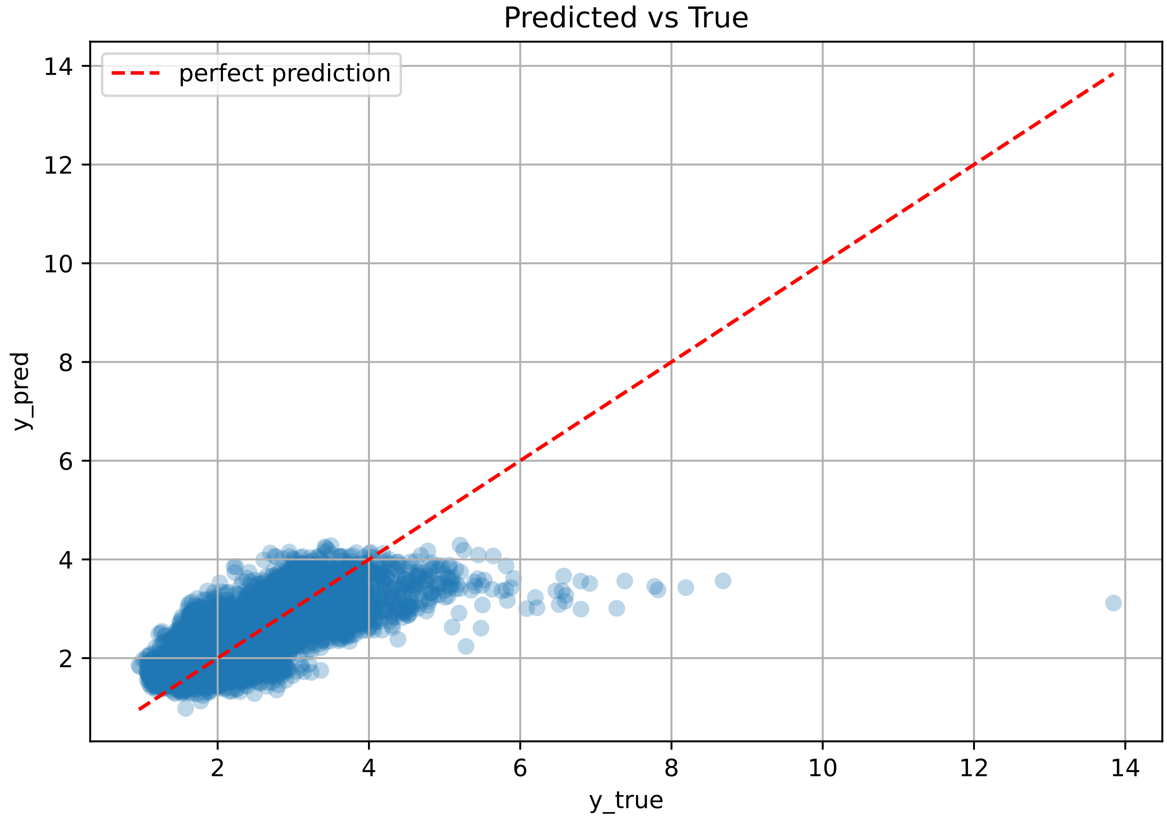
test scores CV: [0.57190033 0.59831718 0.58528624 0.61014038 0.58231006]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5598	0.5896	0.0133	0.5736	0.5742	-0.0154	0.9738	0.9494

Coefficients:

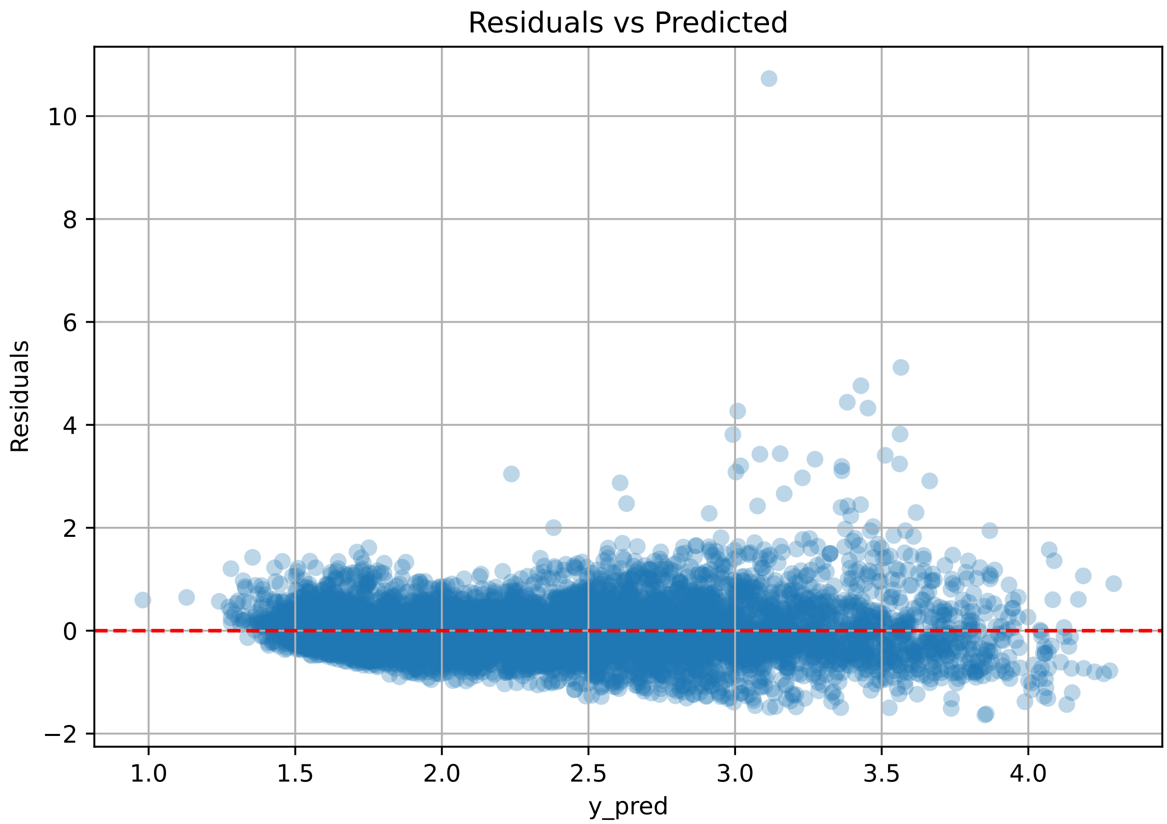
coef_r_CO2 = -0.0600, coef_r_NH3 = 0.2088, coef_r_NH3_wl_2h_A_energy = -0.1841, coef_r_NH3_wl_2h_A_energy_rel = 0.0154, coef_r_NH3_wl_2h_A_rms = -0.0048, coef_r_NH3_wl_2h_D1_energy = -0.0046, coef_r_NH3_wl_8h_A_energy = -0.1419, coef_r_NH3_wl_8h_A_rms = 0.0241, coef_r_NH3_wl_8h_D1_energy = 0.0735, coef_r_NH3_wl_8h_D1_energy_rel = -0.0316, coef_r_NH3_wl_8h_D2_energy = 0.0949, coef_r_NH3_wl_8h_D3_energy = -0.0307, coef_r_NH3_wl_8h_D3_energy_rel = -0.0598, coef_r_THI = 0.5835, coef_r_temperature = -0.4135, coef_r_temperature_wl_2h_A_energy = -0.0414, coef_r_temperature_wl_2h_A_rms = 0.1550, coef_r_temperature_wl_2h_D1_energy_rel = 0.0159, coef_r_temperature_wl_8h_A_energy = -0.1623, coef_r_temperature_wl_8h_A_energy_rel = 0.0007, coef_r_temperature_wl_8h_A_rms = -0.0024, coef_r_temperature_wl_8h_D3_energy = -0.0721, coef_r_umidity = -0.5450, coef_r_umidity_wl_2h_A_energy = 0.1006, coef_r_umidity_wl_2h_A_rms = 0.0364, coef_r_umidity_wl_2h_D1_energy = 0.0505, coef_r_umidity_wl_8h_A_rms = -0.1148, coef_r_umidity_wl_8h_D1_energy_rel = -0.0384, coef_r_umidity_wl_8h_D2_energy_rel = -0.0253, coef_r_umidity_wl_8h_D3_energy = 0.0319, intercept = 2.3427,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

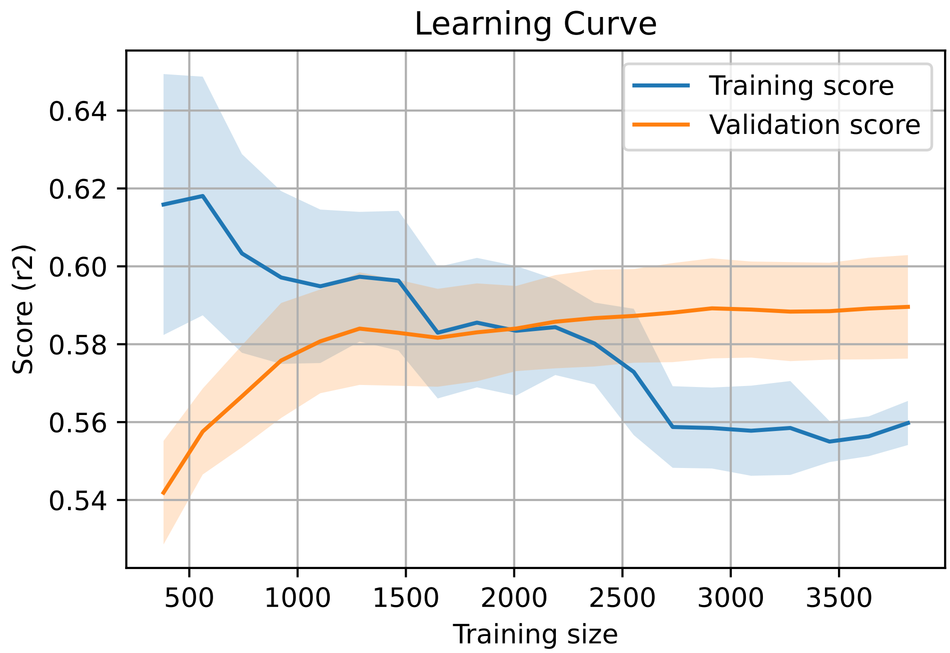
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337

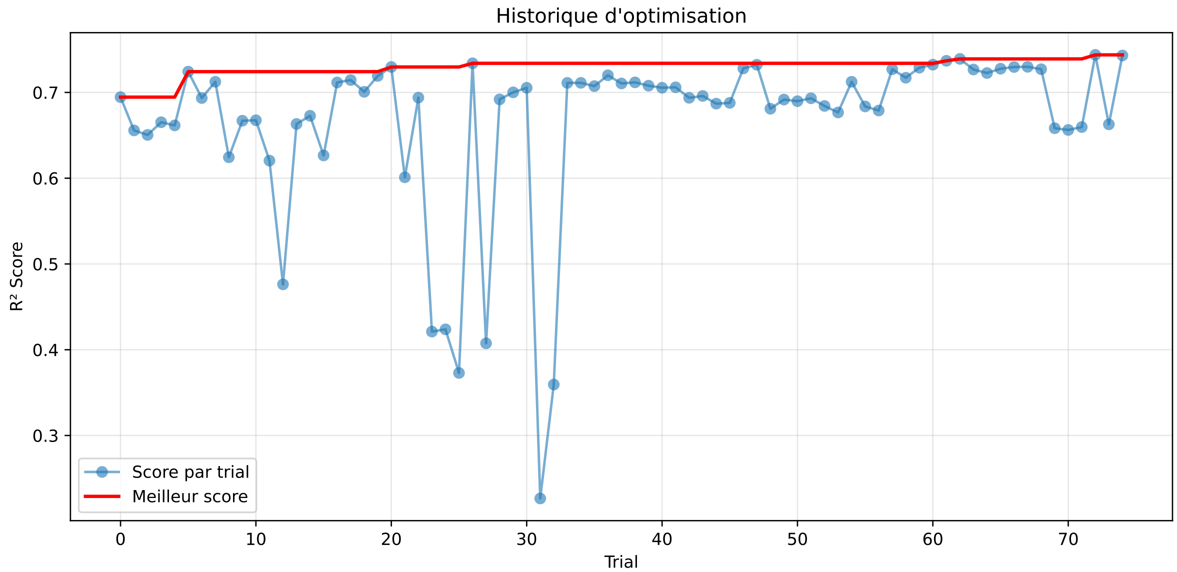


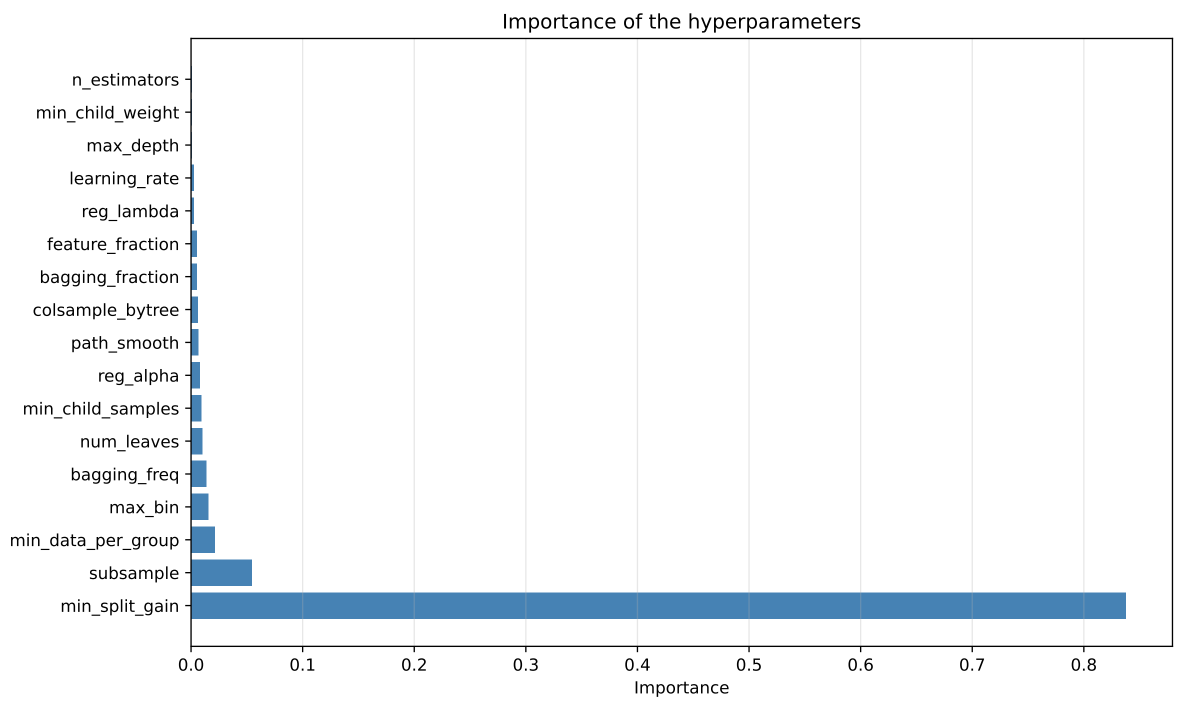
3.5 Model : LGBM

3.5.1 Gridsearch

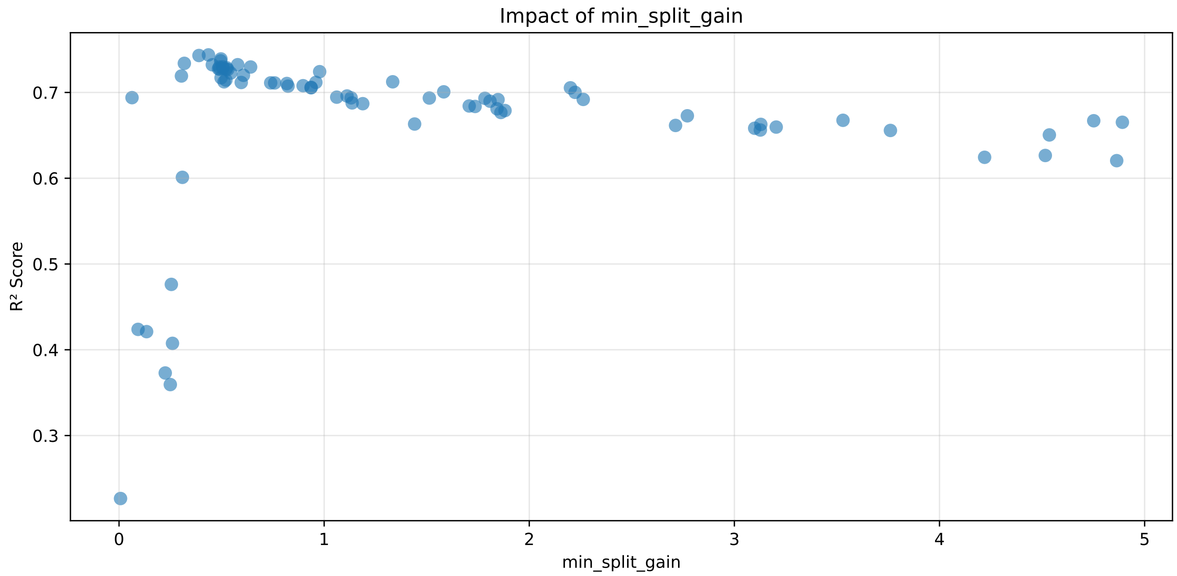
Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

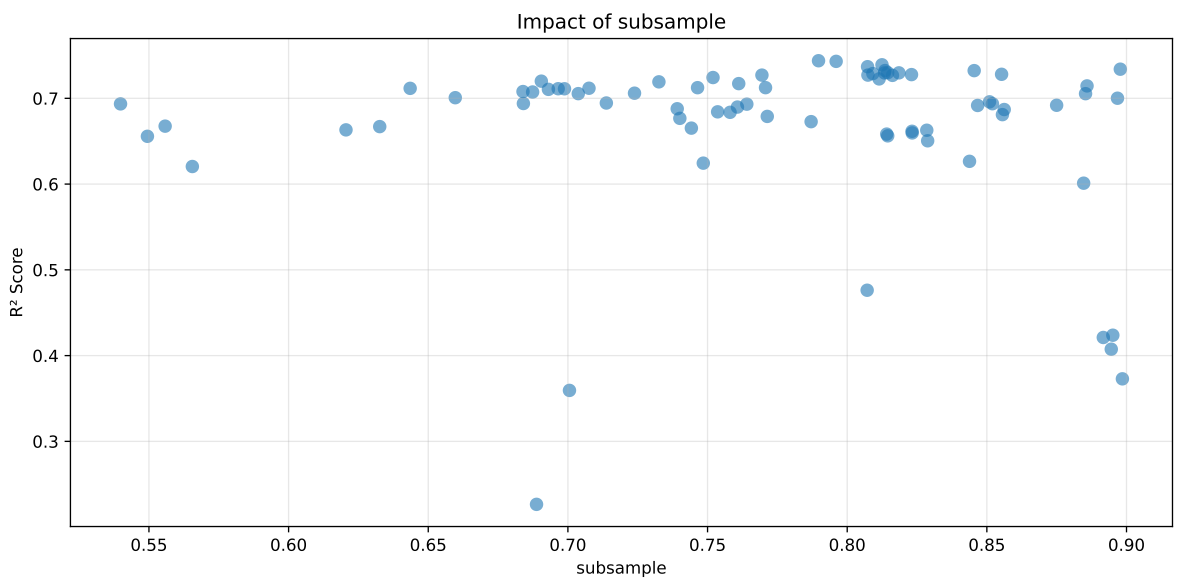
==================== TOP Importance FEATURES ==================== feature importance r_CO2 1132 r_temperature_wl_8h_A_energy 773 r_NH3 742 r_NH3_wl_8h_A_energy 742 r_umidity 723 r_NH3_wl_8h_D3_energy_rel 619 r_temperature 483 r_NH3_wl_2h_A_energy 468 r_THI 413 r_umidity_wl_2h_A_energy 387 r_temperature_wl_2h_A_rms 351 r_NH3_wl_8h_D1_energy 343 r_NH3_wl_8h_D3_energy 339 r_umidity_wl_8h_A_rms 333 r_temperature_wl_8h_A_rms 322 r_temperature_wl_8h_A_energy_rel 287 r_NH3_wl_8h_A_rms 279 r_umidity_wl_2h_D1_energy 265 r_temperature_wl_2h_A_energy 255 r_umidity_wl_2h_A_rms 246 r_NH3_wl_8h_D2_energy 223 r_NH3_wl_2h_A_rms 221 r_NH3_wl_2h_D1_energy 216 r_temperature_wl_8h_D3_energy 215 r_umidity_wl_8h_D2_energy_rel 202 r_umidity_wl_8h_D3_energy 185 r_umidity_wl_8h_D1_energy_rel 163 r_NH3_wl_2h_A_energy_rel 157 r_NH3_wl_8h_D1_energy_rel 140 r_temperature_wl_2h_D1_energy_rel 123





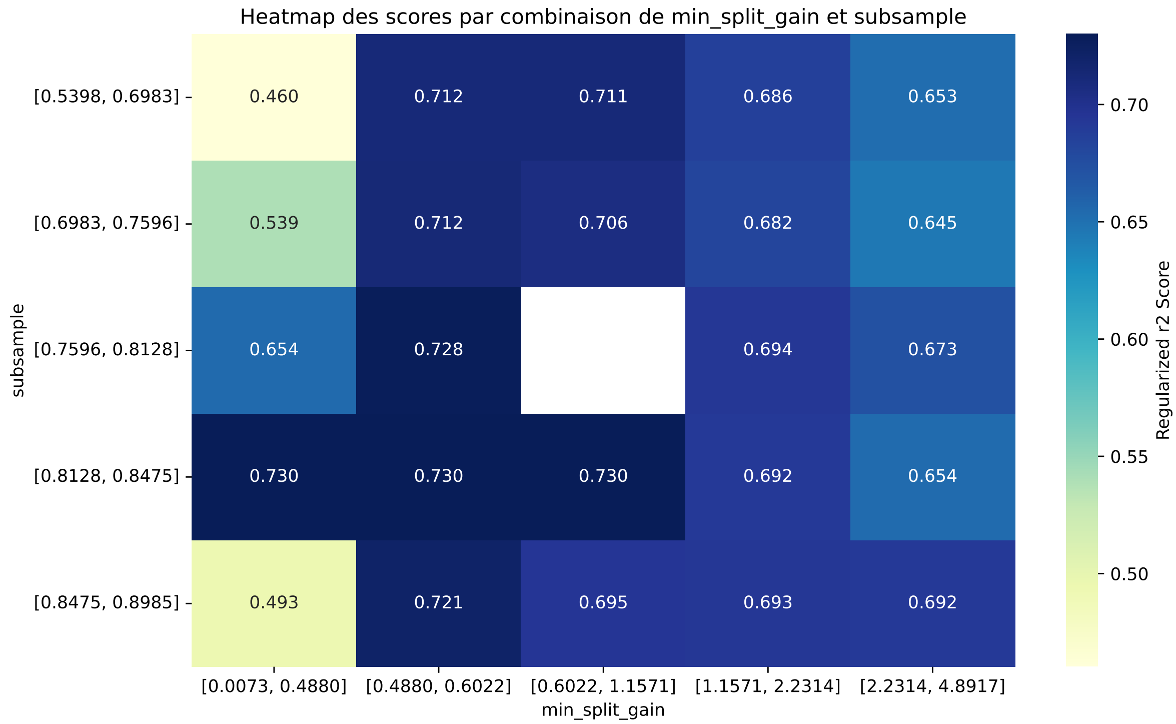
Paramètres avec >5% d'importance : ['min_split_gain', 'subsample']






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main
```

```
The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7799	0.7438	0.7396	0.0079	0.7438	0.7577	0.0139	1.0187	1.0485
2	0.7802	0.7431	0.7383	0.0104	0.7431	0.753	0.0099	1.0133	1.0499
3	0.7672	0.7391	0.7346	0.0097	0.7391	0.7491	0.01	1.0135	1.038
4	0.7635	0.7369	0.7324	0.0081	0.7369	0.7502	0.0133	1.0181	1.0361
5	0.7577	0.734	0.7311	0.0118	0.734	0.7475	0.0136	1.0185	1.0324
6	0.7551	0.7324	0.7279	0.0081	0.7324	0.7432	0.0108	1.0148	1.031
7	0.7588	0.7323	0.7279	0.0075	0.7323	0.743	0.0107	1.0147	1.0362
8	0.7501	0.7299	0.7251	0.0088	0.7299	0.7444	0.0145	1.0199	1.0277
9	0.7521	0.7297	0.7262	0.01	0.7297	0.7466	0.017	1.0232	1.0307
10	0.7484	0.7296	0.7239	0.0108	0.7296	0.7374	0.0077	1.0106	1.0257
---	---	---	---	---	---	---	---	---	---
75	0.9016	0.7891	0.7818	0.0077	0.2266	0.8026	0.0135	1.0171	1.1426

Hyperparameters:

Rank 1: {'reg___n_estimators': 800, 'reg___max_depth': 8, 'reg___learning_rate': 0.014002544536675302, 'reg___num_leaves': 56, 'reg___subsample': 0.7897317468749874, 'reg___colsample_bytree': 0.8933607723742003, 'reg___min_child_samples': 20, 'reg___reg_alpha': 3.579533358429392, 'reg___reg_lambda': 31.41664303228812, 'reg___min_split_gain': 0.435945405862964, 'reg___path_smooth': 3.641984970302704, 'reg___min_child_weight':

2.0382983252072684, 'reg___feature_fraction': 0.7668773166699461, 'reg___bagging_fraction': 0.6138747332116783, 'reg___bagging_freq': 5, 'reg___max_bin': 123, 'reg___min_data_per_group': 87}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 4, 'reg___learning_rate': 0.03540799775060347, 'reg___num_leaves': 51, 'reg___subsample': 0.7960415670075334, 'reg___colsample_bytree': 0.8971498606822725, 'reg___min_child_samples': 20, 'reg___reg_alpha': 3.4219377954599386, 'reg___reg_lambda': 36.15634860030508, 'reg___min_split_gain': 0.389645933885855, 'reg___path_smooth': 4.286926728285013, 'reg___min_child_weight': 2.3000291887033115, 'reg___feature_fraction': 0.5437277369737562, 'reg___bagging_fraction': 0.6655767024276757, 'reg___bagging_freq': 4, 'reg___max_bin': 123, 'reg___min_data_per_group': 102}

Rank 3: {'reg___n_estimators': 400, 'reg___max_depth': 8, 'reg___learning_rate': 0.038236972459725495, 'reg___num_leaves': 48, 'reg___subsample': 0.8124539686630756, 'reg___colsample_bytree': 0.8016470358482967, 'reg___min_child_samples': 27, 'reg___reg_alpha': 3.4486143917359144, 'reg___reg_lambda': 38.90452019063983, 'reg___min_split_gain': 0.4968211900111336, 'reg___path_smooth': 0.3283671320254994, 'reg___min_child_weight': 1.716980362691308, 'reg___feature_fraction': 0.524488852668408, 'reg___bagging_fraction': 0.645044395790917, 'reg___bagging_freq': 4, 'reg___max_bin': 109, 'reg___min_data_per_group': 98}

Rank 4: {'reg___n_estimators': 600, 'reg___max_depth': 8, 'reg___learning_rate': 0.02051059883202887, 'reg___num_leaves': 57, 'reg___subsample': 0.8072899546792739, 'reg___colsample_bytree': 0.806113675073099, 'reg___min_child_samples': 27, 'reg___reg_alpha': 3.544044740655587, 'reg___reg_lambda': 38.43780363865816, 'reg___min_split_gain': 0.4952512349830645, 'reg___path_smooth': 3.6059487504307635, 'reg___min_child_weight': 1.9866071711339903, 'reg___feature_fraction': 0.5321930920706197, 'reg___bagging_fraction': 0.632742559597498, 'reg___bagging_freq': 4, 'reg___max_bin': 110, 'reg___min_data_per_group': 86}

Rank 5: {'reg___n_estimators': 700, 'reg___max_depth': 6, 'reg___learning_rate': 0.058458324499934926, 'reg___num_leaves': 41, 'reg___subsample': 0.8977844914450209, 'reg___colsample_bytree': 0.6227657588297059, 'reg___min_child_samples': 22, 'reg___reg_alpha': 6.4855599826271915, 'reg___reg_lambda': 48.98086189091329, 'reg___min_split_gain': 0.3183936756401079, 'reg___path_smooth': 3.008595000526351, 'reg___min_child_weight': 0.45725951686150684, 'reg___feature_fraction': 0.6433772746574297, 'reg___bagging_fraction': 0.7326822767943479, 'reg___bagging_freq': 2, 'reg___max_bin': 63, 'reg___min_data_per_group': 127}

Rank 6: {'reg___n_estimators': 400, 'reg___max_depth': 8, 'reg___learning_rate': 0.019157207981744015, 'reg___num_leaves': 49, 'reg___subsample': 0.8135532129693579, 'reg___colsample_bytree': 0.7863121668059214, 'reg___min_child_samples': 14, 'reg___reg_alpha': 3.3967630416780747, 'reg___reg_lambda': 38.26942604724215, 'reg___min_split_gain': 0.578523360518306, 'reg___path_smooth': 1.4869937135979898, 'reg___min_child_weight':

5.665021400172414, 'reg__feature_fraction': 0.7550921660527354, 'reg__bagging_fraction': 0.6081455310606226, 'reg__bagging_freq': 4, 'reg__max_bin': 106, 'reg__min_data_per_group': 102}

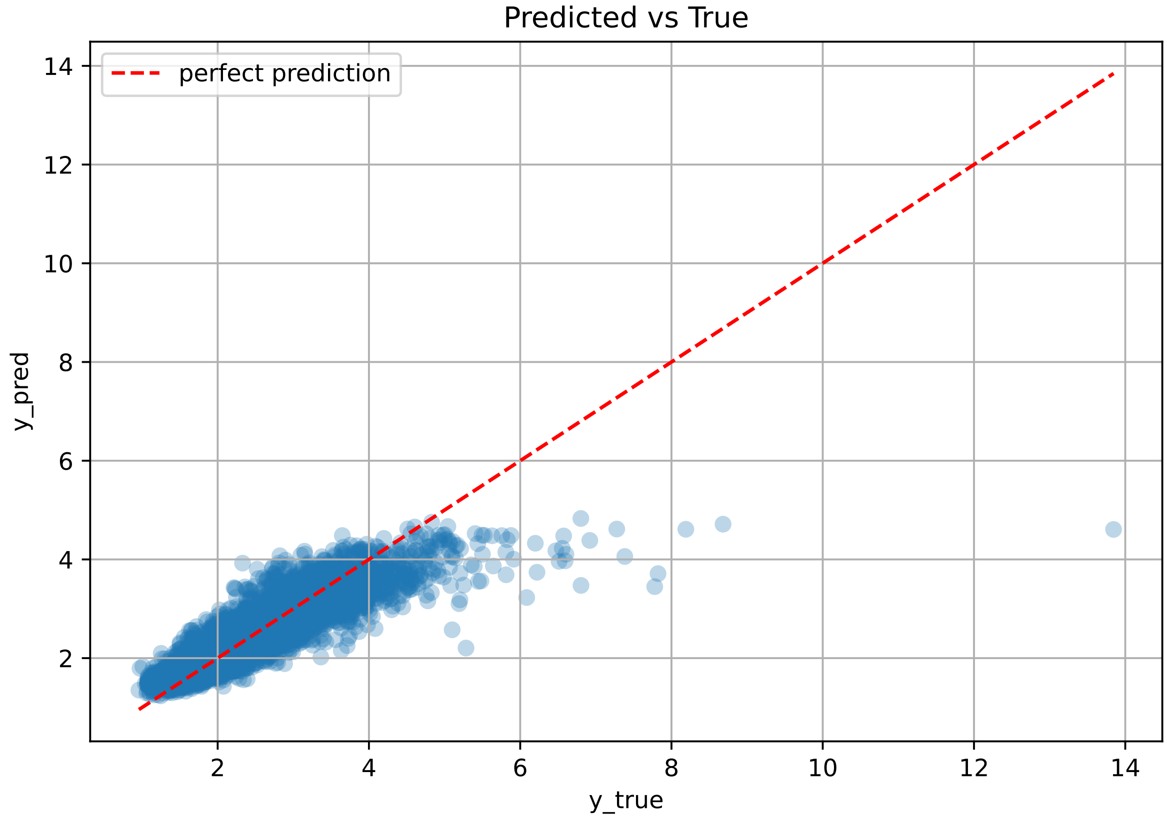
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.07031042438088461, 'reg__num_leaves': 52, 'reg__subsample': 0.8454626788386319, 'reg__colsample_bytree': 0.86537968603412, 'reg__min_child_samples': 17, 'reg__reg_alpha': 3.9701611038444344, 'reg__reg_lambda': 34.02881190199803, 'reg__min_split_gain': 0.45480241200811566, 'reg__path_smooth': 2.7779813131293736, 'reg__min_child_weight': 7.284509279485822, 'reg__feature_fraction': 0.5033351796618053, 'reg__bagging_fraction': 0.6918918899423833, 'reg__bagging_freq': 3, 'reg__max_bin': 87, 'reg__min_data_per_group': 77}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.03091694527522028, 'reg__num_leaves': 48, 'reg__subsample': 0.8144965577389078, 'reg__colsample_bytree': 0.7948809771728689, 'reg__min_child_samples': 15, 'reg__reg_alpha': 5.031581158608894, 'reg__reg_lambda': 31.19627798944448, 'reg__min_split_gain': 0.5054968597814647, 'reg__path_smooth': 2.357557909297738, 'reg__min_child_weight': 2.0086873848565108, 'reg__feature_fraction': 0.5372620949524844, 'reg__bagging_fraction': 0.6440887779277489, 'reg__bagging_freq': 4, 'reg__max_bin': 124, 'reg__min_data_per_group': 100}

Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.012911968687311118, 'reg__num_leaves': 63, 'reg__subsample': 0.8185153234871418, 'reg__colsample_bytree': 0.6949645589738889, 'reg__min_child_samples': 26, 'reg__reg_alpha': 4.714618928623631, 'reg__reg_lambda': 41.55121586022105, 'reg__min_split_gain': 0.6415235442045414, 'reg__path_smooth': 4.1492085419705695, 'reg__min_child_weight': 0.07321265481667083, 'reg__feature_fraction': 0.7380248001209534, 'reg__bagging_fraction': 0.7831468878929274, 'reg__bagging_freq': 2, 'reg__max_bin': 143, 'reg__min_data_per_group': 112}

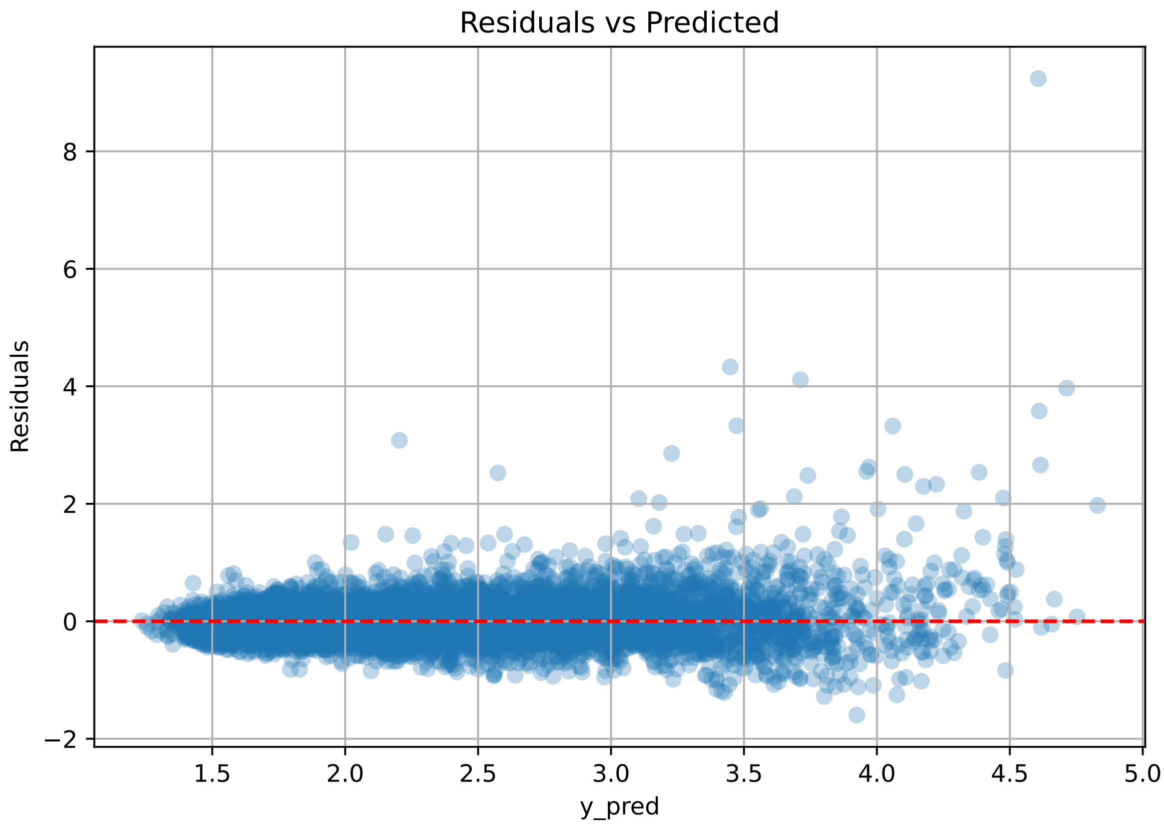
Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.0295388528970714, 'reg__num_leaves': 50, 'reg__subsample': 0.8133732965070346, 'reg__colsample_bytree': 0.7951387705194458, 'reg__min_child_samples': 14, 'reg__reg_alpha': 4.860991145791974, 'reg__reg_lambda': 36.3901129371259, 'reg__min_split_gain': 0.48878564328578916, 'reg__path_smooth': 2.3522500987254933, 'reg__min_child_weight': 1.810420279837668, 'reg__feature_fraction': 0.5356076650004116, 'reg__bagging_fraction': 0.6407652314963119, 'reg__bagging_freq': 4, 'reg__max_bin': 113, 'reg__min_data_per_group': 100}

3.5.2 Scatter



Predicted vs True
perfect prediction
14
12
10
8
6
4
2
y_pred
2
4
6
8
10
12
14
y_true

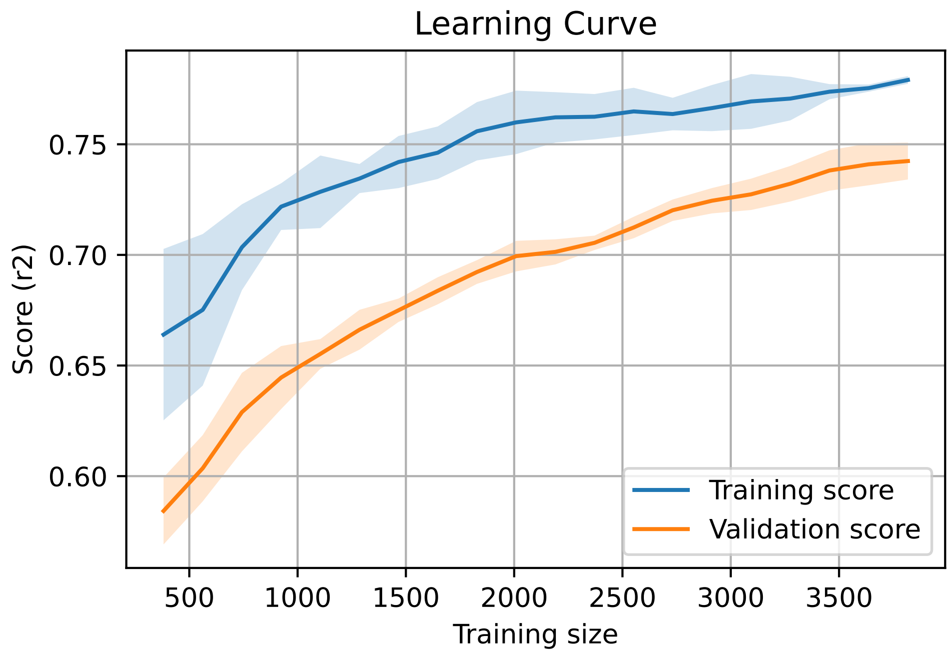
3.5.3 Scatter residuals



3.5.4 Learning curve

$\textbf{Len(group)} : 7793, \textbf{Len(training)} : 5456,$

$\textbf{Len(training\_real)} : 3820, \textbf{Len(test\_of\_training)} : 1636, \textbf{Len(test)} : 2337$

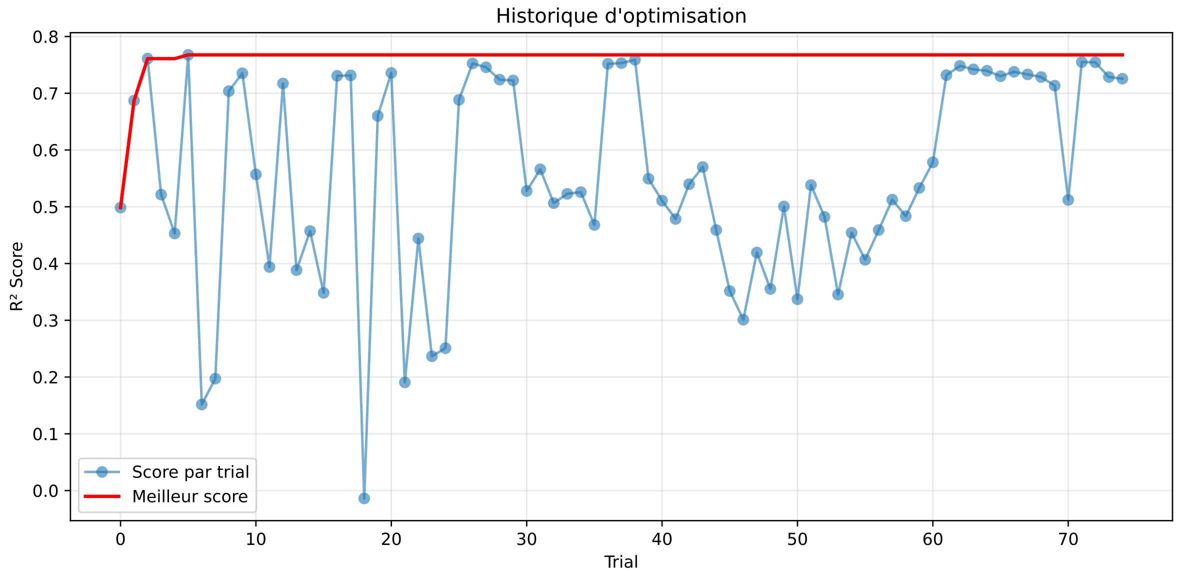


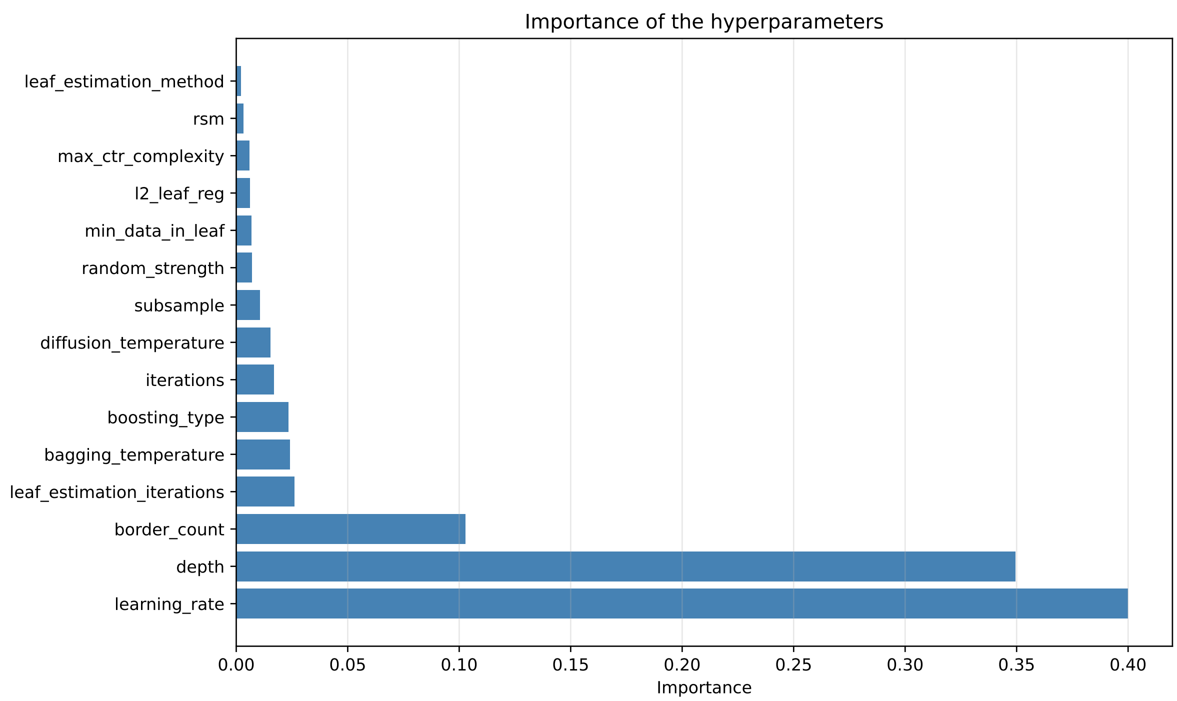
3.6 Model : CatBoost

3.6.1 Gridsearch

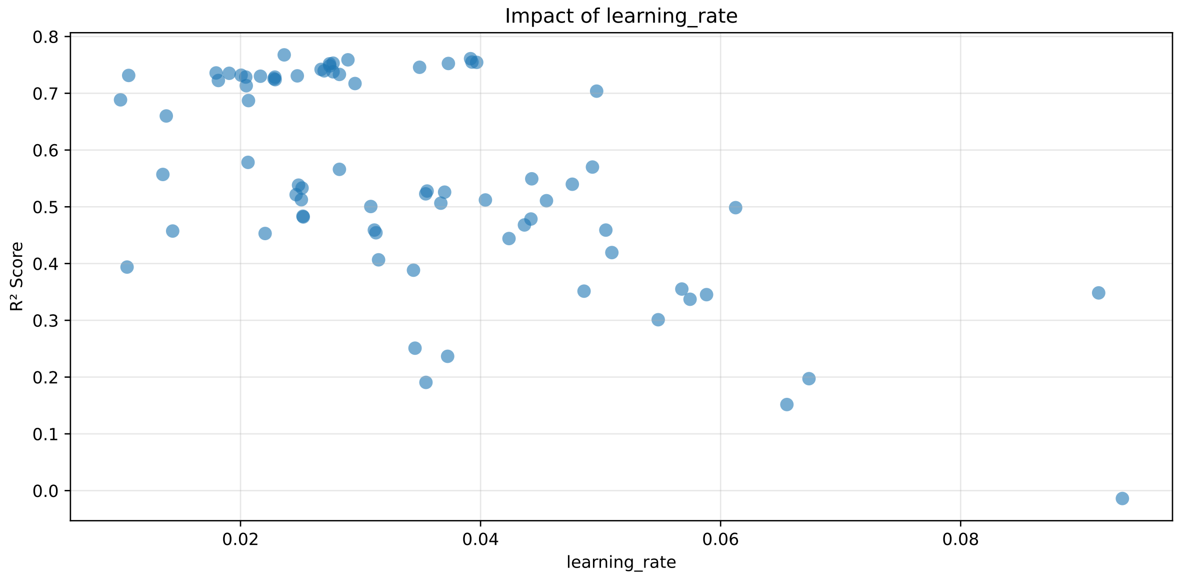
Distribution y_train vs y_test: y_train: mean=2.343, std=0.800, min=0.959, max=13.845 y_test: mean=2.367, std=0.801, min=0.973, max=7.822

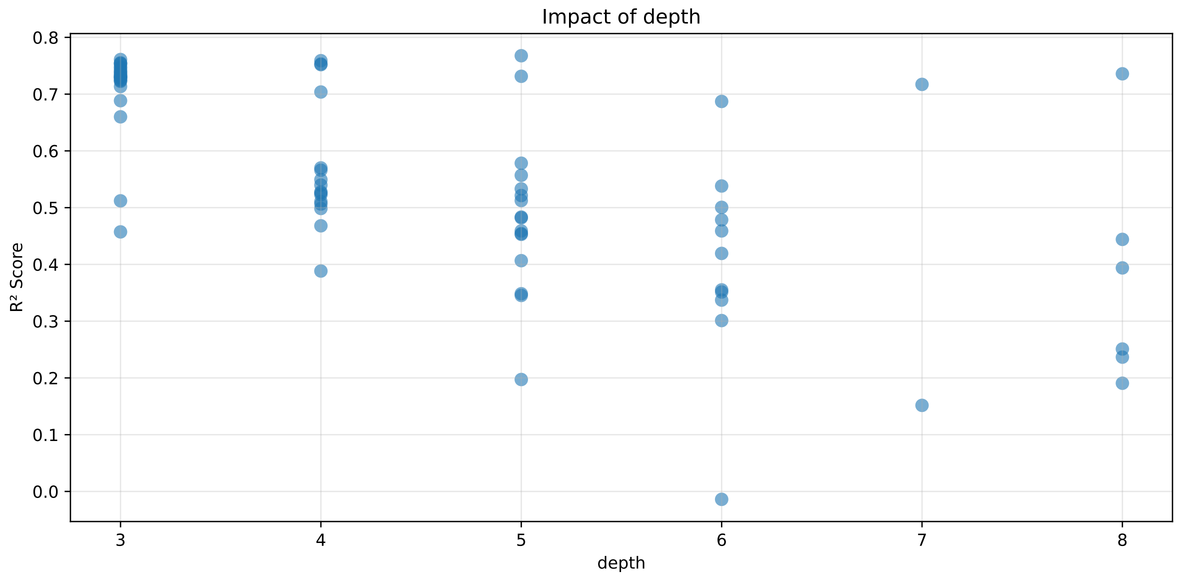
===================== TOP Importance FEATURES ===================== feature importance r_umidity 18.584165 r_temperature_wl_8h_A_energy 10.080632 r_temperature_wl_2h_A_rms 7.919864 r_CO2 6.984155 r_temperature_wl_8h_A_energy_rel 6.922484 r_temperature 5.130937 r_NH3 3.786205 r_THI 3.276275 r_NH3_wl_8h_D1_energy 3.259000 r_NH3_wl_8h_A_energy 3.112752 r_NH3_wl_8h_D3_energy 2.927578 r_NH3_wl_2h_A_energy 2.645962 r_temperature_wl_2h_D1_energy_rel 2.582373 r_umidity_wl_2h_A_energy 2.239115 r_NH3_wl_8h_D2_energy 2.132267 r_umidity_wl_2h_A_rms 1.996728 r_NH3_wl_8h_D3_energy_rel 1.866412 r_umidity_wl_8h_A_rms 1.724373 r_NH3_wl_8h_A_rms 1.529392 r_NH3_wl_2h_D1_energy 1.369775 r_temperature_wl_2h_A_energy 1.358080 r_temperature_wl_8h_D3_energy 1.270312 r_NH3_wl_2h_A_energy_rel 1.131523 r_NH3_wl_8h_D1_energy_rel 1.106327 r_temperature_wl_8h_A_rms 1.091297 r_NH3_wl_2h_A_rms 1.058504 r_umidity_wl_2h_D1_energy 0.820607 r_umidity_wl_8h_D1_energy_rel 0.810453 r_umidity_wl_8h_D3_energy 0.696968 r_umidity_wl_8h_D2_energy_rel 0.585486

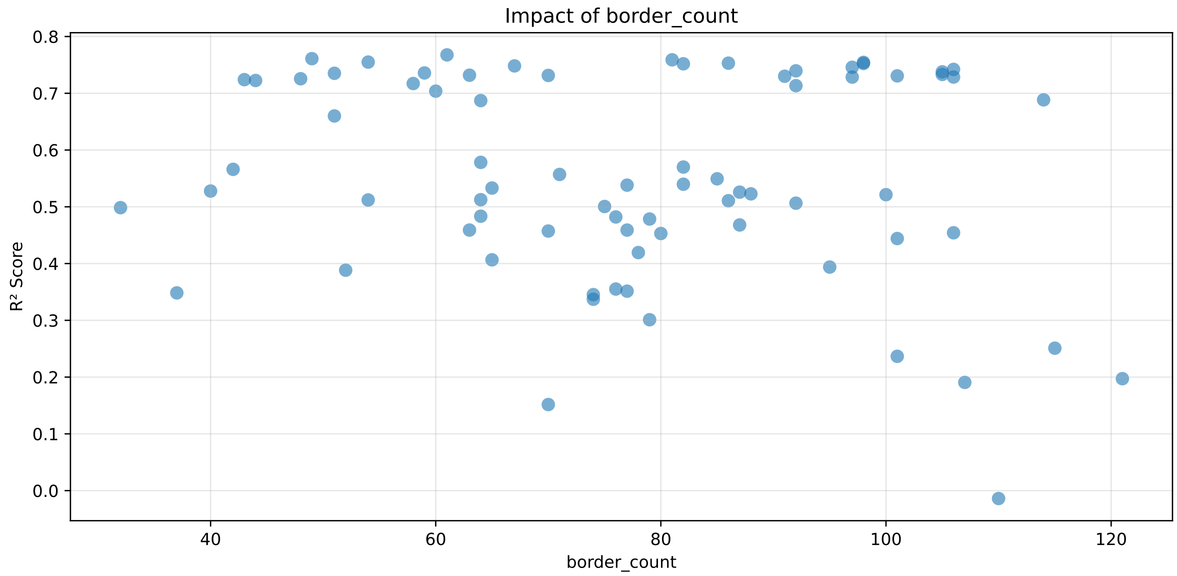




Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'border_count']


Impact of learning_rate
R² Score
learning_rate
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.02
0.04
0.06
0.08


Impact of depth
R² Score
depth
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
3
4
5
6
7
8



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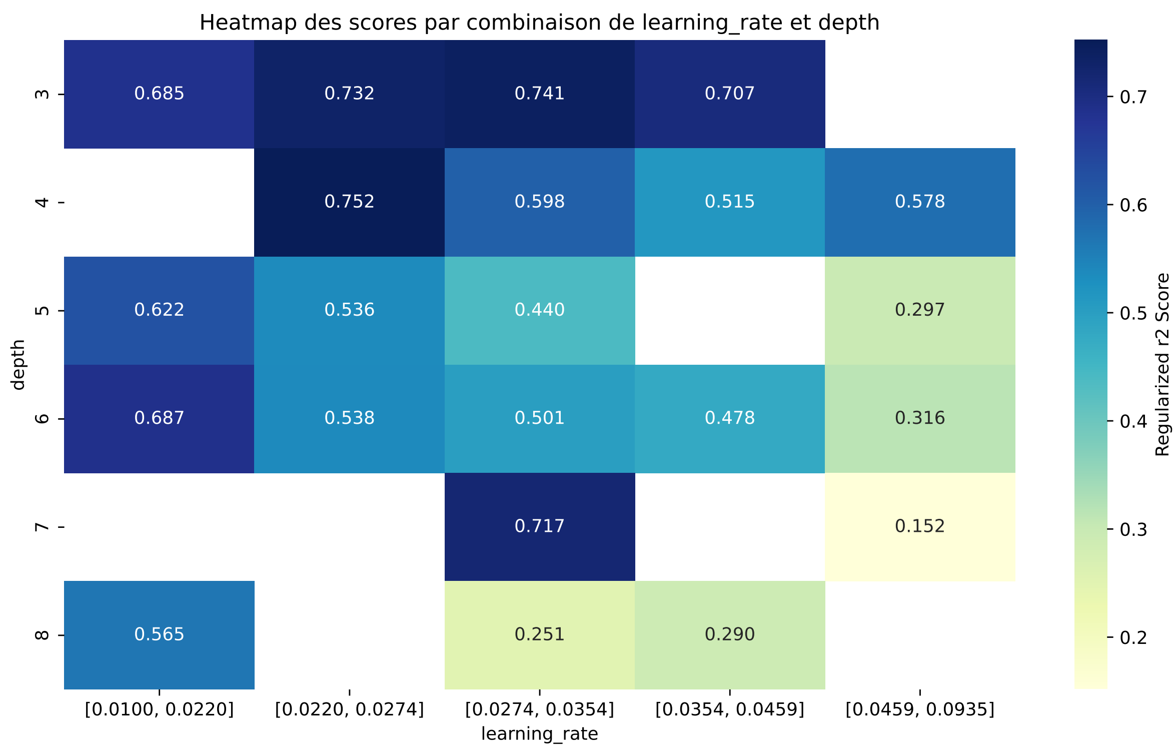
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.8027	0.7676	0.7587	0.006	0.7676	0.7656	-0.0021	0.9973	1.0457
2	0.7966	0.761	0.7549	0.0081	0.761	0.7582	-0.0028	0.9964	1.0469
3	0.7894	0.759	0.7534	0.0066	0.759	0.7592	0.0003	1.0004	1.0402
4	0.786	0.755	0.75	0.0076	0.755	0.7518	-0.0031	0.9958	1.0411
5	0.79	0.7545	0.7496	0.0051	0.7545	0.7552	0.0007	1.0009	1.0472
6	0.7772	0.7531	0.7465	0.0068	0.7531	0.7542	0.0011	1.0015	1.032
7	0.7866	0.7525	0.7493	0.0071	0.7525	0.7532	0.0007	1.001	1.0454
8	0.7794	0.7519	0.7475	0.006	0.7519	0.7542	0.0022	1.003	1.0365
9	0.7741	0.7482	0.7442	0.0069	0.7482	0.7483	0.0001	1.0001	1.0345
10	0.7693	0.7458	0.7419	0.0068	0.7458	0.745	-0.0008	0.999	1.0315
---	---	---	---	---	---	---	---	---	---
75	0.9821	0.8161	0.8028	0.0042	-0.0139	0.818	0.0019	1.0023	1.2034

Hyperparameters:

Rank 1: {'reg___iterations': 900, 'reg___depth': 5, 'reg___learning_rate': 0.023641295416644705, 'reg___l2_leaf_reg': 17.140900072092425, 'reg___bagging_temperature': 4.778545941698777, 'reg___random_strength': 1.6306058718662126, 'reg___subsample': 0.5999109277385857, 'reg___min_data_in_leaf': 49, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.5901771427395555, 'reg___border_count': 61, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 3095.6115288969777}

Rank 2: {'reg___iterations': 1000, 'reg___depth': 3, 'reg___learning_rate': 0.03916627788223819, 'reg___l2_leaf_reg': 42.0143285867659, 'reg___bagging_temperature': 1.0543137361724835, 'reg___random_strength': 0.3336403398543719, 'reg___subsample': 0.7187462365094158, 'reg___min_data_in_leaf': 15, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.7149996354118832, 'reg___border_count': 49, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 4, 'reg___diffusion_temperature': 5572.885967225966}

Rank 3: {'reg___iterations': 700, 'reg___depth': 4, 'reg___learning_rate': 0.028951334532561268, 'reg___l2_leaf_reg': 44.25158312906309, 'reg___bagging_temperature': 2.9143709027329505, 'reg___random_strength': 0.557901557416331, 'reg___subsample': 0.5830885167864497, 'reg___min_data_in_leaf': 43, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.5024083833948462, 'reg___border_count': 81, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 6776.966849126478}

Rank 4: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.03928434690562324, 'reg___l2_leaf_reg': 27.314023568072194, 'reg___bagging_temperature': 3.297342545655969, 'reg___random_strength': 0.31434539333827755, 'reg___subsample': 0.807467472813765, 'reg___min_data_in_leaf': 39, 'reg___leaf_estimation_iterations': 8, 'reg___rsm':

0.802771411912242, ’reg___border_count’: 54, ’reg___leaf_estimation_method’: ‘Newton’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 4121.918279942921}

Rank 5: {’reg___iterations’: 800, ’reg___depth’: 3, ’reg___learning_rate’: 0.039677668645949966, ’reg___l2_leaf_reg’: 36.04099771584629, ’reg___bagging_temperature’: 3.582129481592662, ’reg___random_strength’: 0.005807006845245155, ’reg___subsample’: 0.8064622581353678, ’reg___min_data_in_leaf’: 40, ’reg___leaf_estimation_iterations’: 8, ’reg___rsm’: 0.7983091454043802, ’reg___border_count’: 98, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 5978.009656041451}

Rank 6: {’reg___iterations’: 900, ’reg___depth’: 4, ’reg___learning_rate’: 0.027715031572729795, ’reg___l2_leaf_reg’: 34.5986134080955, ’reg___bagging_temperature’: 2.636780179626464, ’reg___random_strength’: 0.5614207839802855, ’reg___subsample’: 0.7339969322935284, ’reg___min_data_in_leaf’: 46, ’reg___leaf_estimation_iterations’: 1, ’reg___rsm’: 0.5424404386256632, ’reg___border_count’: 86, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 3, ’reg___diffusion_temperature’: 6570.8593546505745}

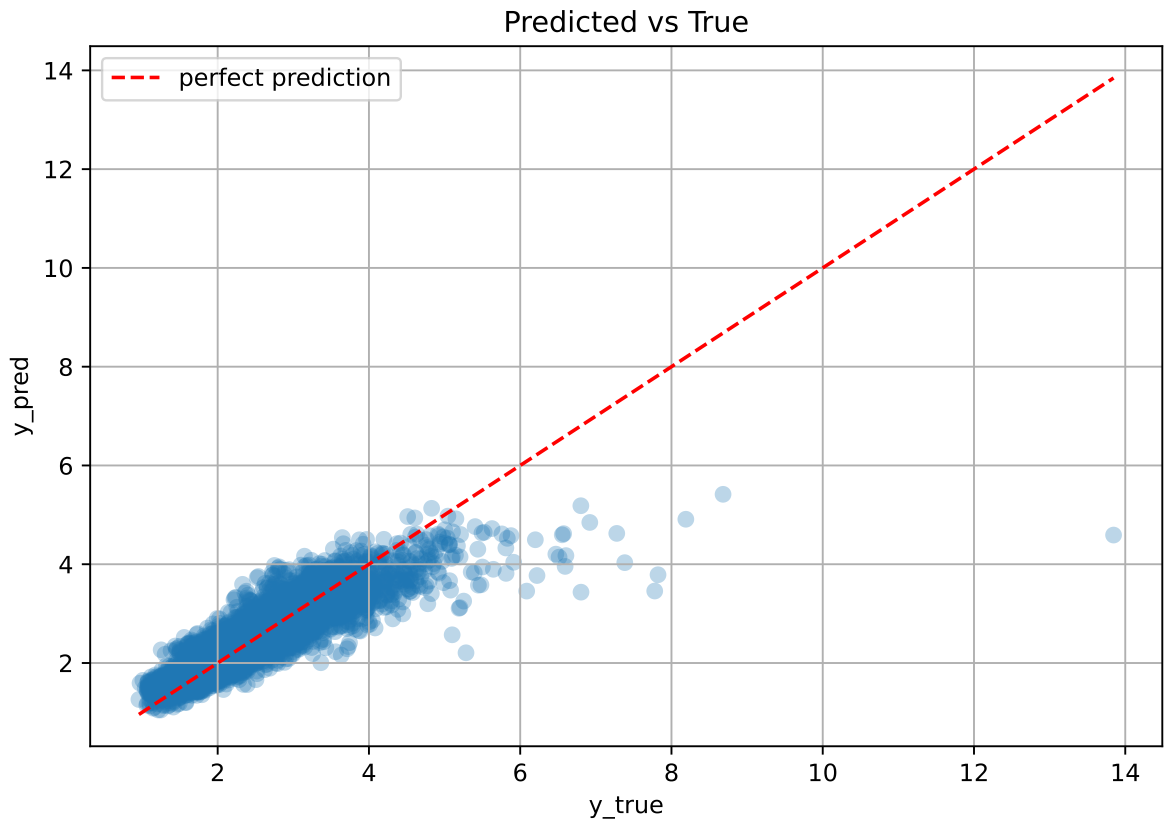
Rank 7: {’reg___iterations’: 800, ’reg___depth’: 3, ’reg___learning_rate’: 0.03731988673304314, ’reg___l2_leaf_reg’: 38.0890450714912, ’reg___bagging_temperature’: 1.5098005629290747, ’reg___random_strength’: 0.11838086126710878, ’reg___subsample’: 0.6488934037120763, ’reg___min_data_in_leaf’: 50, ’reg___leaf_estimation_iterations’: 10, ’reg___rsm’: 0.8937621660135759, ’reg___border_count’: 98, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 5522.610384713074}

Rank 8: {’reg___iterations’: 900, ’reg___depth’: 4, ’reg___learning_rate’: 0.027415820147577524, ’reg___l2_leaf_reg’: 34.852221926083, ’reg___bagging_temperature’: 2.8615453674715456, ’reg___random_strength’: 0.4033534930726874, ’reg___subsample’: 0.5793334644472776, ’reg___min_data_in_leaf’: 46, ’reg___leaf_estimation_iterations’: 1, ’reg___rsm’: 0.5020901362455208, ’reg___border_count’: 82, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 2, ’reg___diffusion_temperature’: 6692.737574997012}

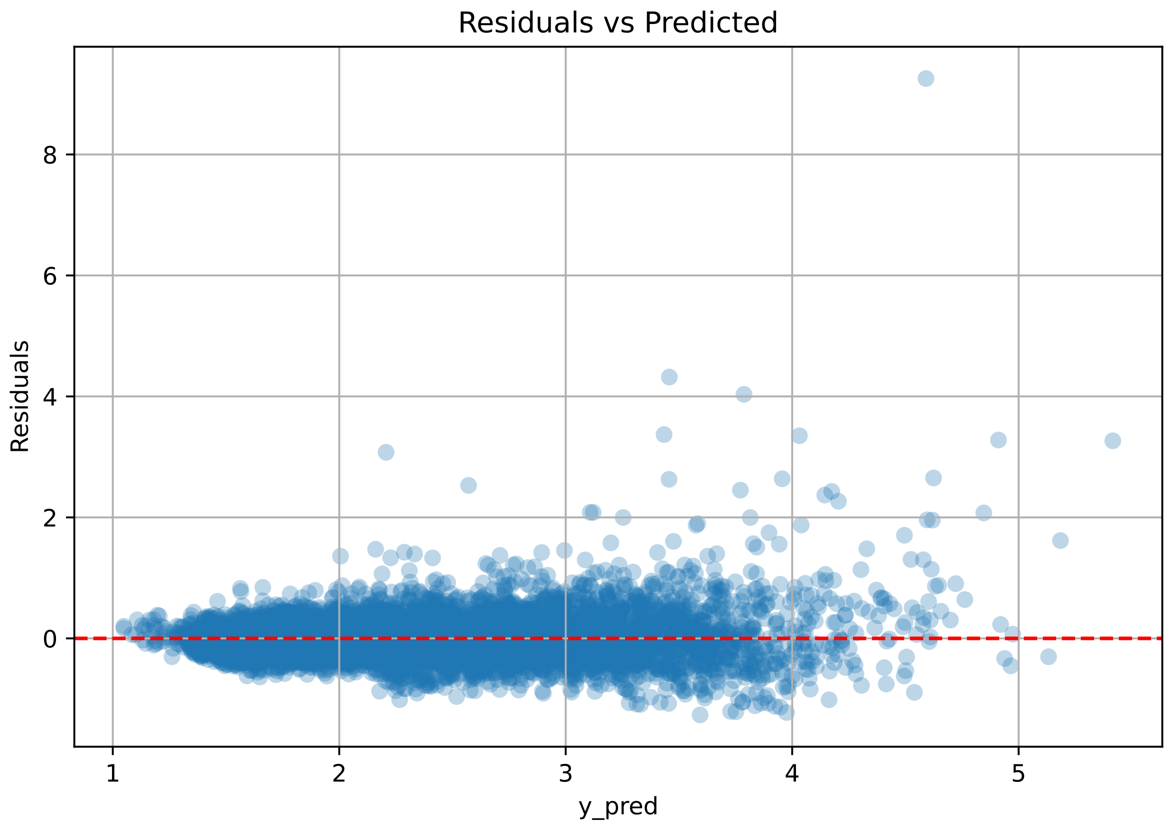
Rank 9: {’reg___iterations’: 1000, ’reg___depth’: 3, ’reg___learning_rate’: 0.02744048516219981, ’reg___l2_leaf_reg’: 29.843024317219147, ’reg___bagging_temperature’: 2.4523497215729417, ’reg___random_strength’: 0.2792363608619748, ’reg___subsample’: 0.6899241276417505, ’reg___min_data_in_leaf’: 39, ’reg___leaf_estimation_iterations’: 4, ’reg___rsm’: 0.622017028942361, ’reg___border_count’: 67, ’reg___leaf_estimation_method’: ‘Gradient’, ’reg___boosting_type’: ‘Ordered’, ’reg___max_ctr_complexity’: 3, ’reg___diffusion_temperature’: 3672.773290871043}

Rank 10: {’reg___iterations’: 800, ’reg___depth’: 3, ’reg___learning_rate’: 0.034920974245154376, ’reg___l2_leaf_reg’: 39.452088670754044, ’reg___bagging_temperature’: 1.6629671426470842, 'reg___random_strength': 1.9391269526046604, 'reg___subsample': 0.614365128321272, 'reg___min_data_in_leaf': 11, 'reg___leaf_estimation_iterations': 10, 'reg___rsm': 0.7315221597331361, 'reg___border_count': 97, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 5238.223514153906}

3.6.2 Scatter



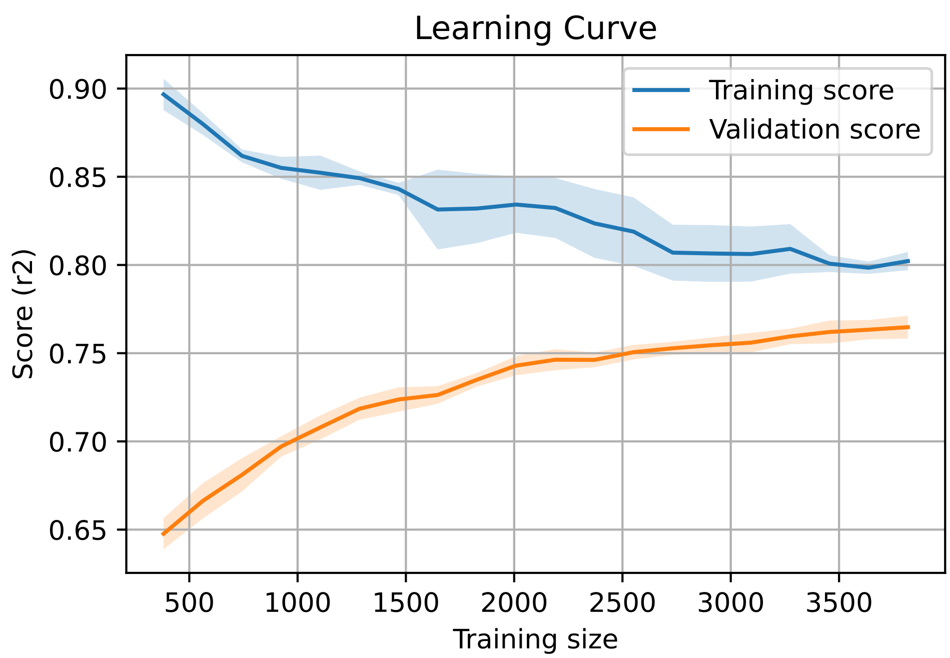
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 7793, **Len(training)** : 5456,

Len(training_real) : 3820, **Len(test_of_training)** : 1636, **Len(test)** : 2337



3.7 Sumup-table

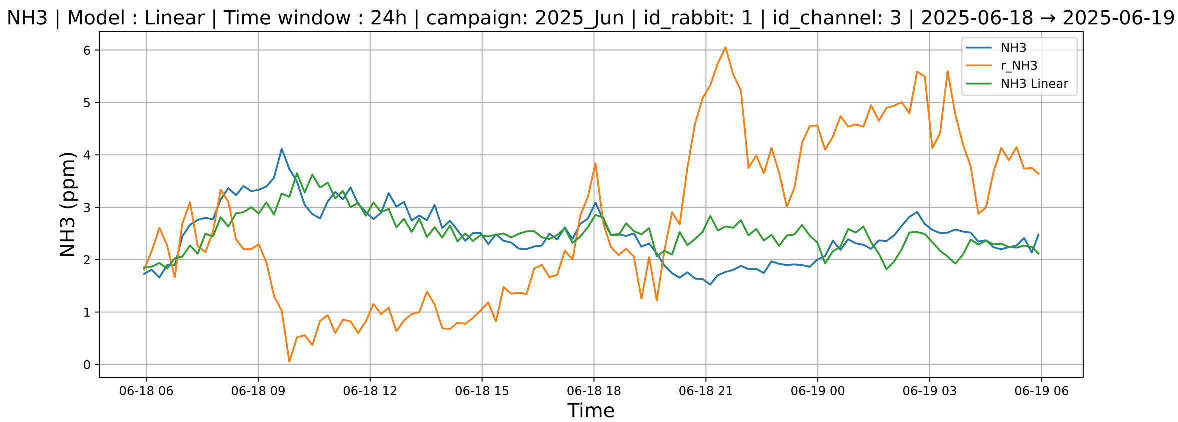
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.56	0.59	0.013	0.574	0.574	-0.015	0.974	0.949	nan
LGBM	0.78	0.744	0.008	0.74	0.758	0.014	1.019	1.048	0.7438
CatBoost	0.803	0.768	0.006	0.759	0.766	-0.002	0.997	1.046	0.7676

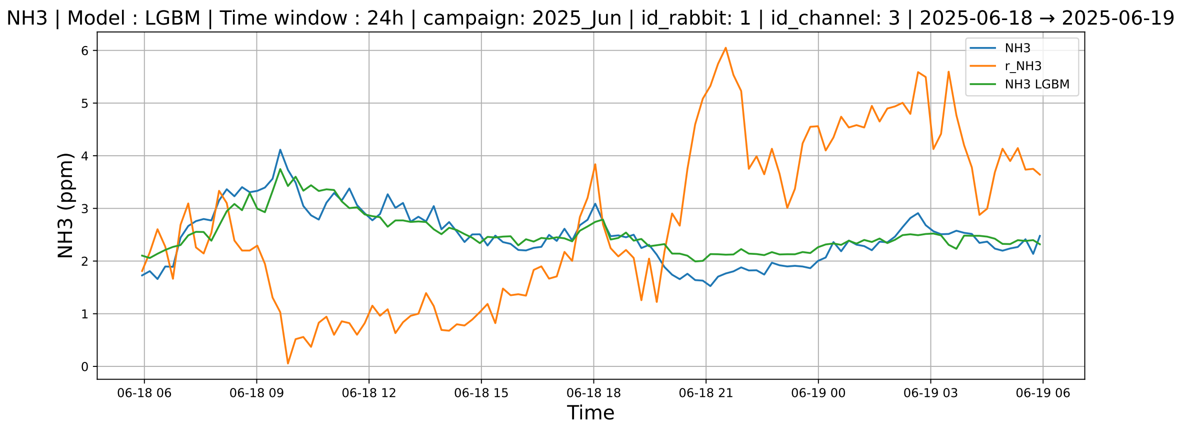
4 Plot

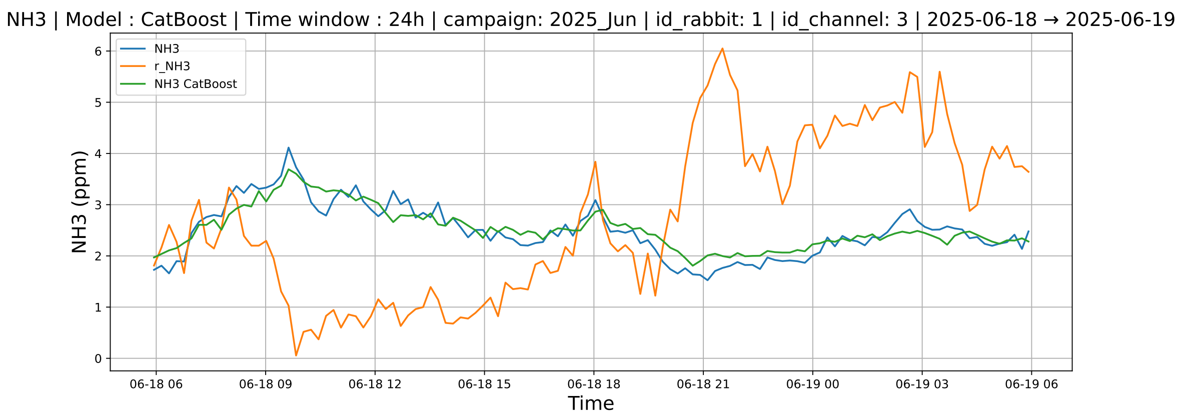
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

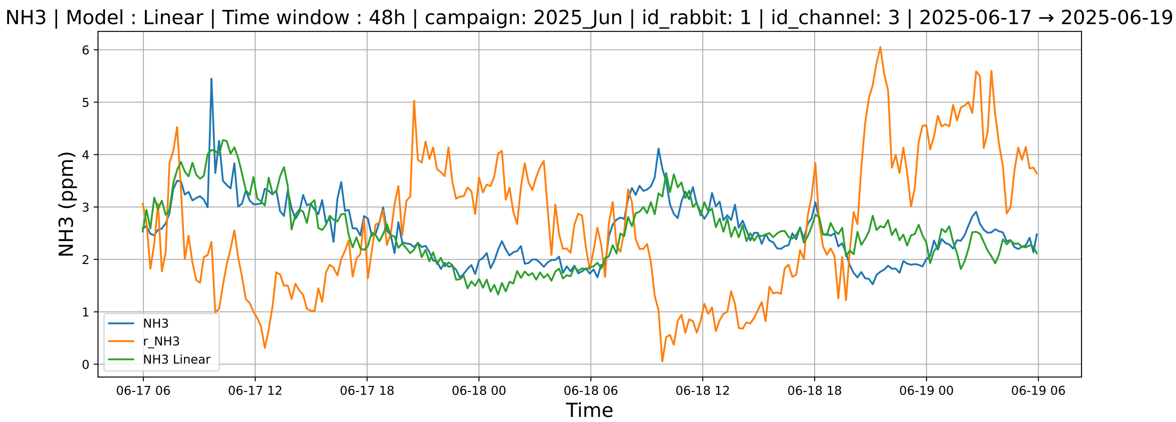
4.1.1.1 Time window : 24

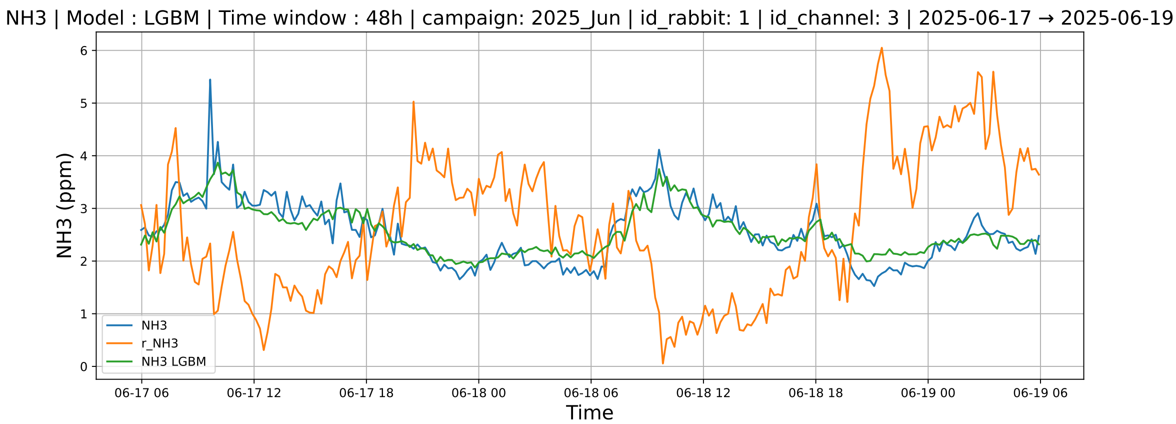


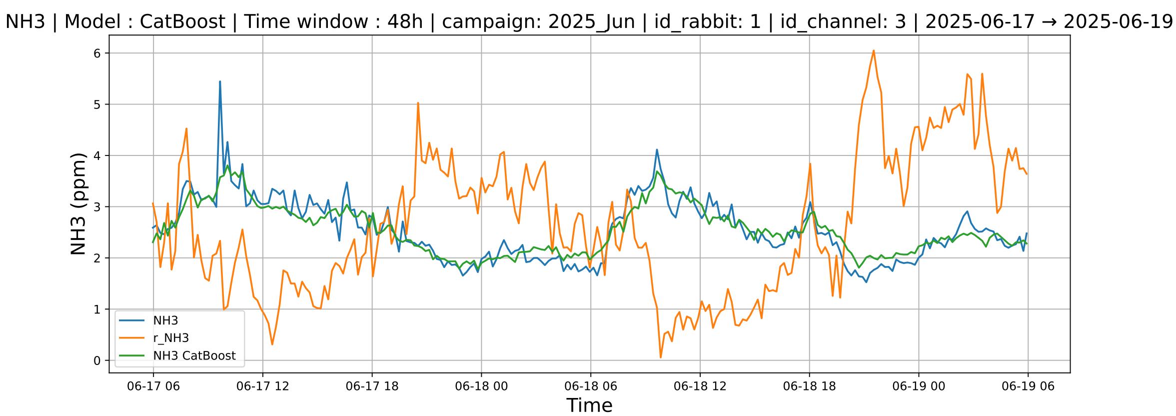




4.1.1.2 Time window : 48

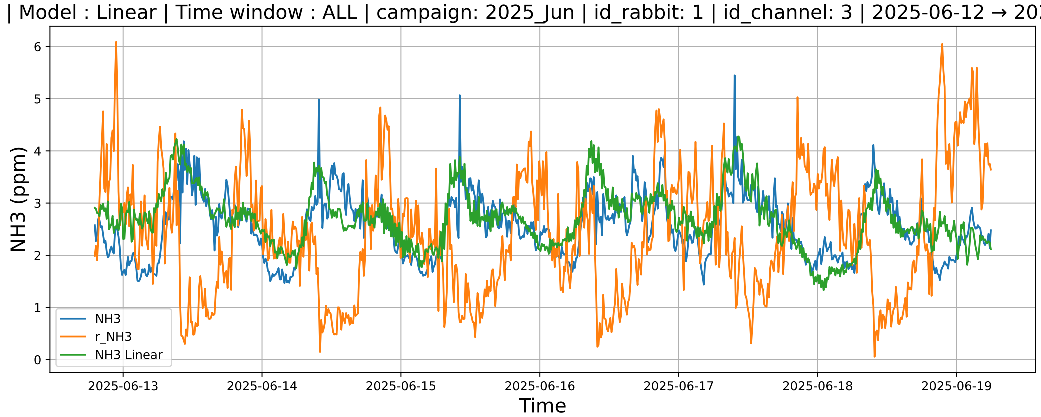




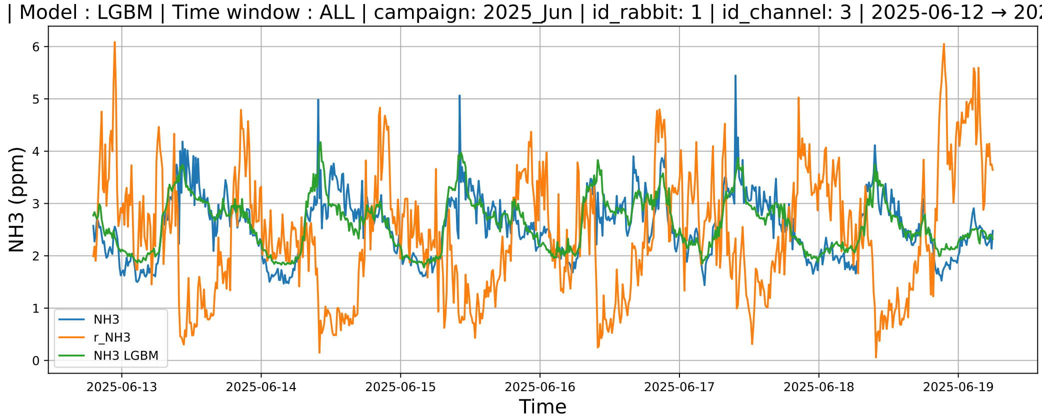


4.1.1.3 Time window : ALL

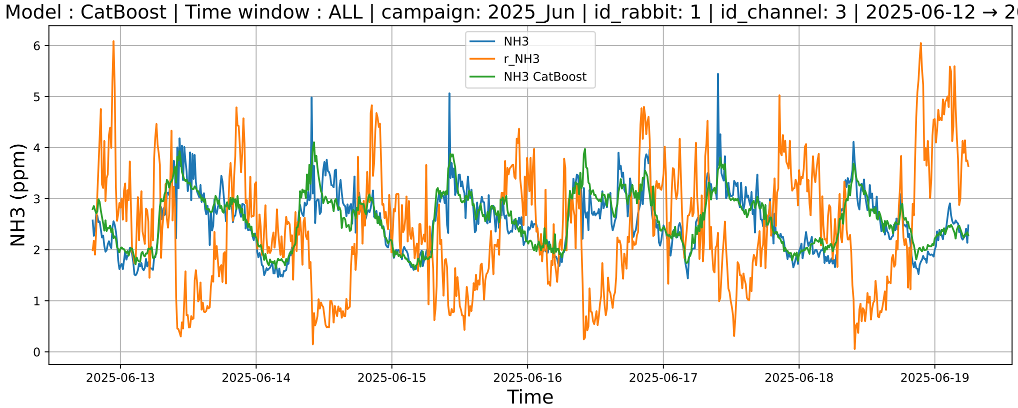
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

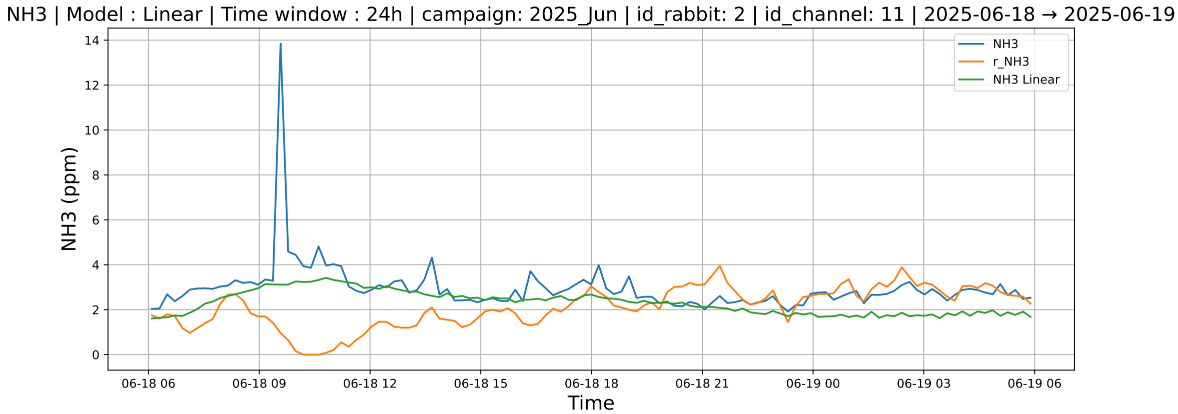


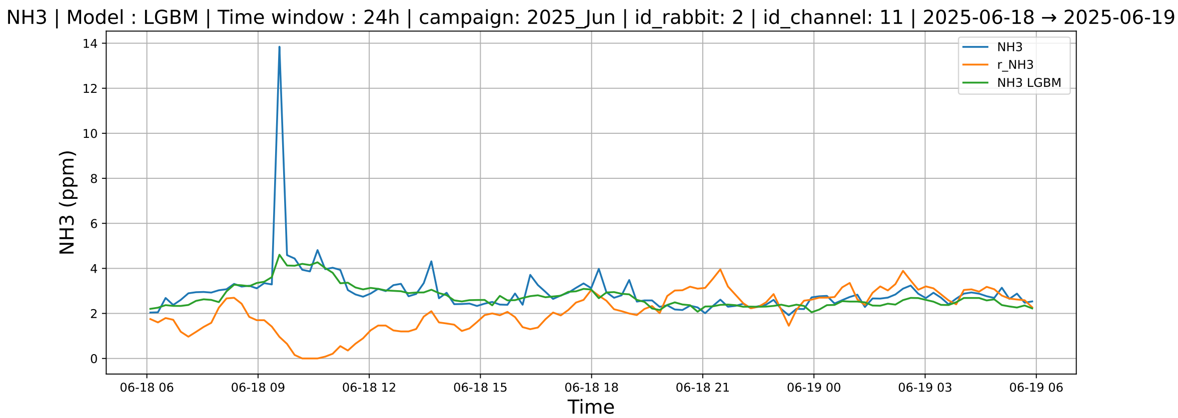
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

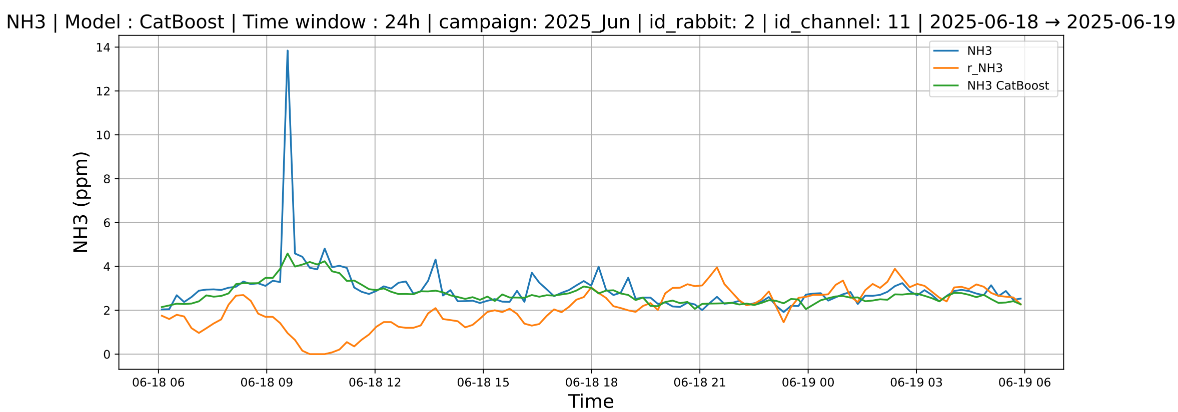


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

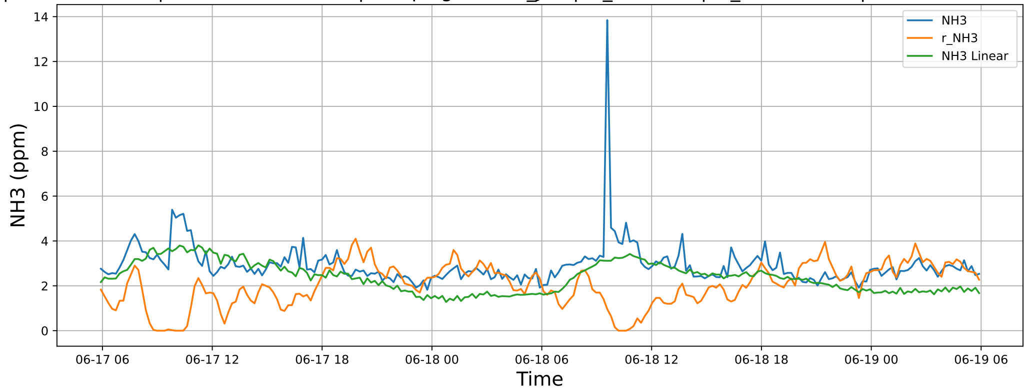




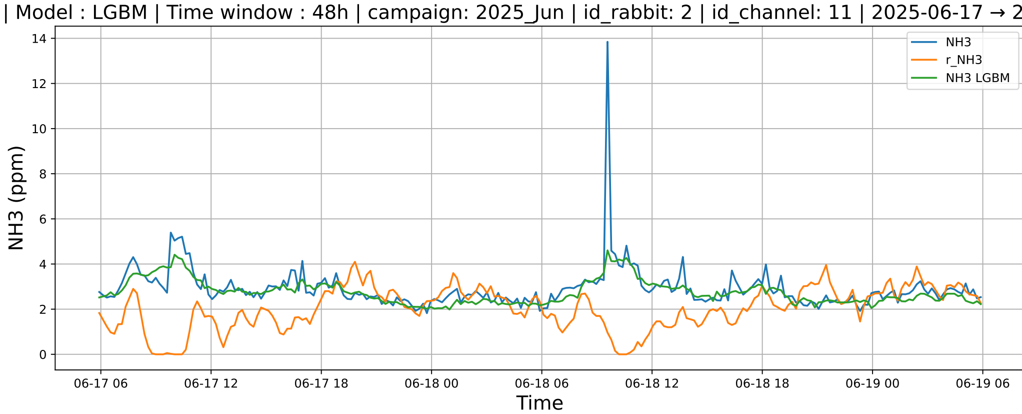


4.1.2.2 Time window : 48

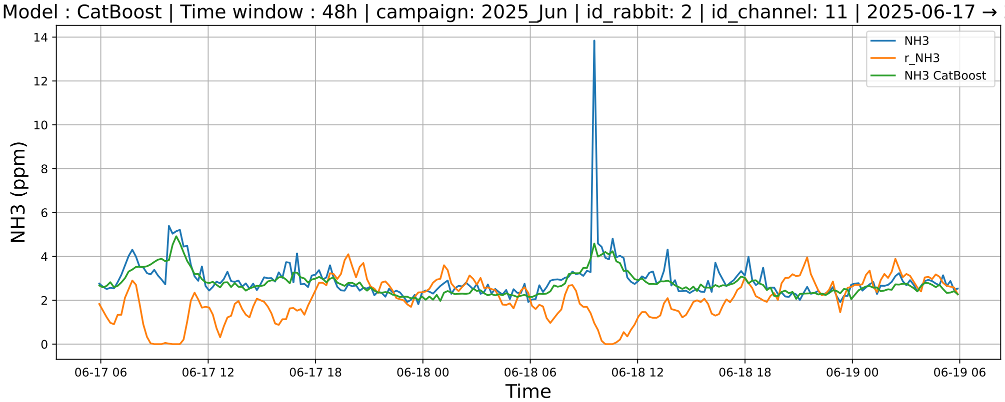
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



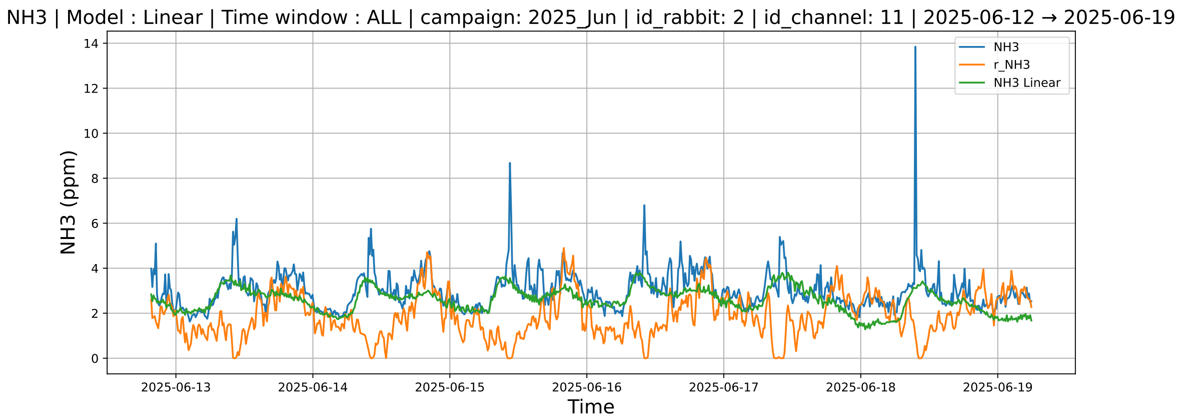
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

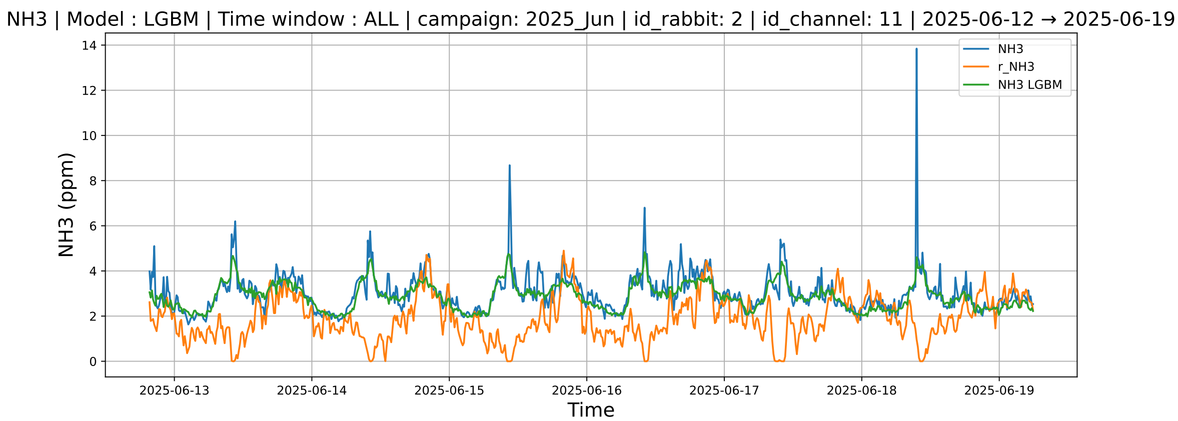


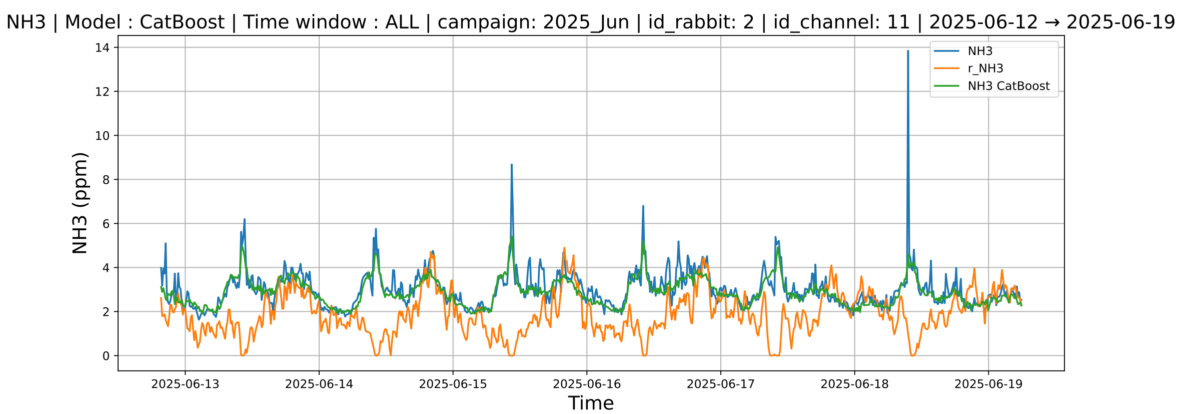
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.1.2.3 Time window : ALL

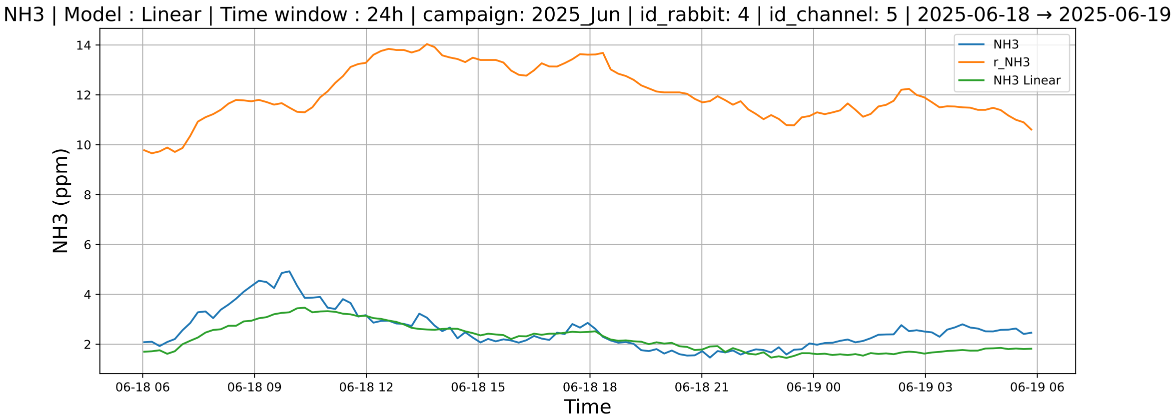


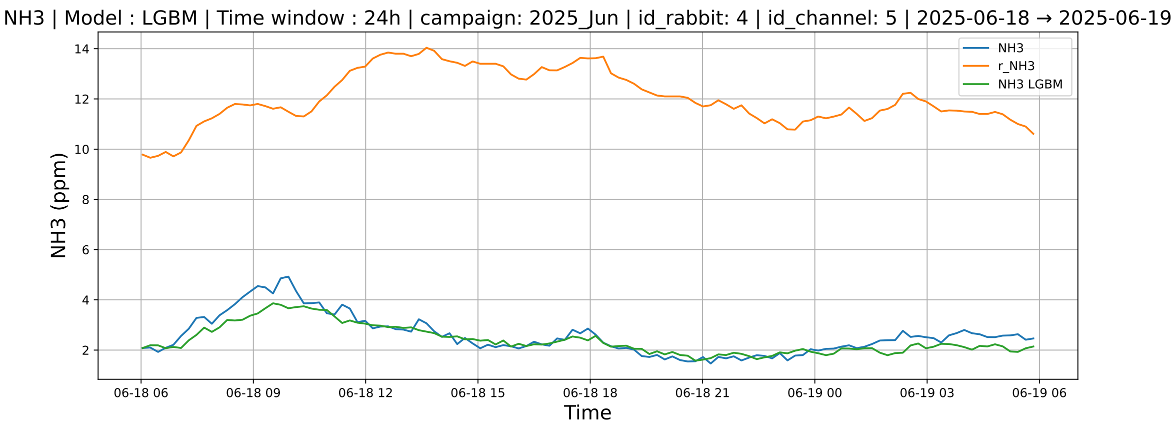


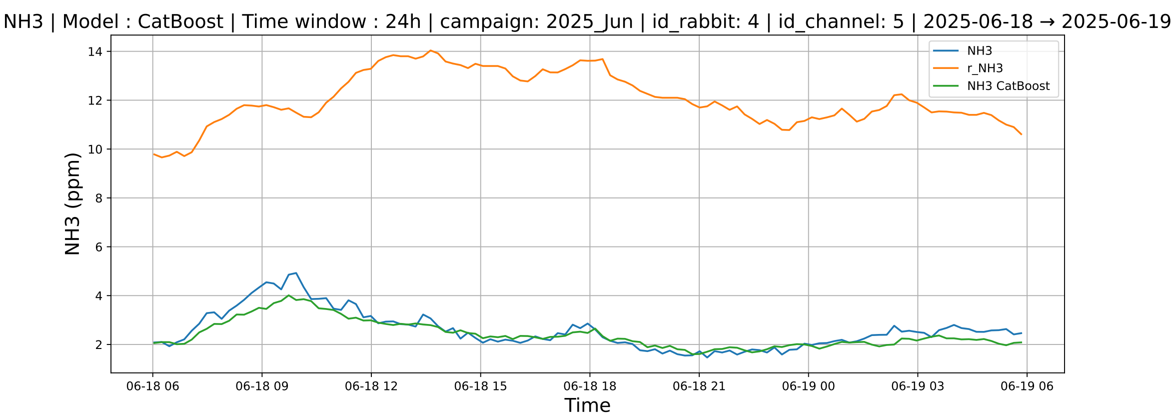


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

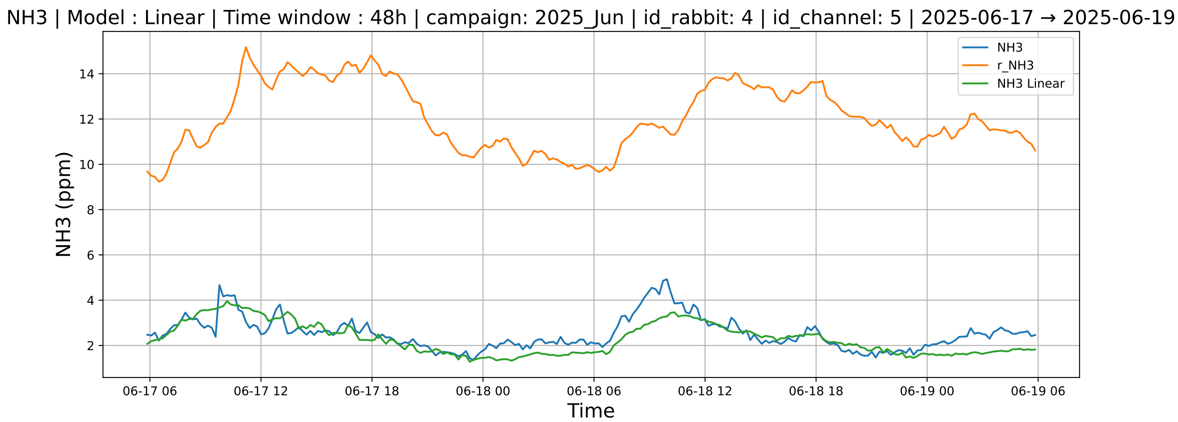
4.1.3.1 Time window : 24

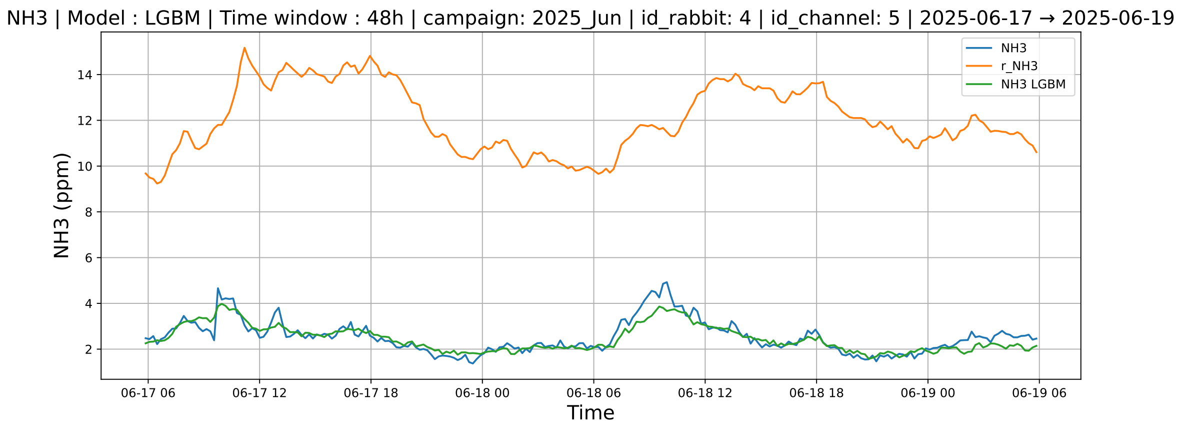


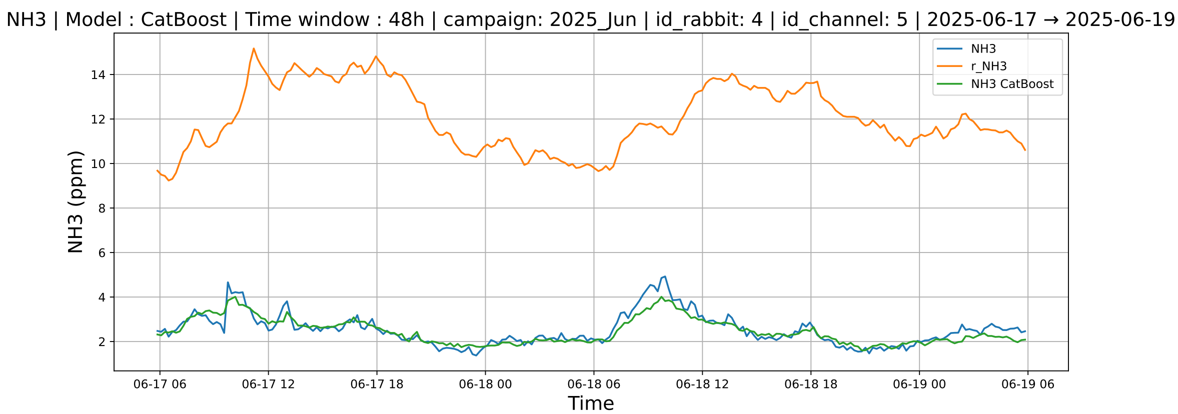




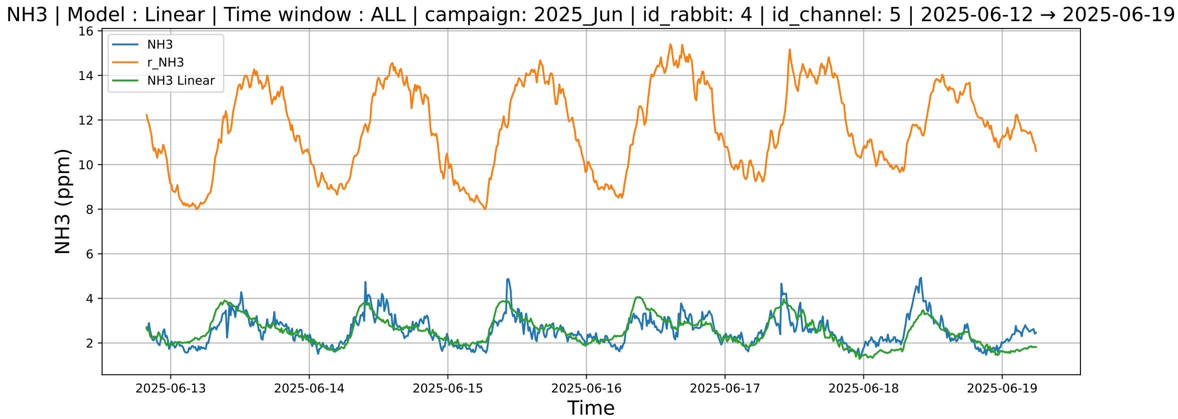
4.1.3.2 Time window : 48

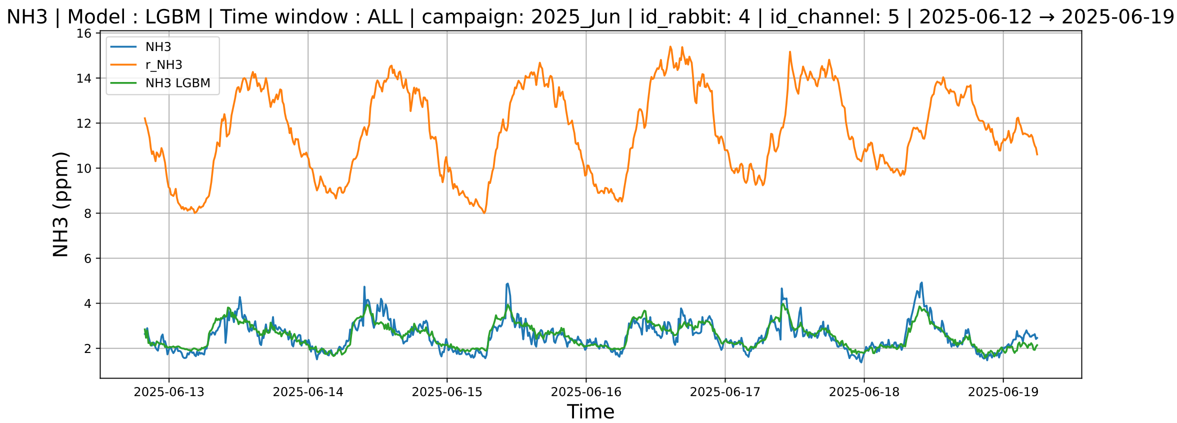


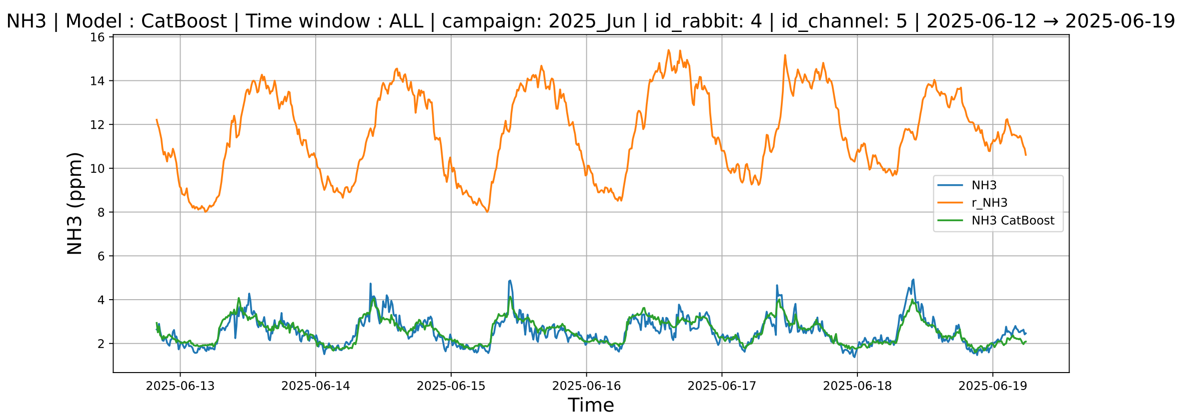




4.1.3.3 Time window : ALL

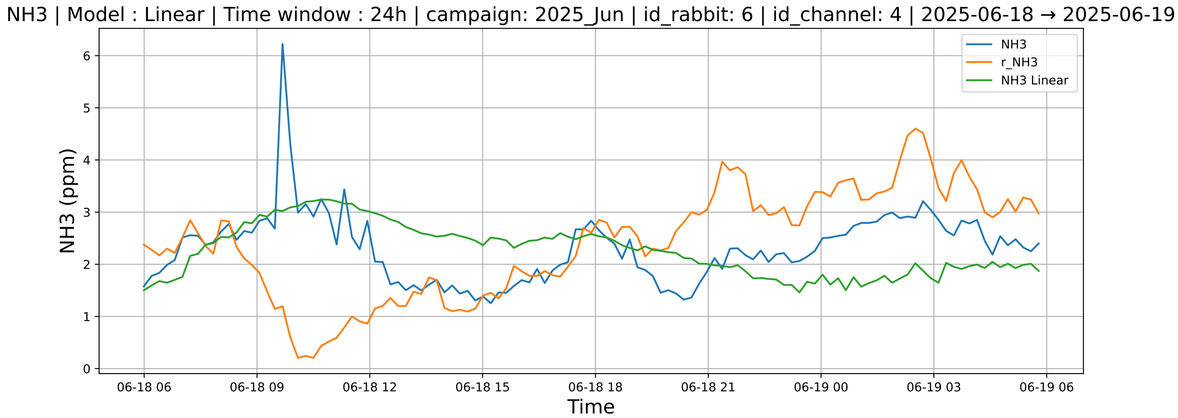


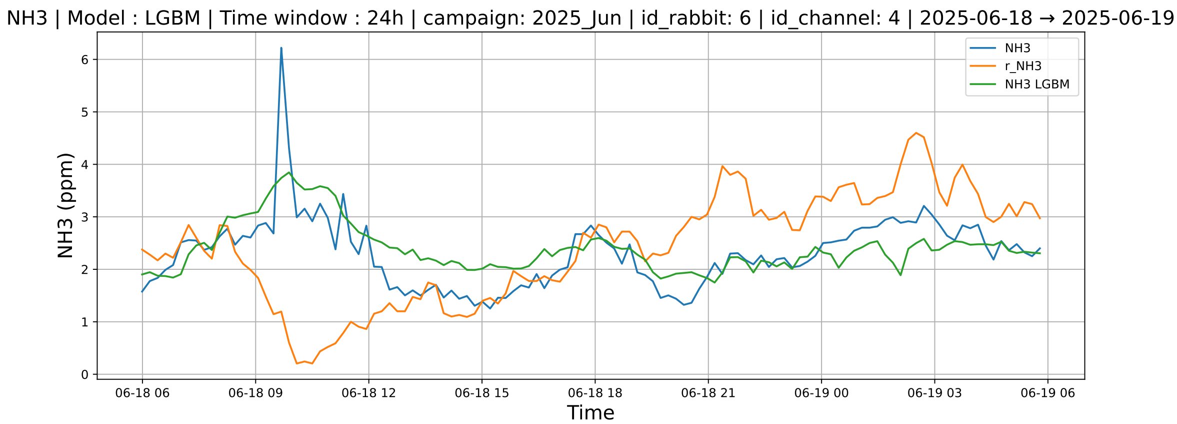


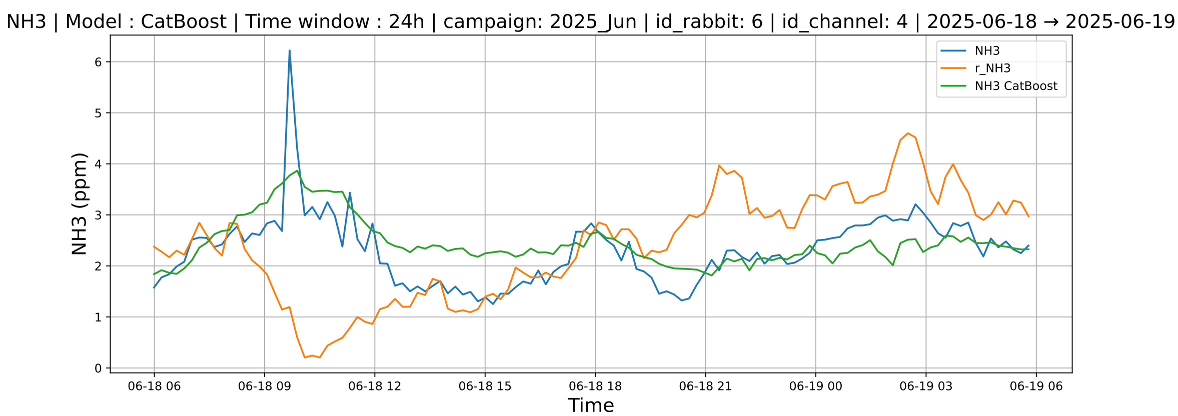


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

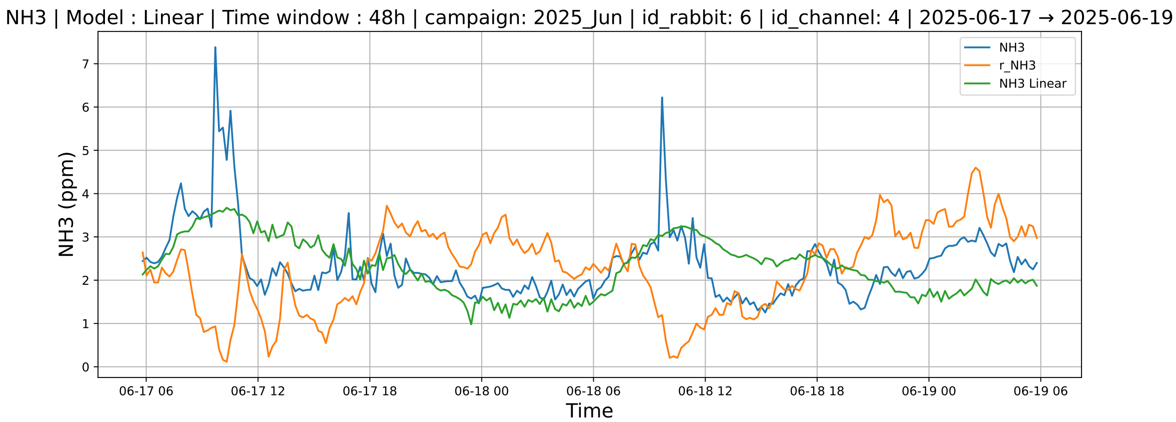
4.1.4.1 Time window : 24

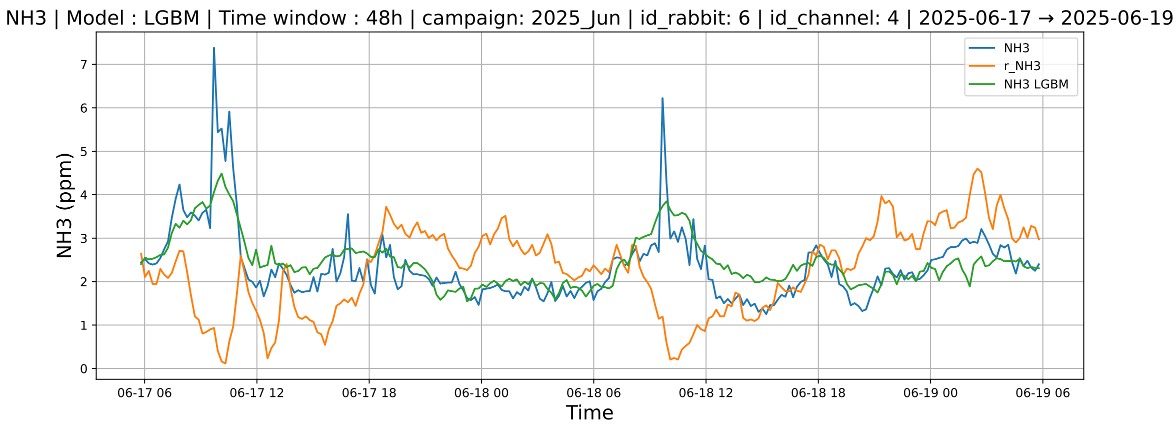


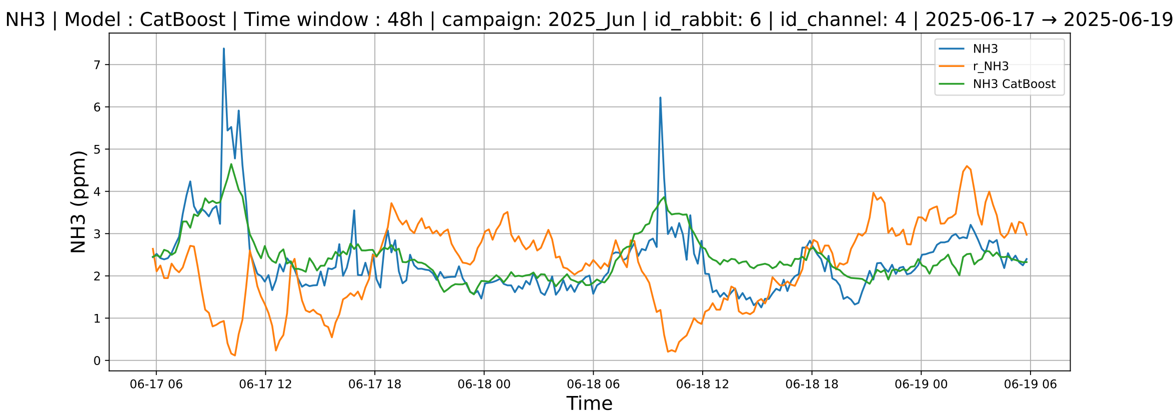




4.1.4.2 Time window : 48

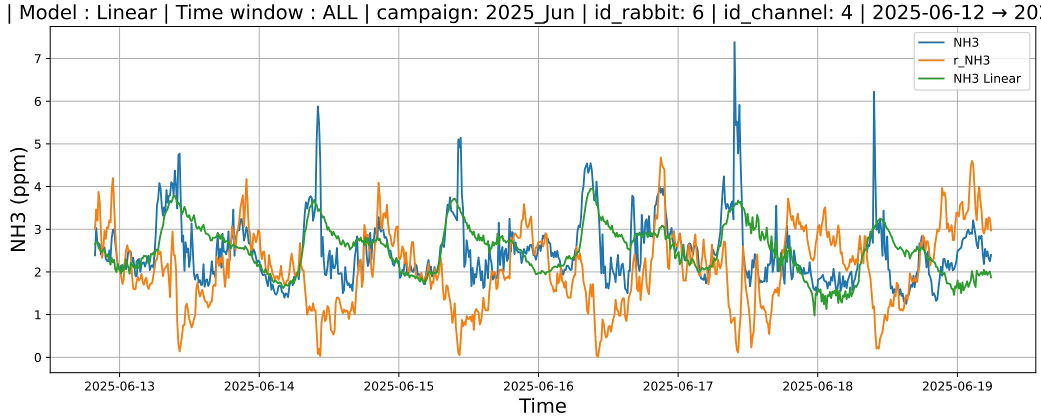




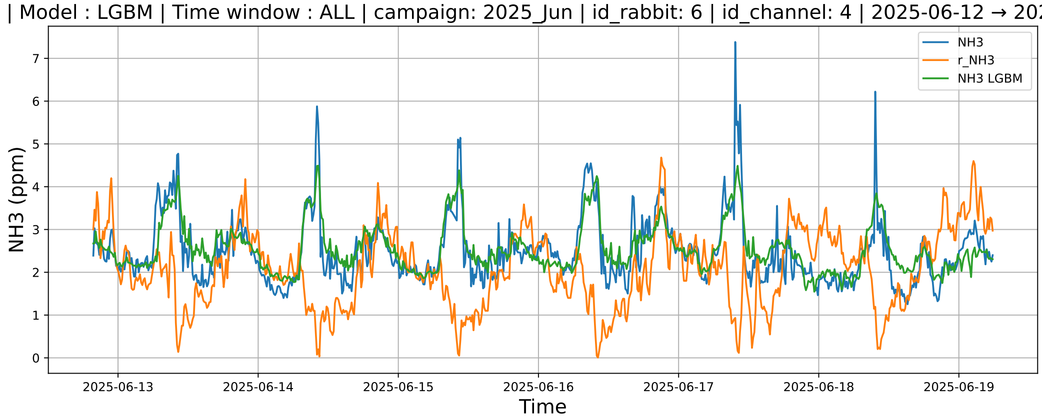


4.1.4.3 Time window : ALL

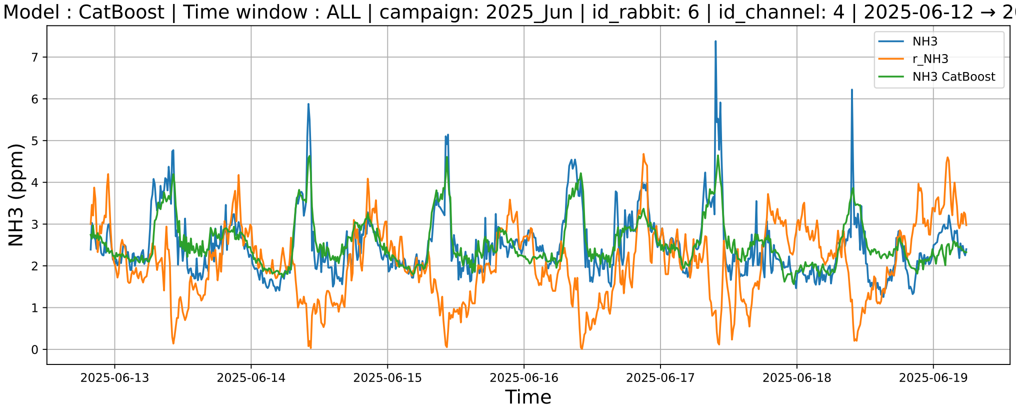
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

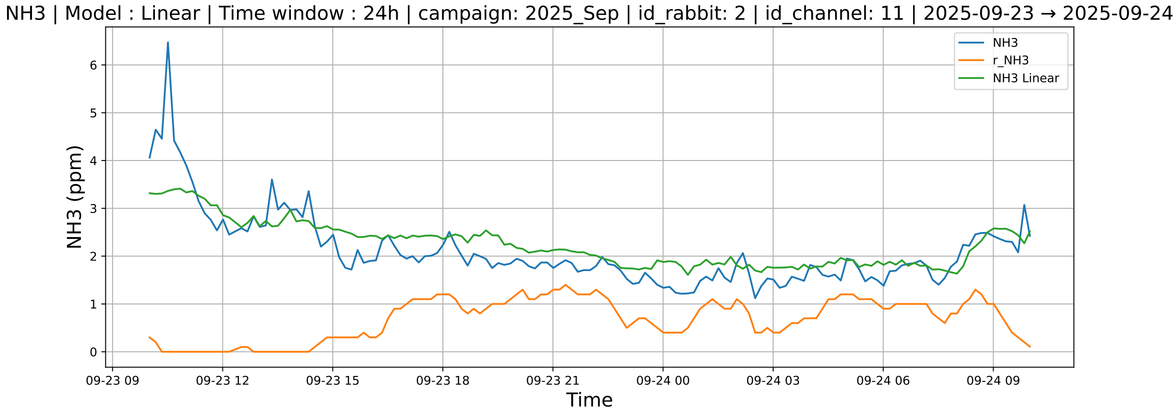


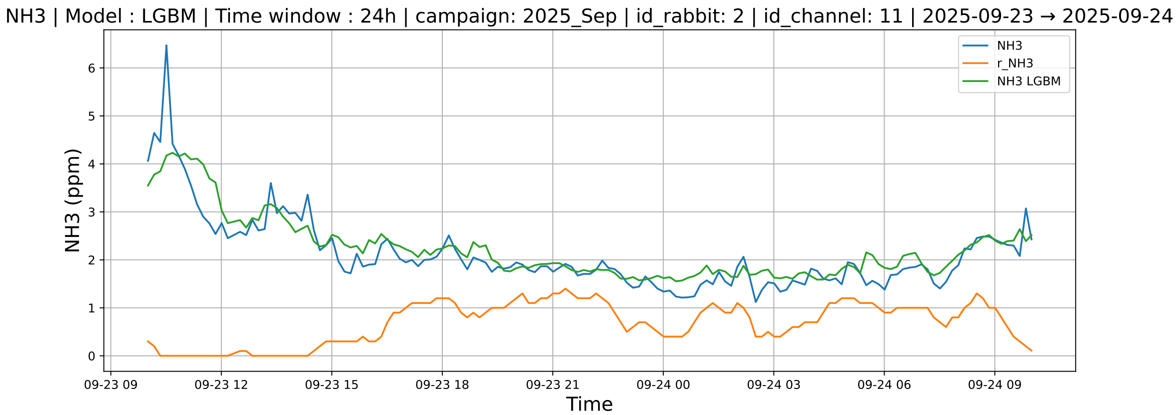
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

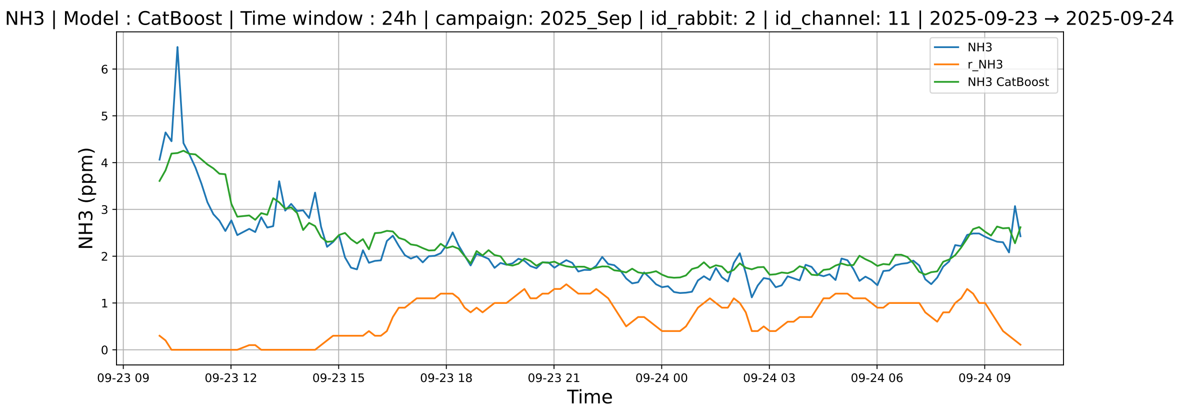


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

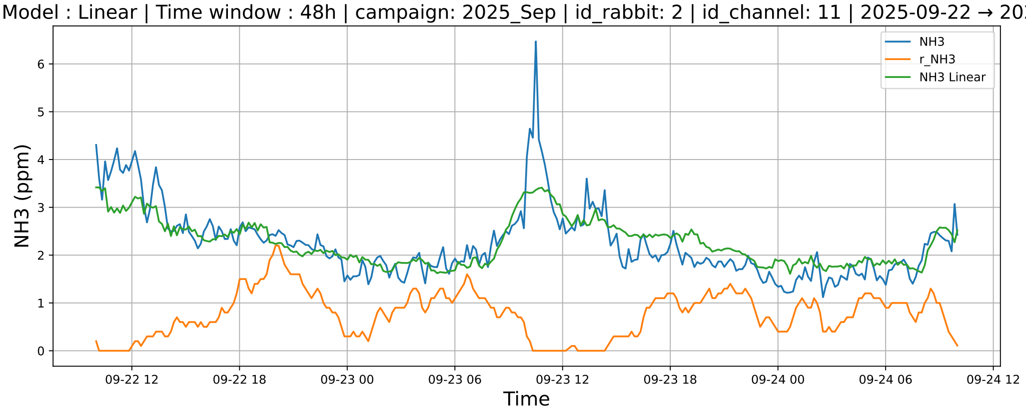




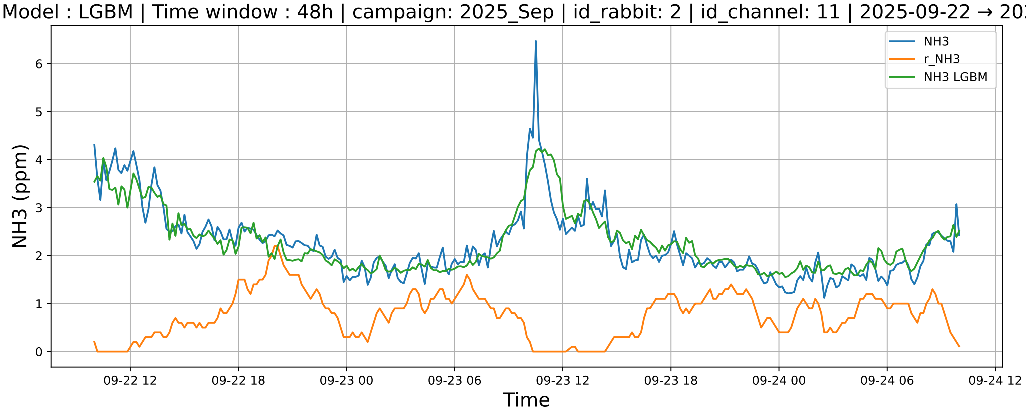


4.1.5.2 Time window : 48

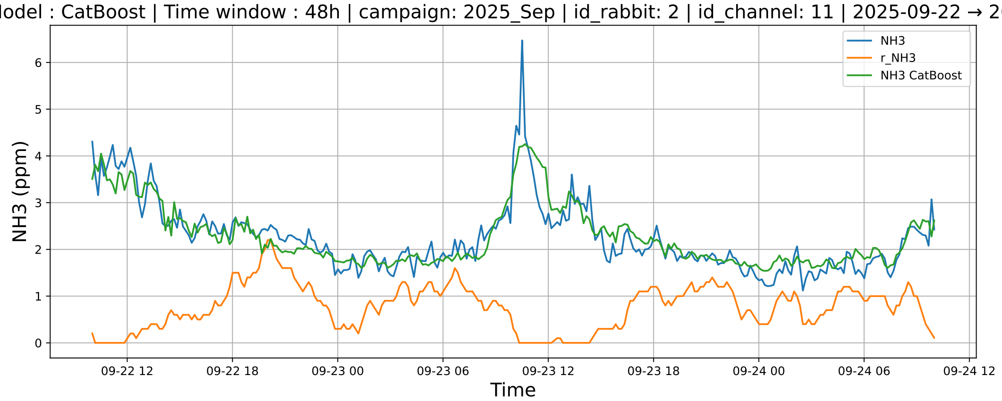
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

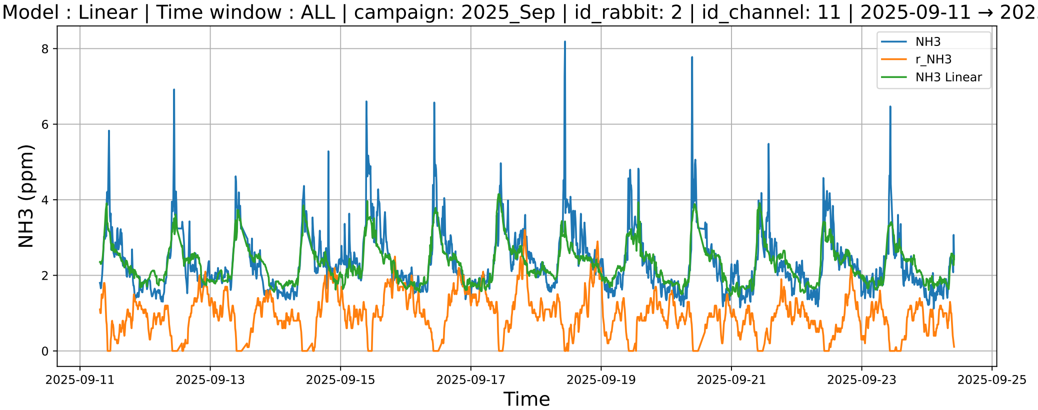


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

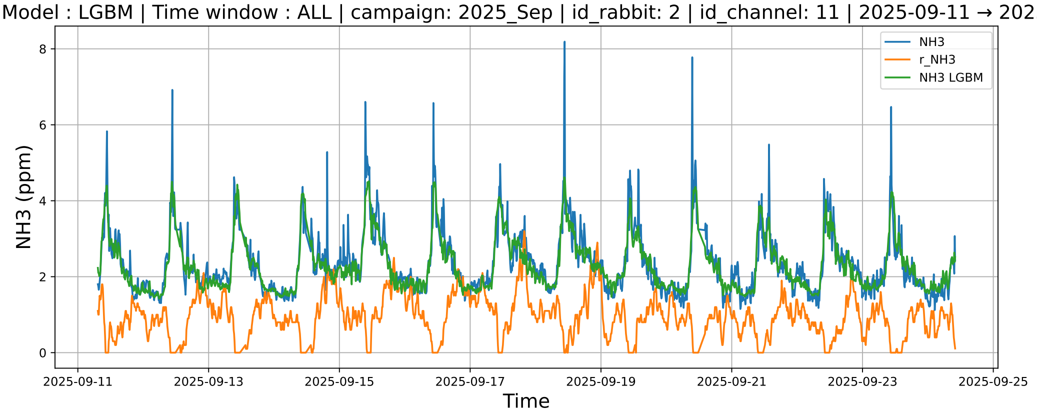


4.1.5.3 Time window : ALL

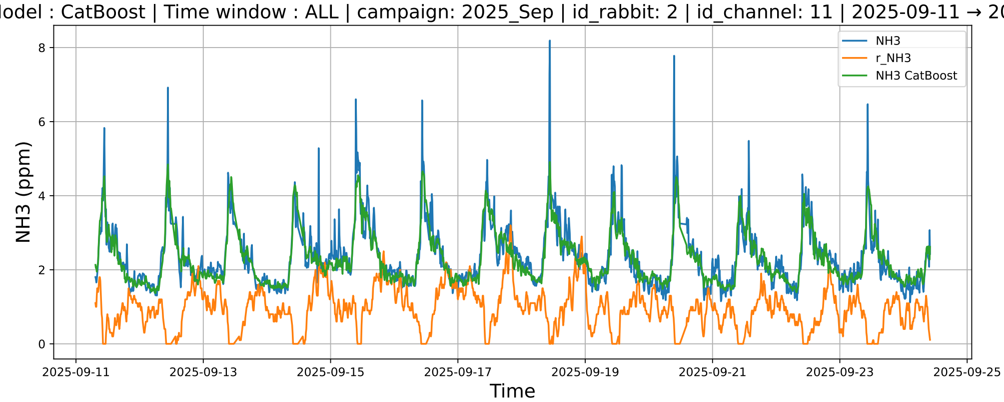
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



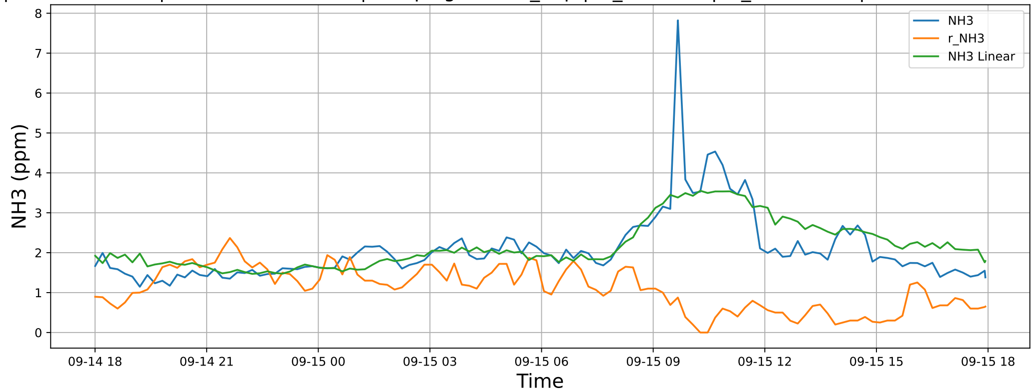
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



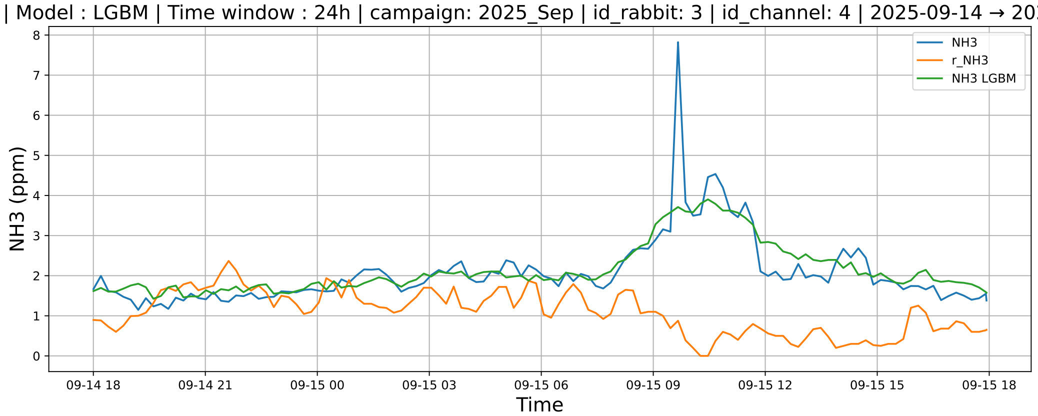
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

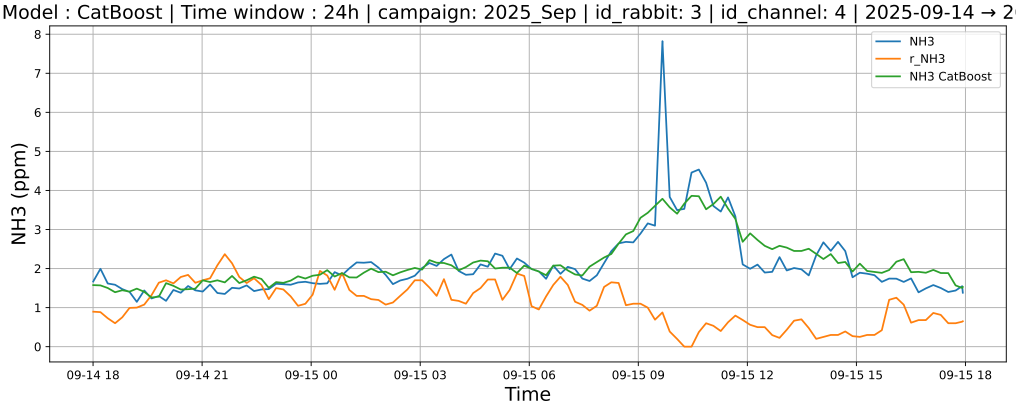
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



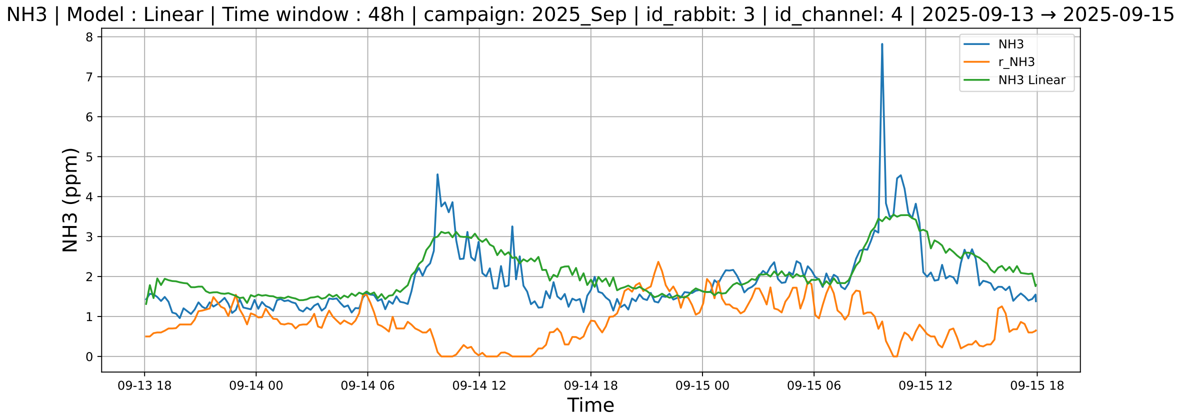
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

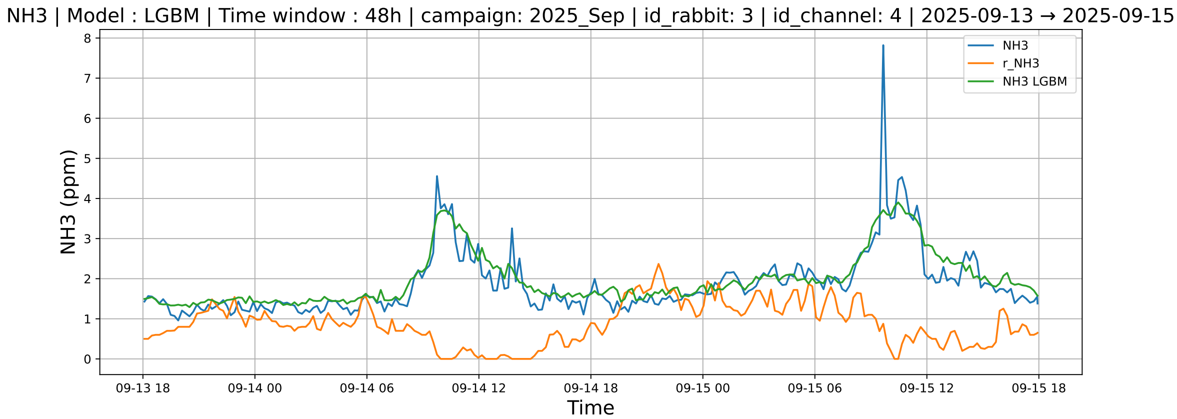


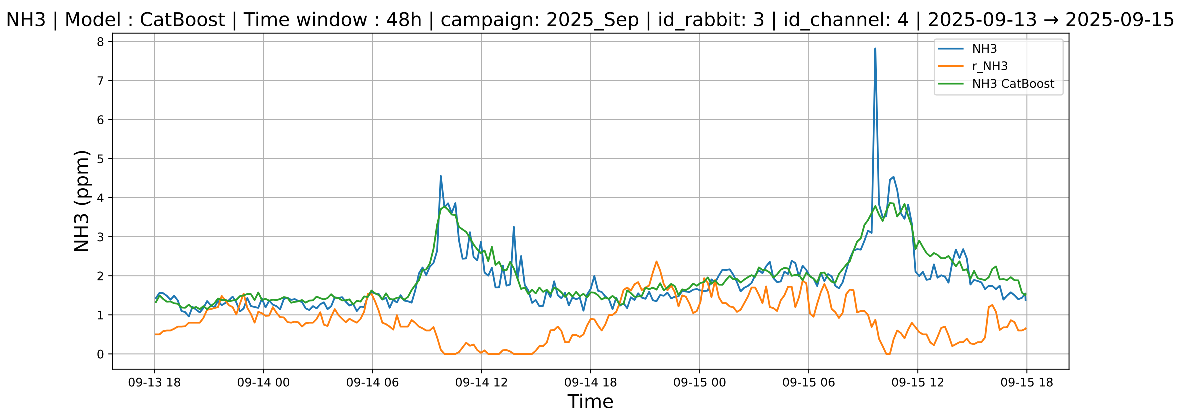
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



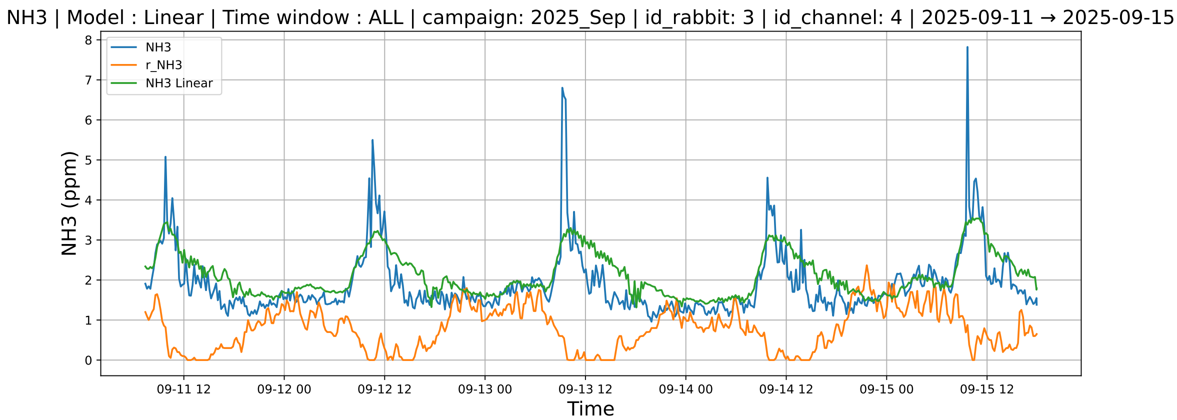
4.1.6.2 Time window : 48

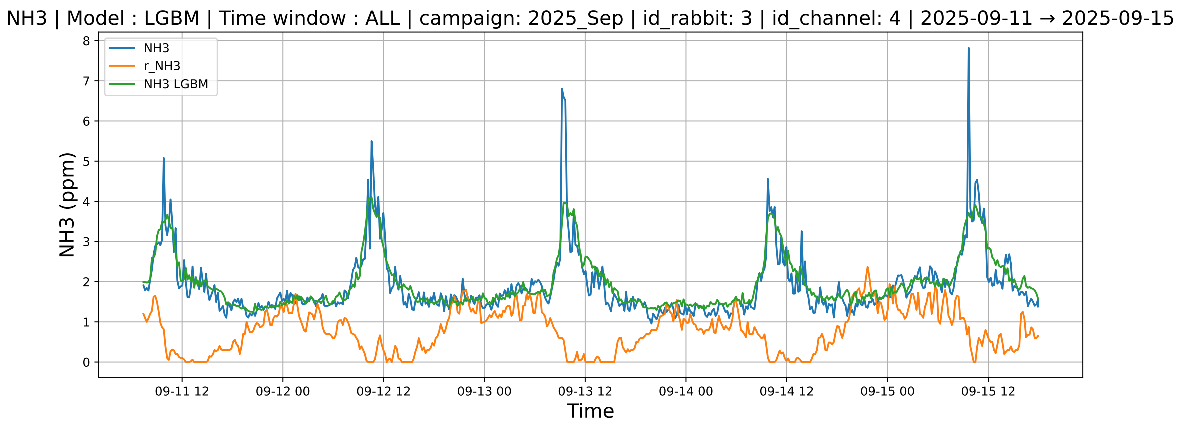


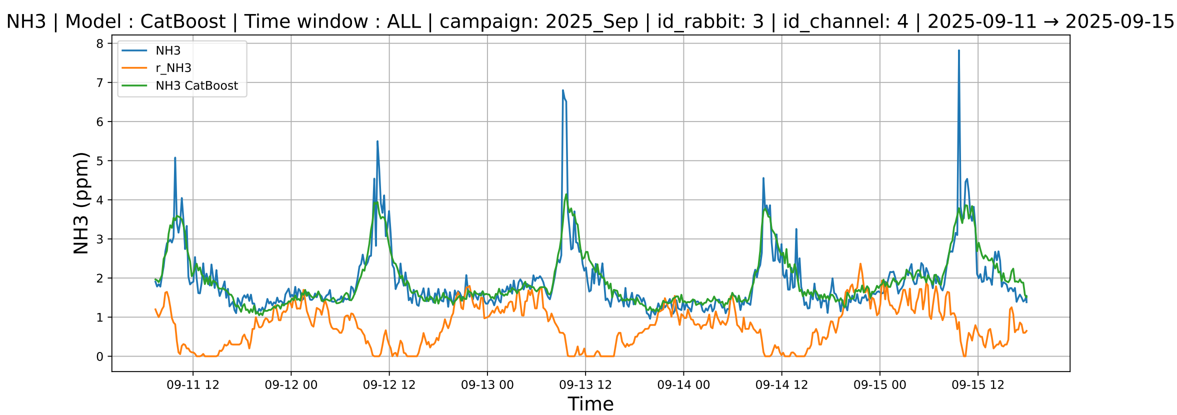




4.1.6.3 Time window : ALL

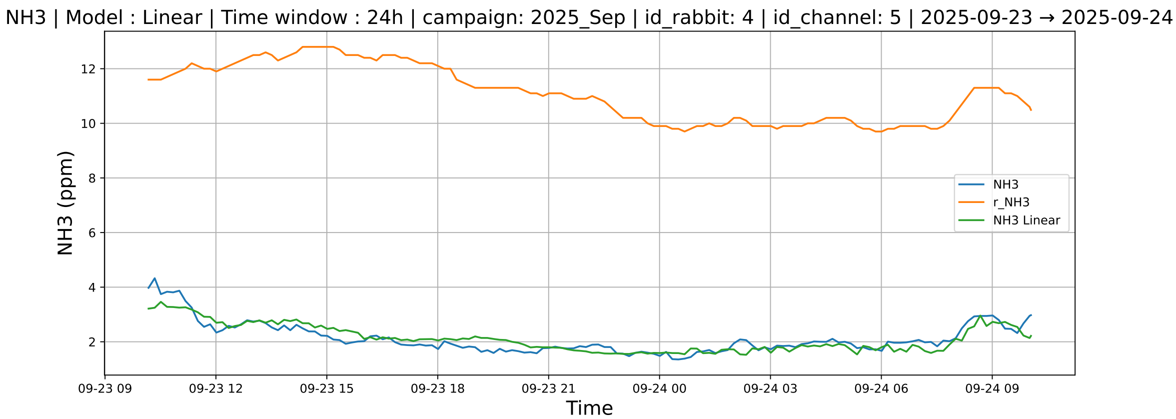


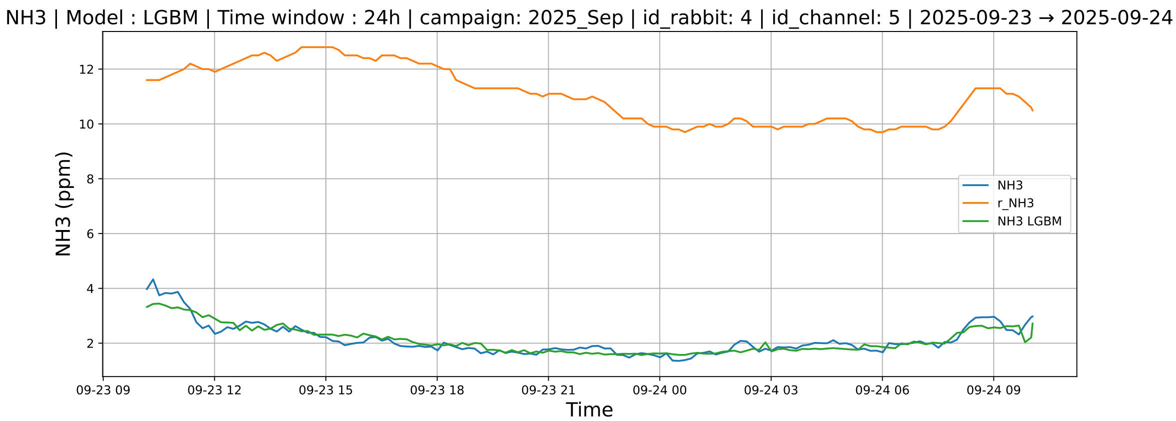


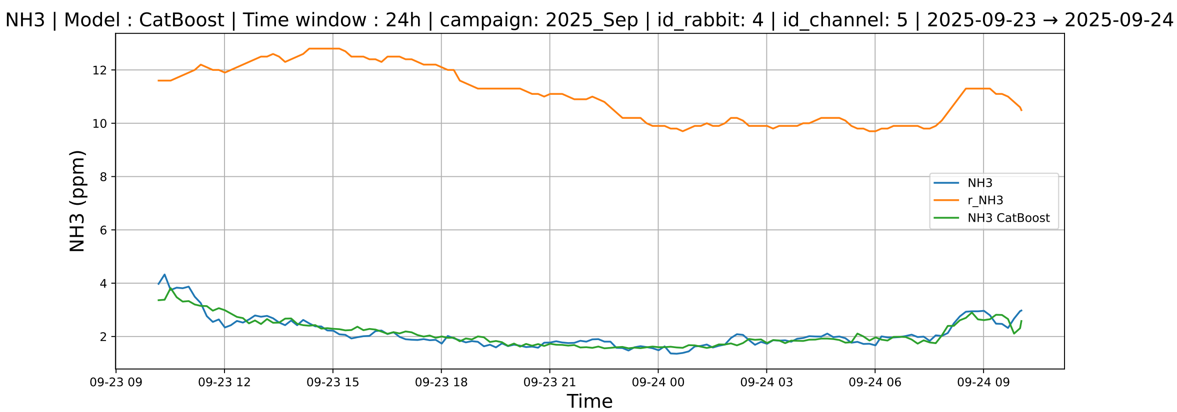


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

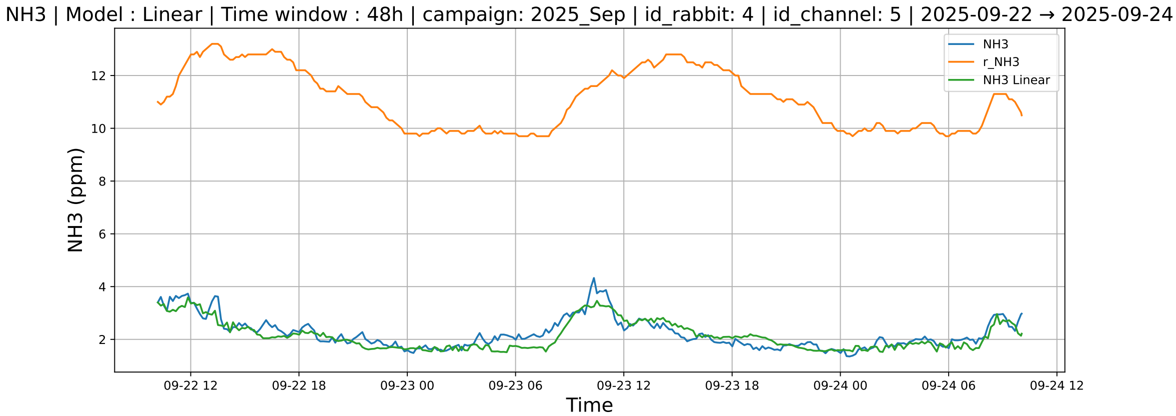
4.1.7.1 Time window : 24

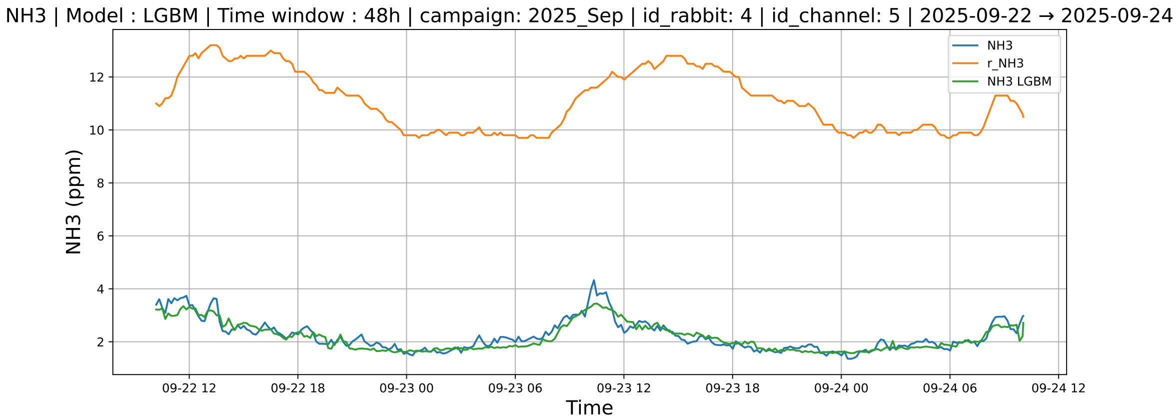


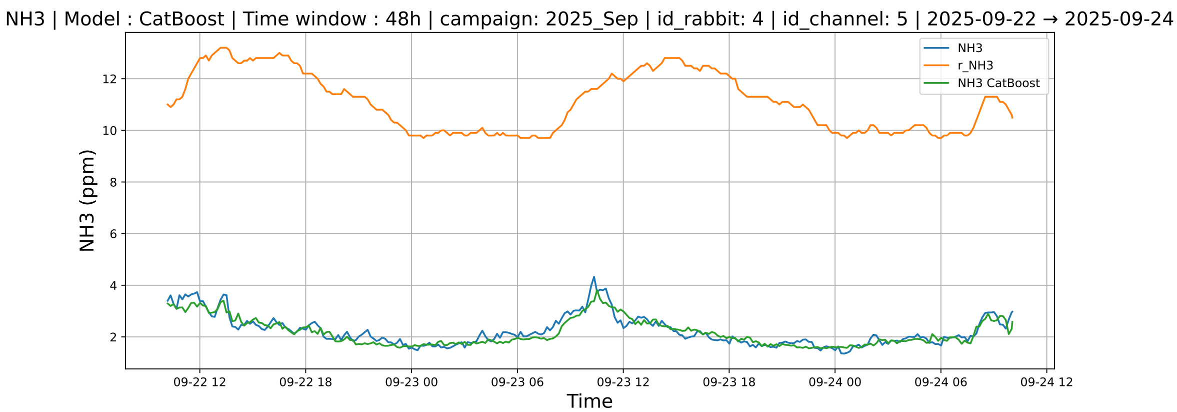




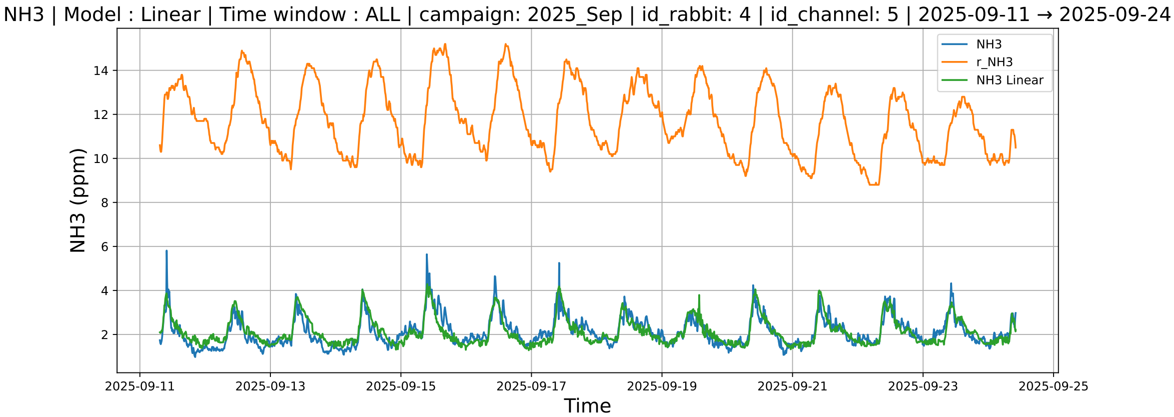
4.1.7.2 Time window : 48

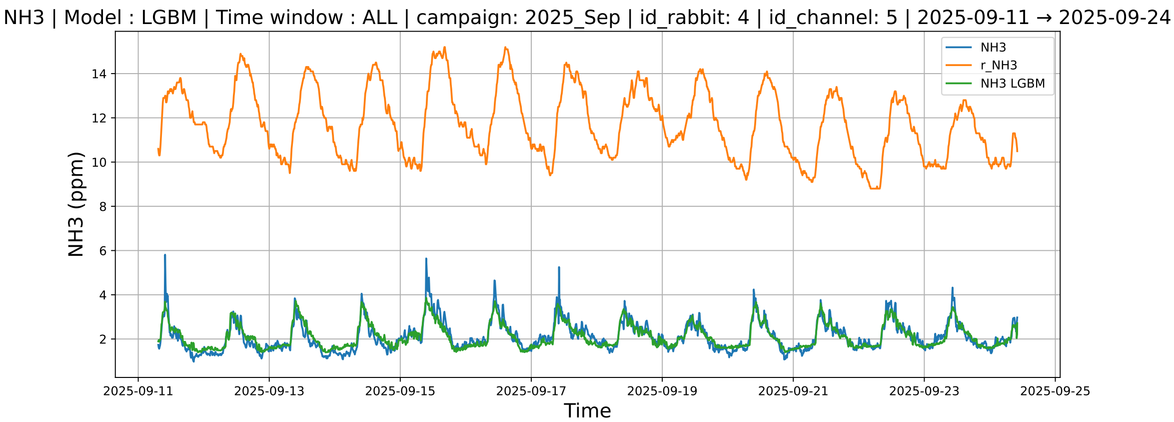


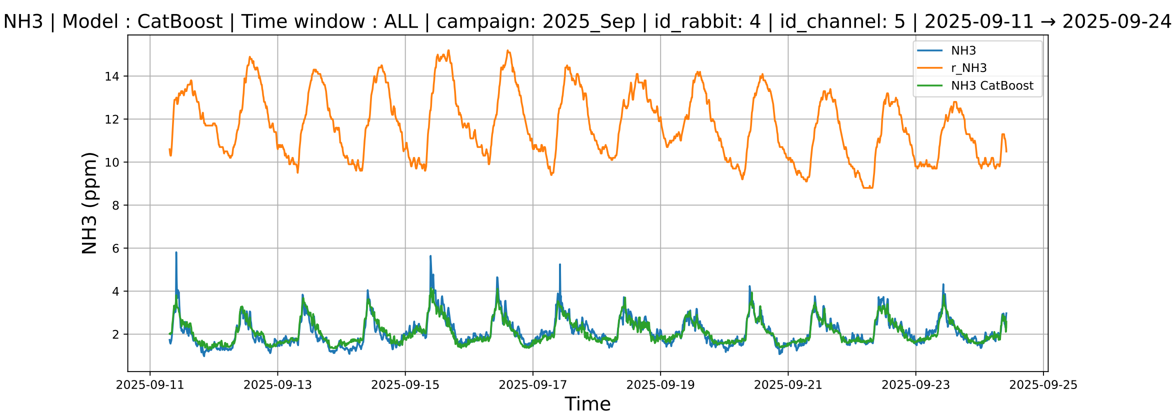




4.1.7.3 Time window : ALL

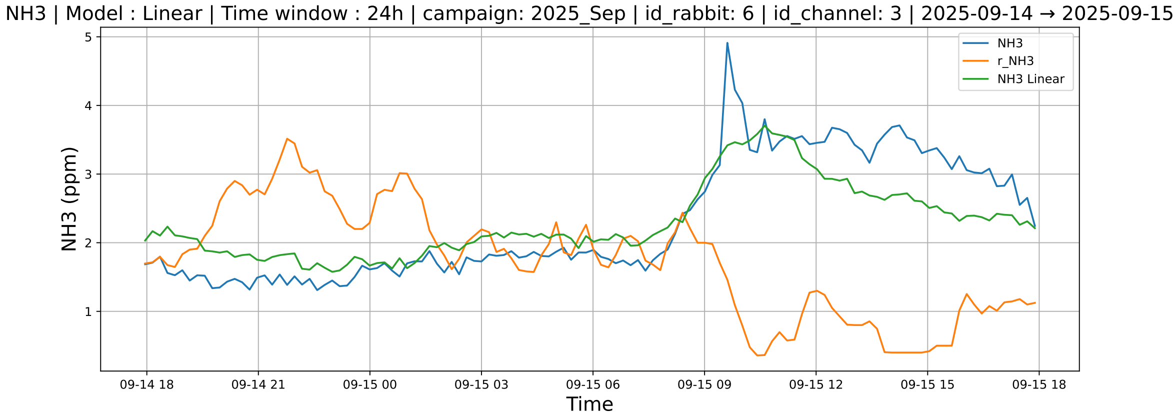


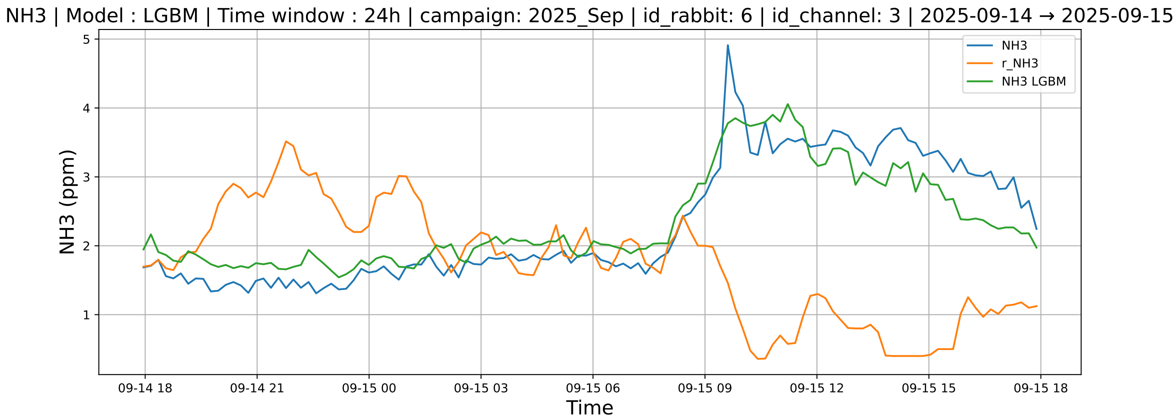


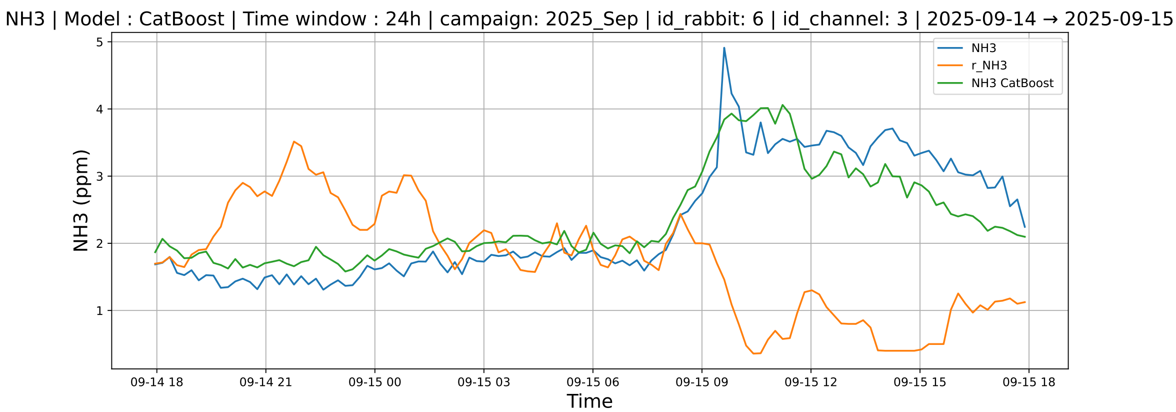


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

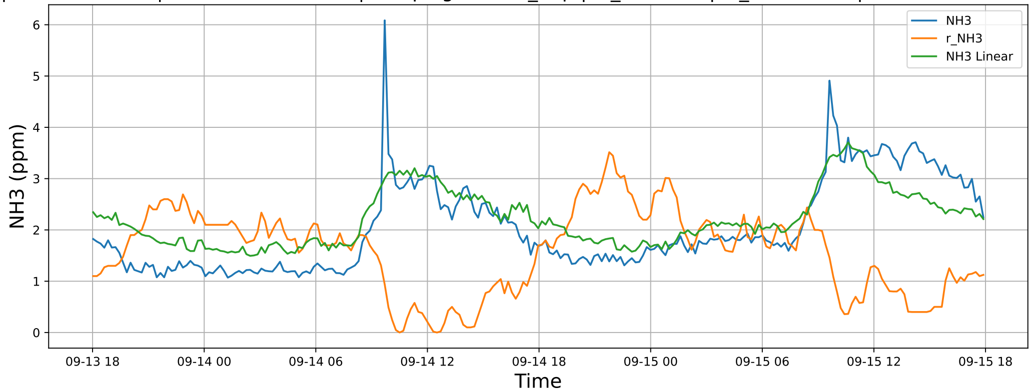




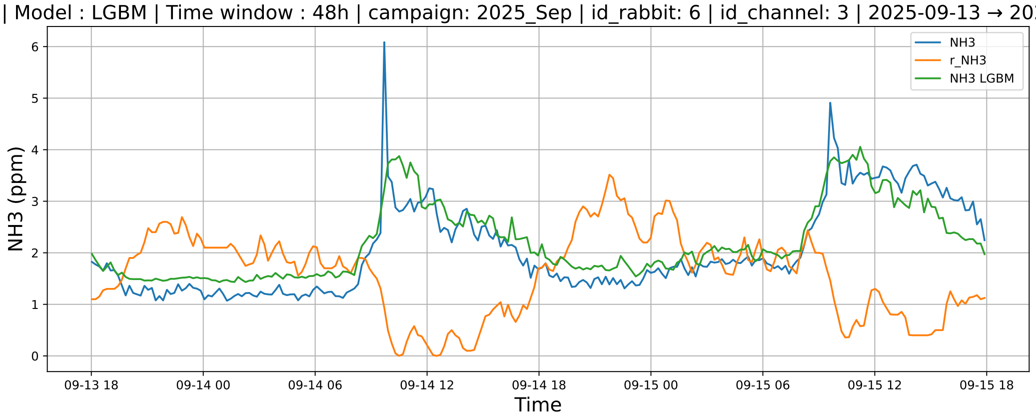


4.1.8.2 Time window : 48

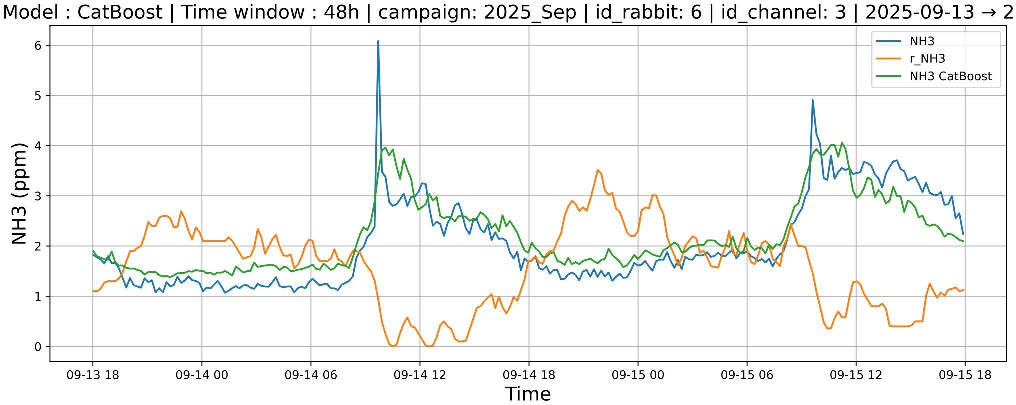
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



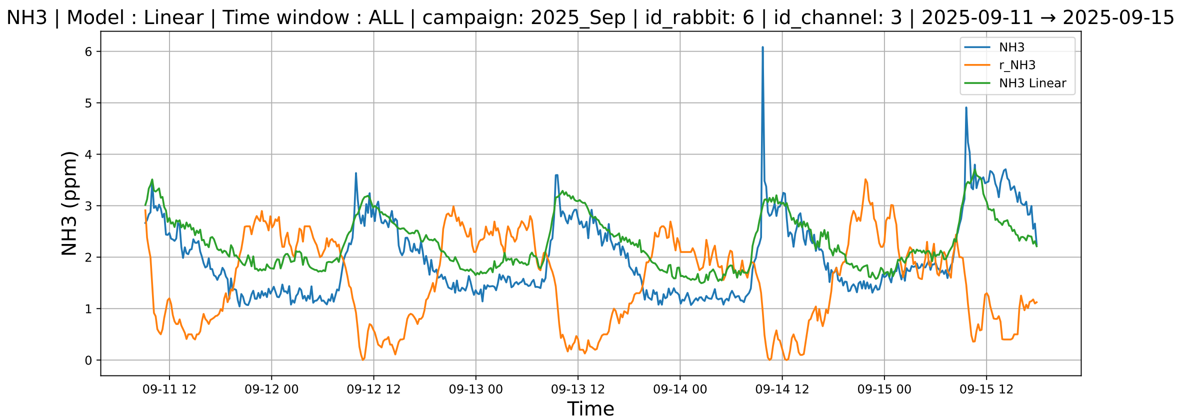
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

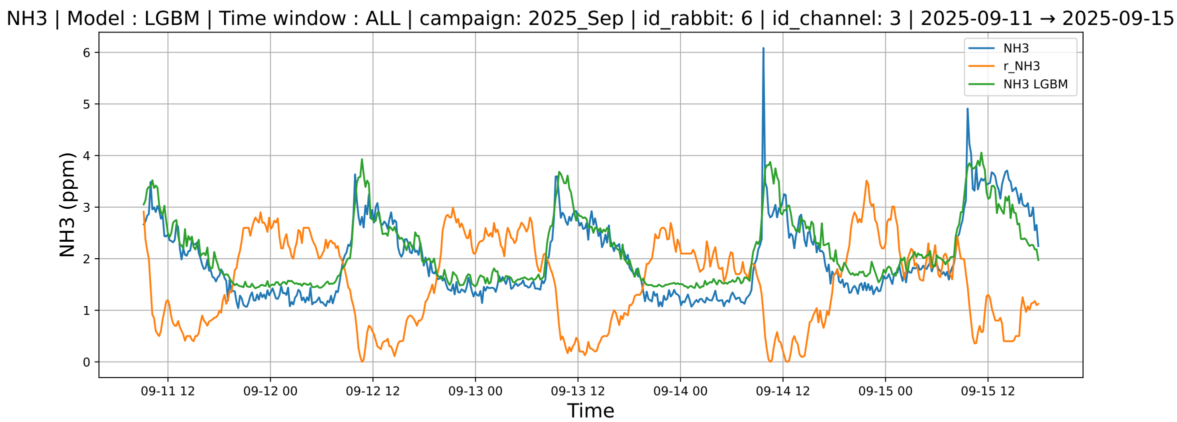


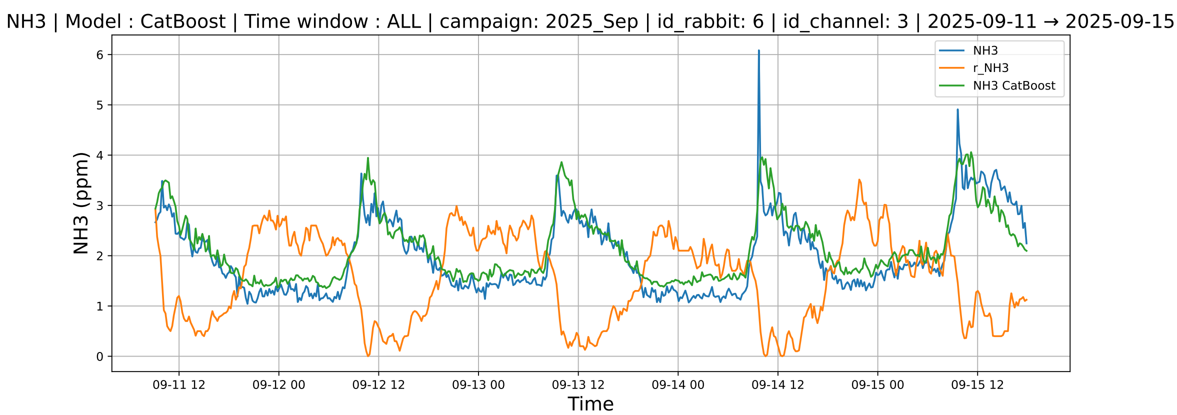
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.1.8.3 Time window : ALL





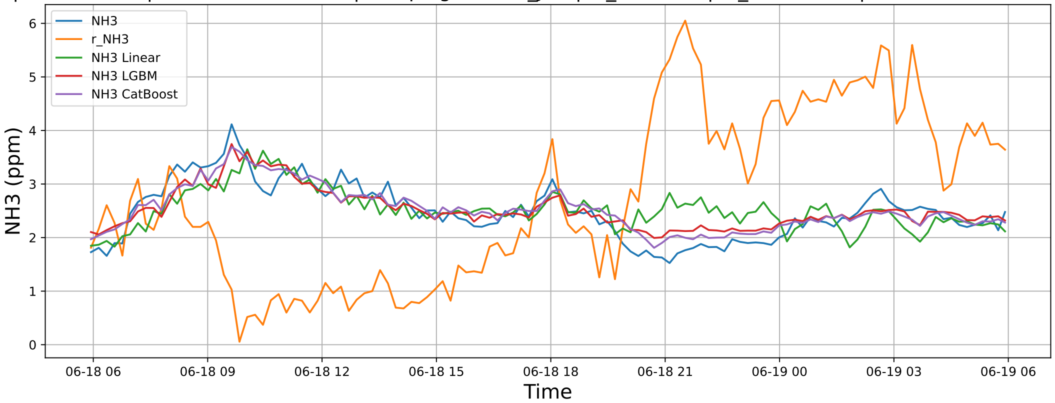


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

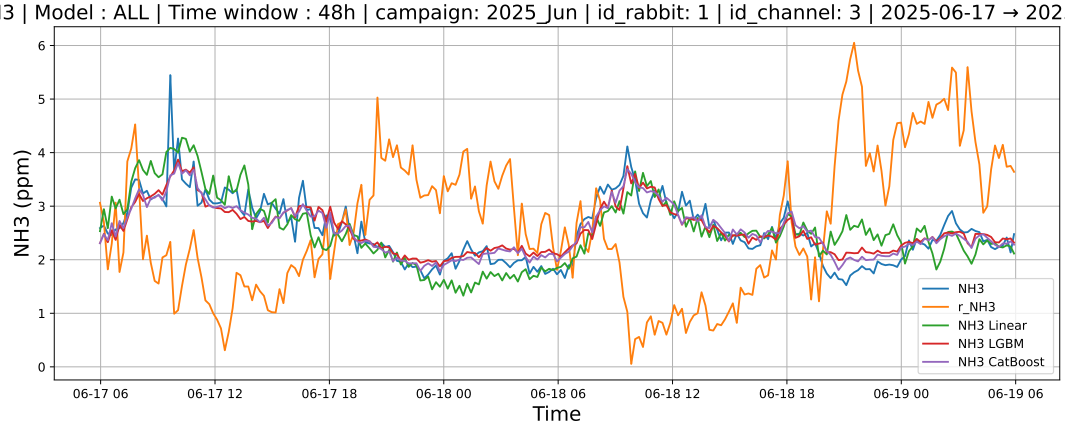
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



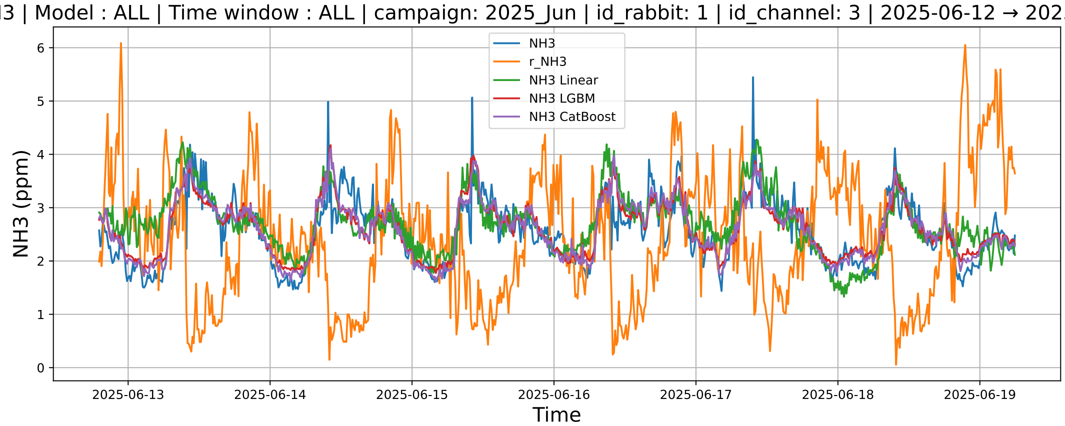
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



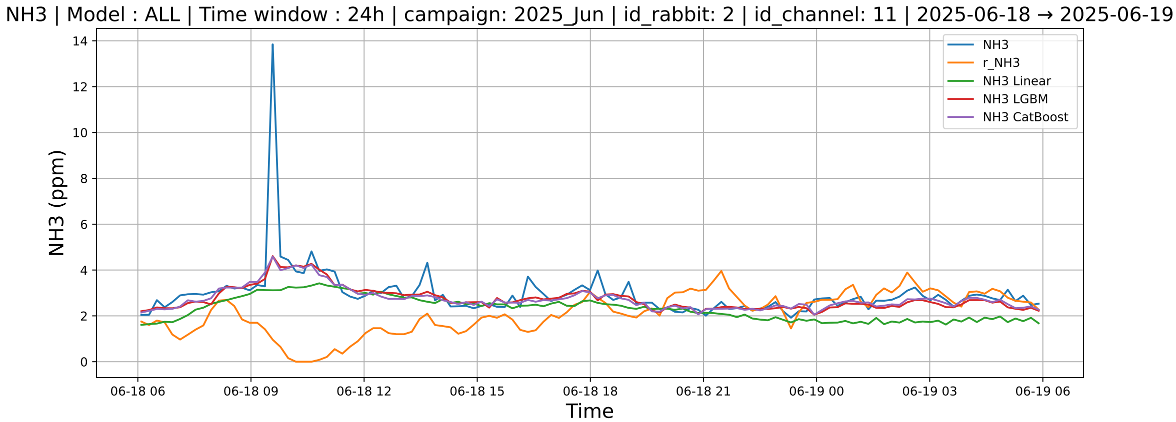
4.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

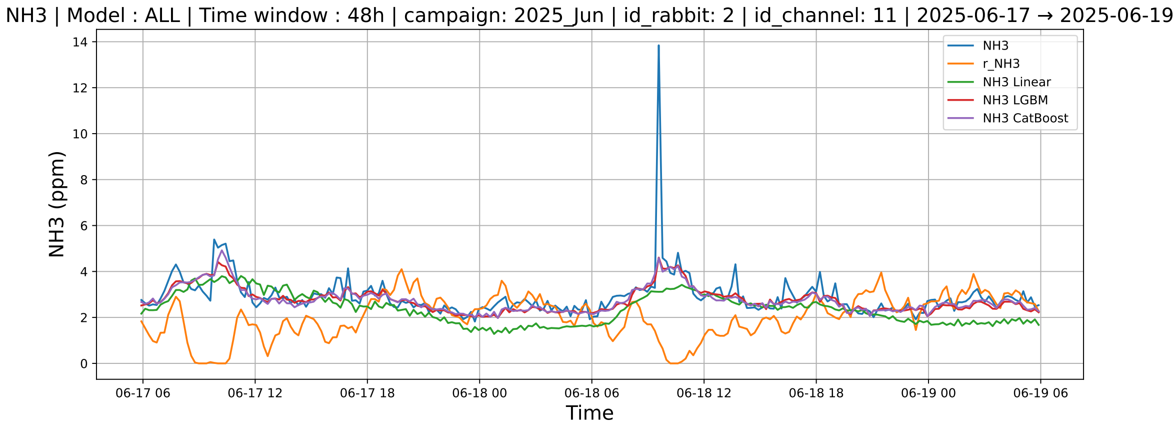


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

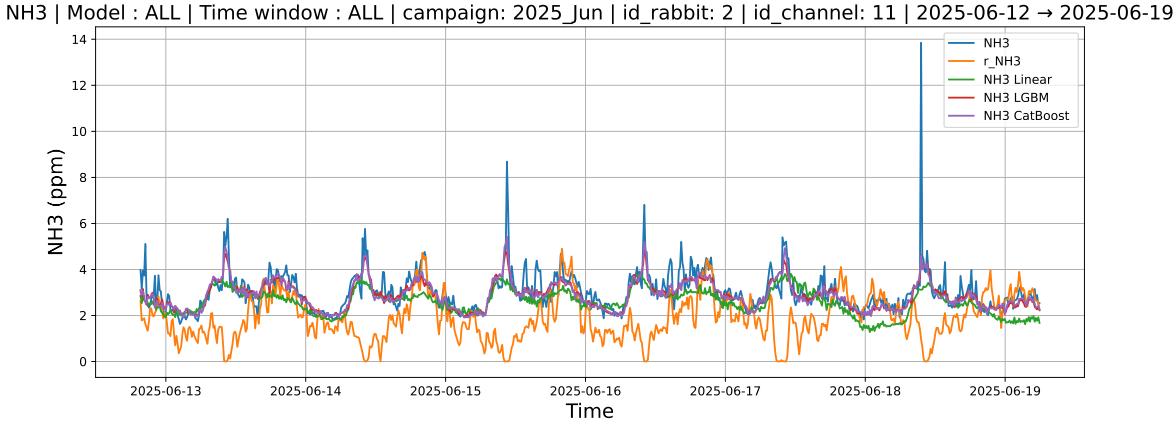
4.2.2.1 Time window : 24



4.2.2.2 Time window : 48

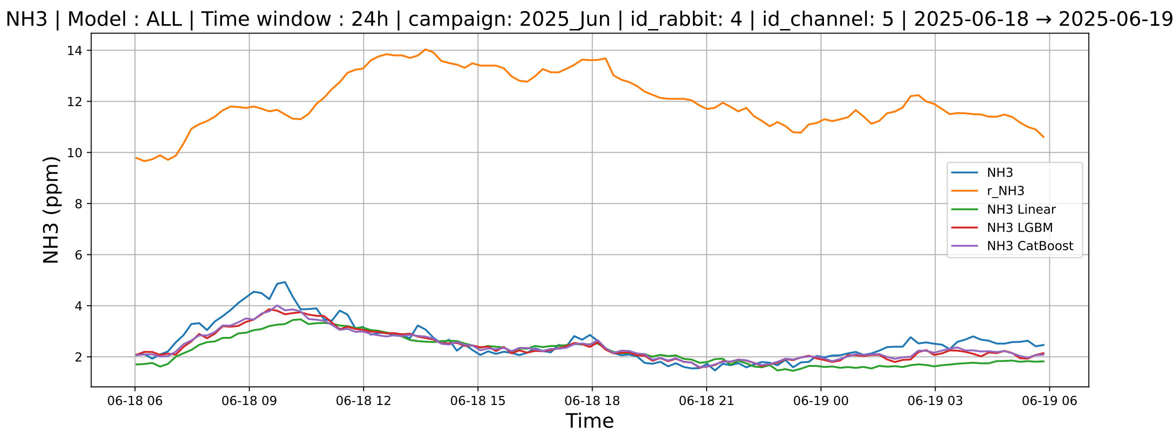


4.2.2.3 Time window : ALL

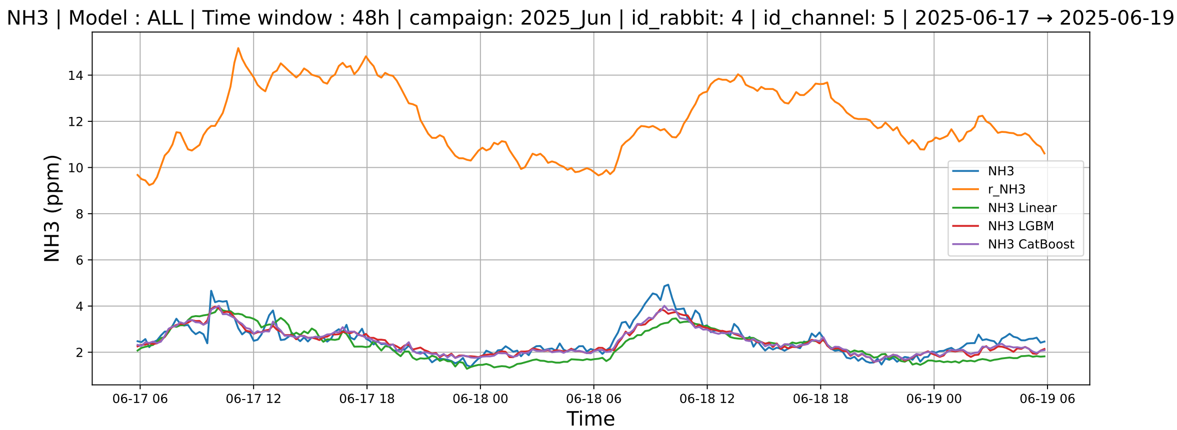


4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

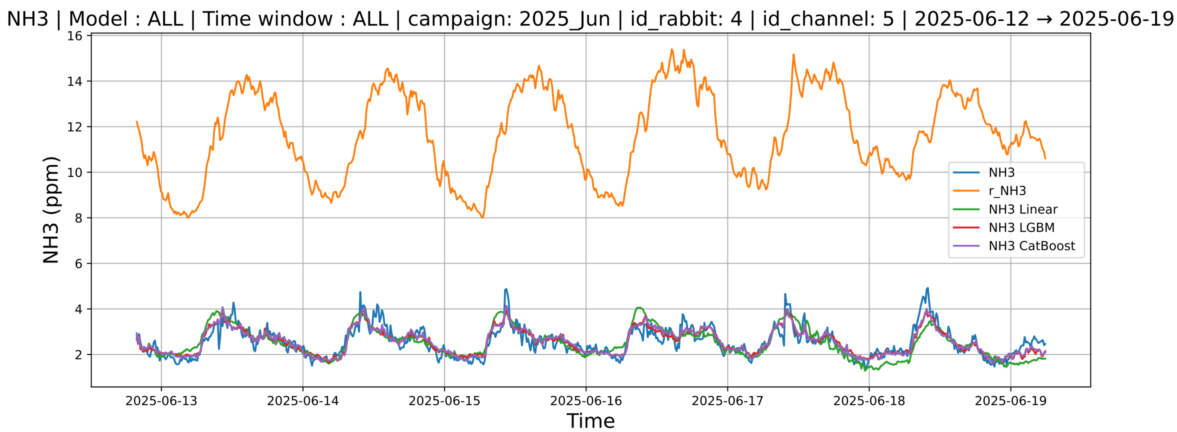
4.2.3.1 Time window : 24



4.2.3.2 Time window : 48



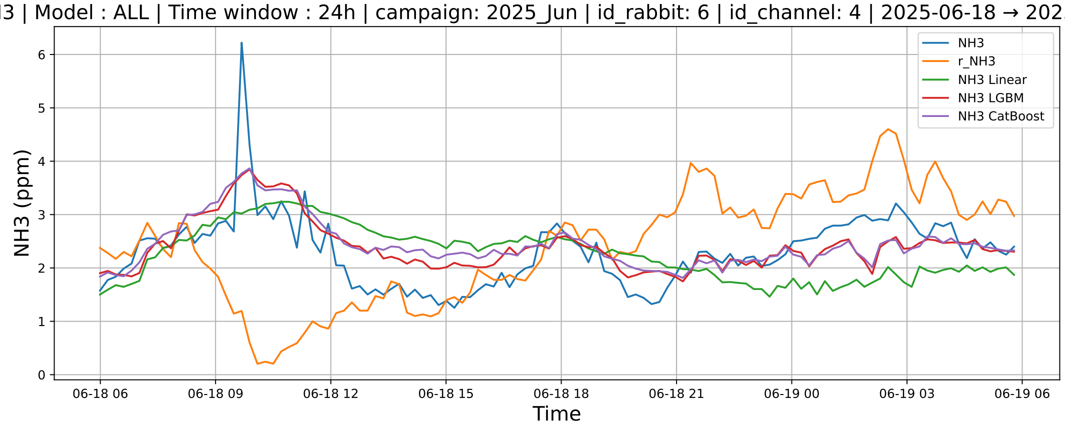
4.2.3.3 Time window : ALL



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

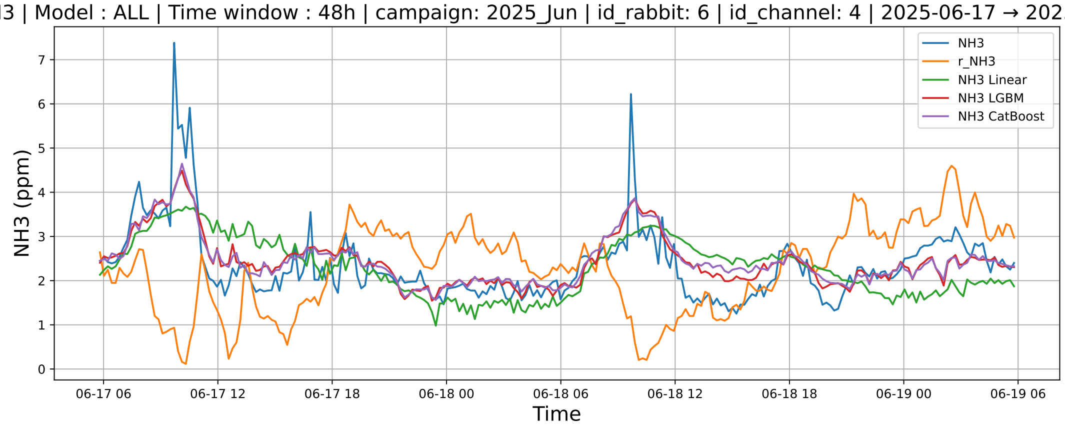
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



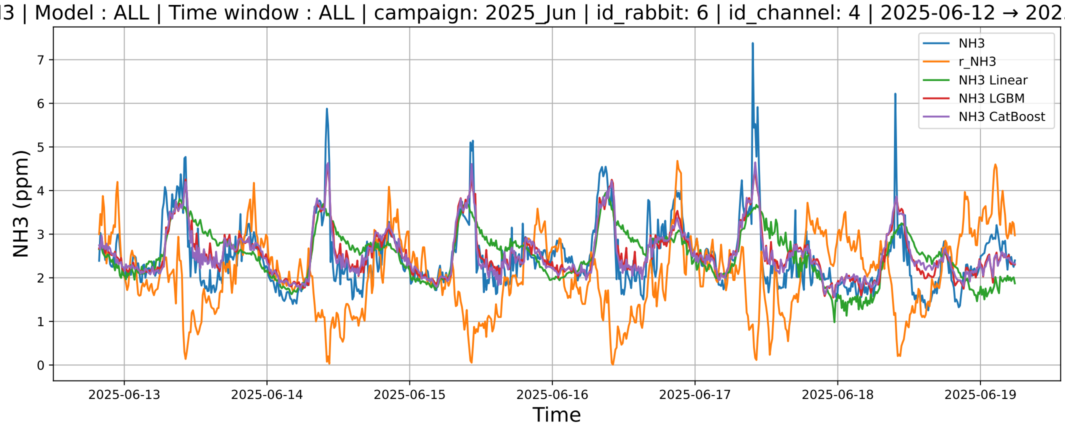
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



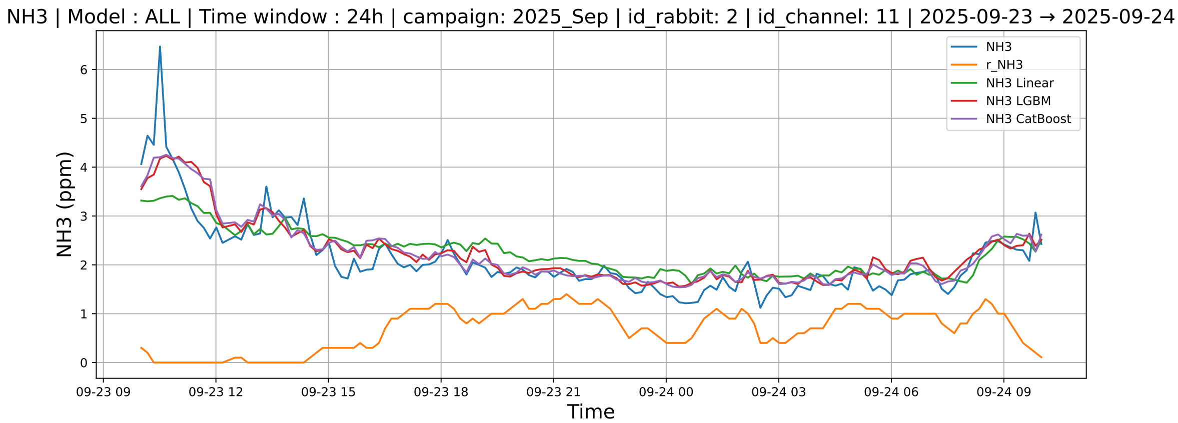
4.2.4.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

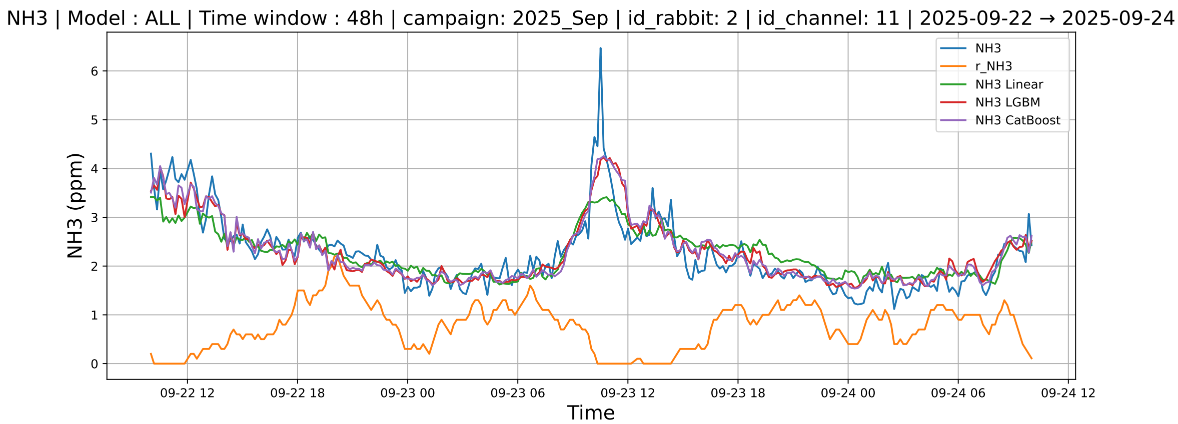


4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

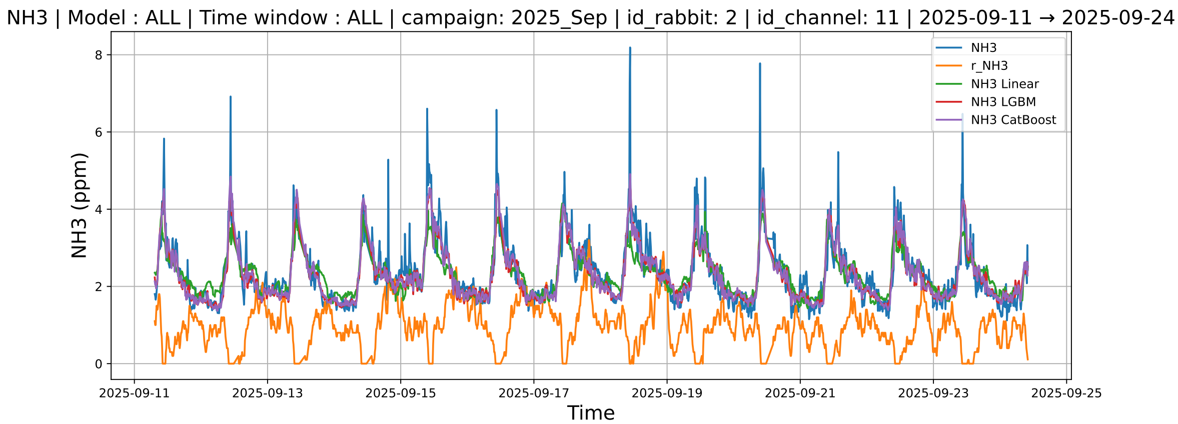
4.2.5.1 Time window : 24



4.2.5.2 Time window : 48

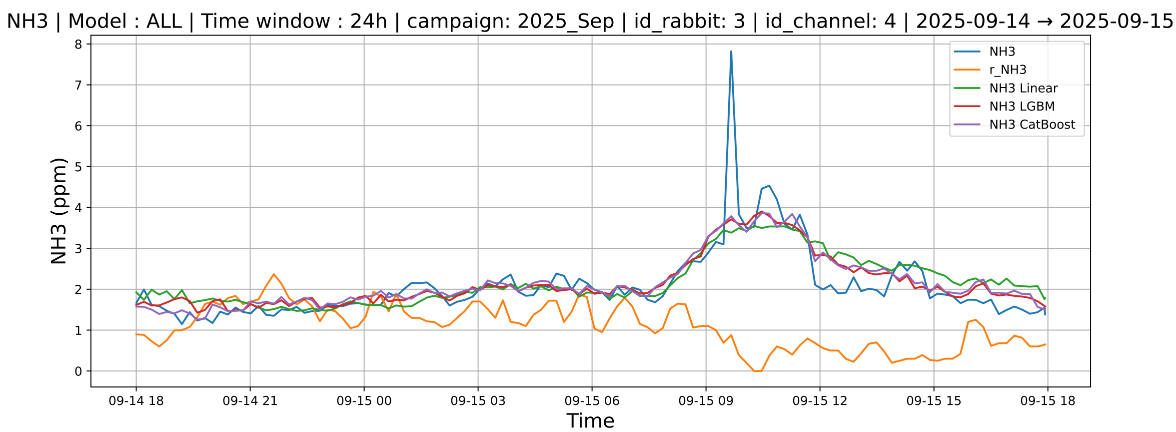


4.2.5.3 Time window : ALL

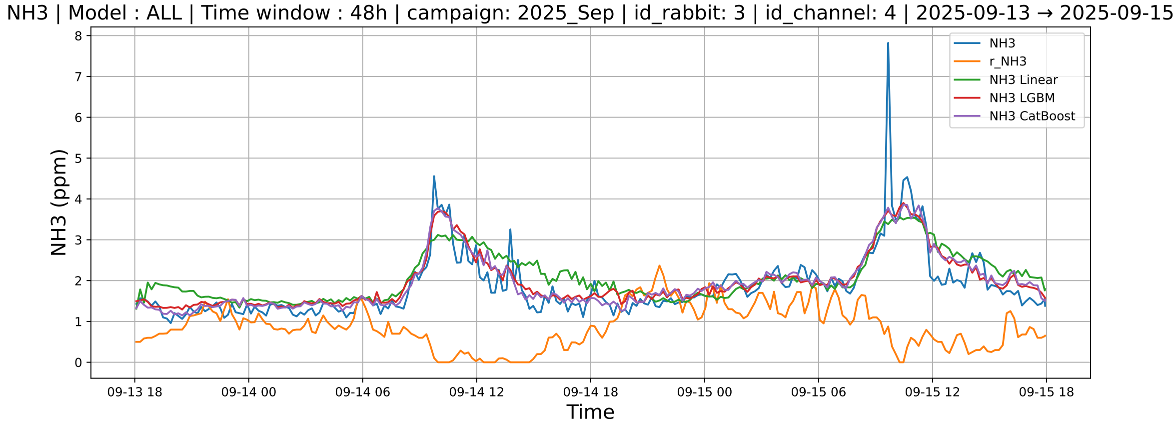


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

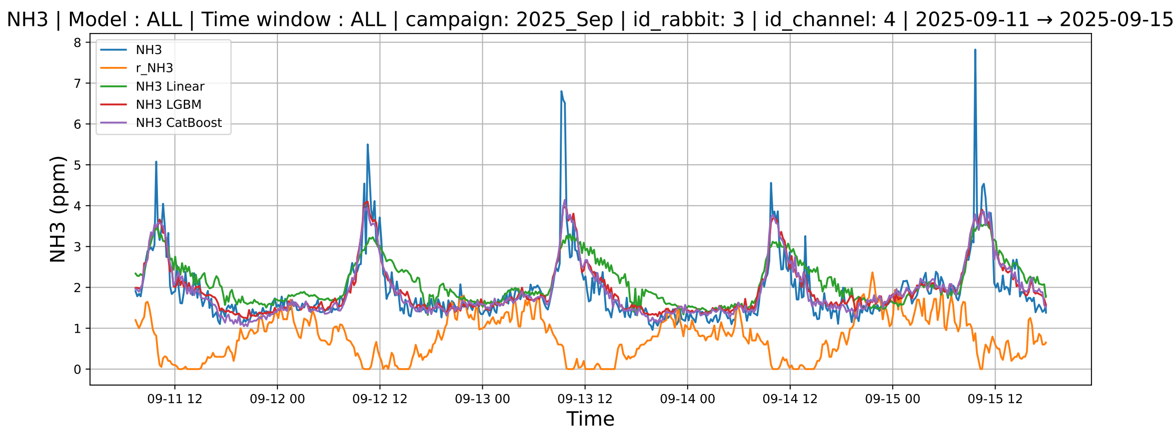
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48



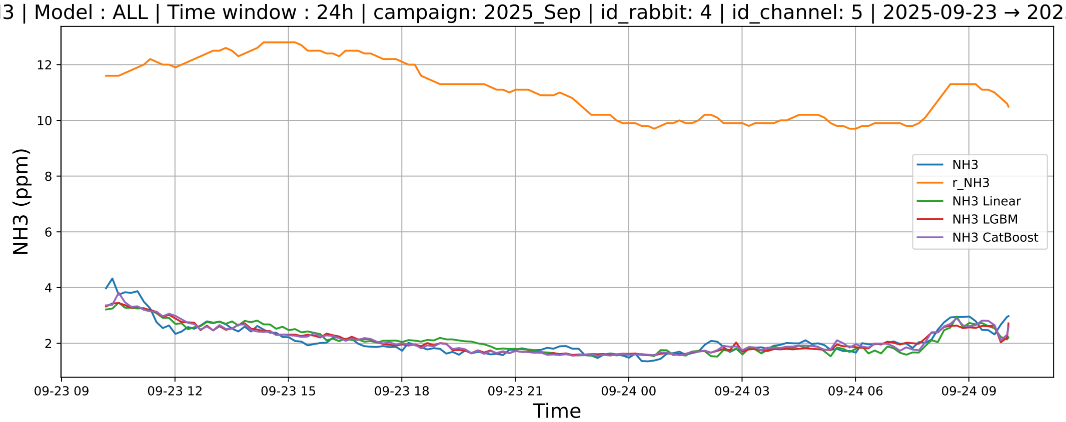
4.2.6.3 Time window : ALL



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

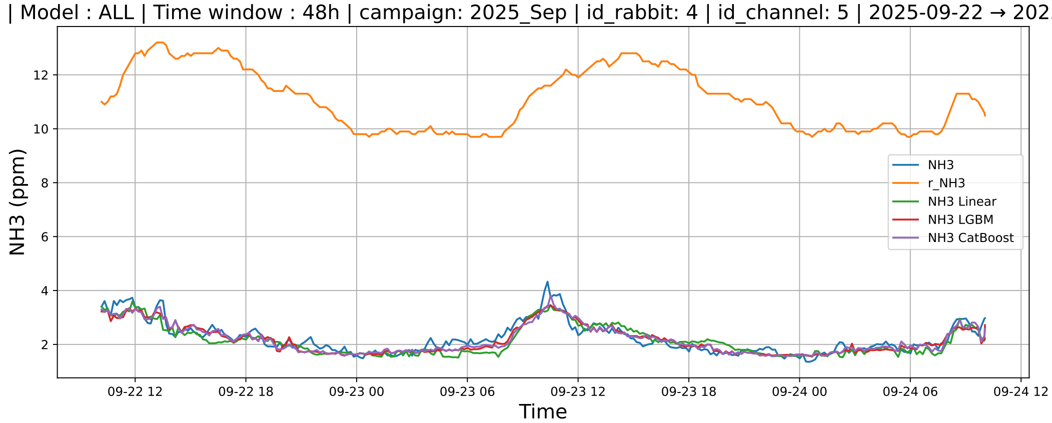
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



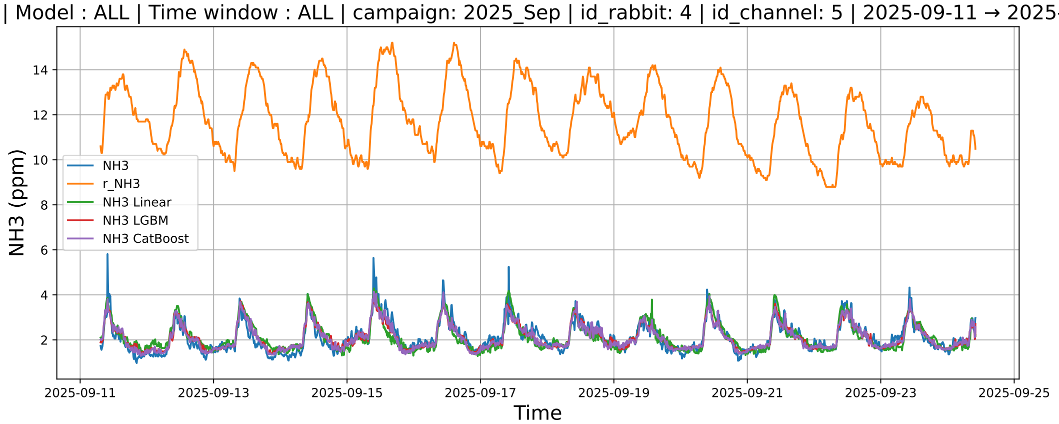
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



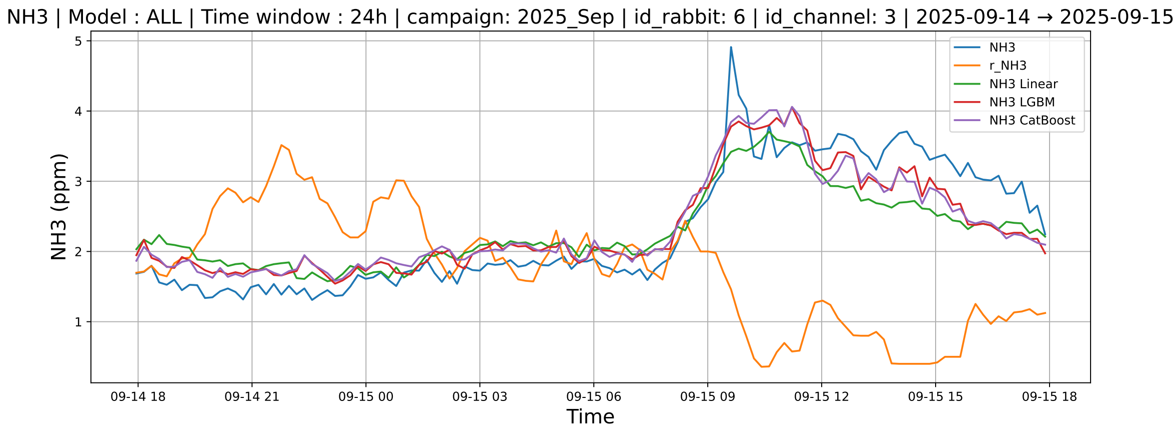
4.2.7.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

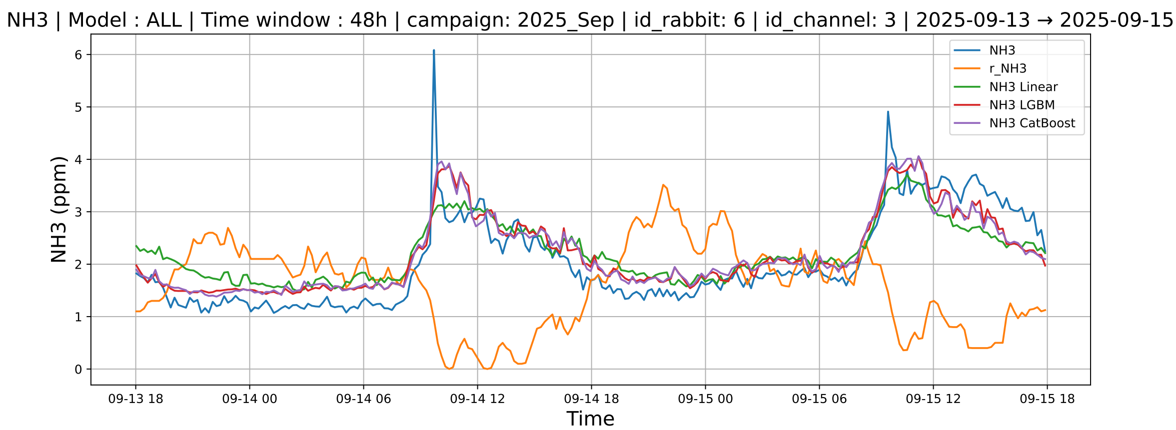


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

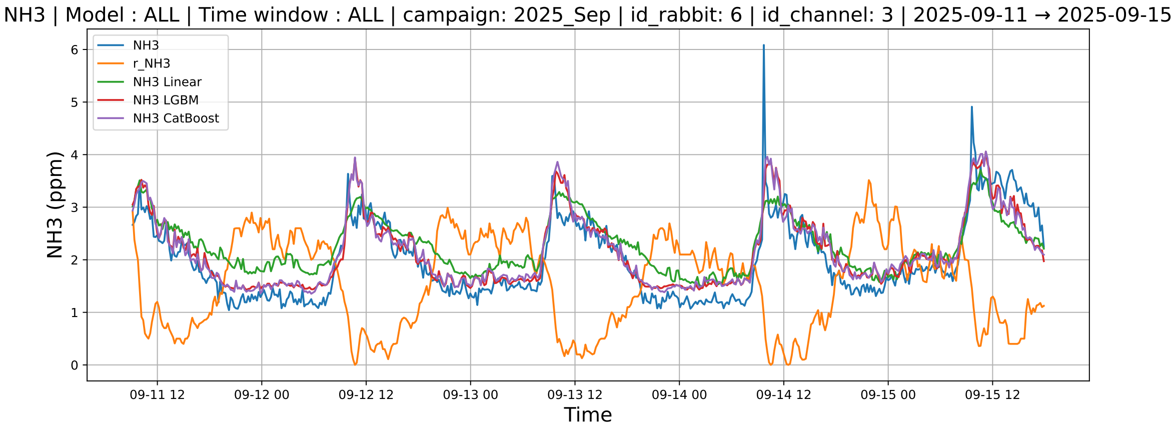
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12
Time